# Class 8
# Physics
# Solutions

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# 1. MATHEMATICS FOR PHYSICS

## Worksheet – 1

## Level – I

1) B 2) C 3) A 4) B 5) A 6) B 7) C 8) B
9) B 10) C 11) A 12) B 13) A 14) B 15) D 16) B
17) C

1. $\cos\theta = \frac{\text{Adjacent side}}{\text{Hypoten side}} = \frac{3}{5}$
2. $\sec\theta = \frac{\text{Hypotenuse side}}{\text{Adjacent side}} = \frac{5}{3}$
3. $\sin\theta = \frac{\text{Opposite side}}{\text{Hypotenuse side}} = \frac{4}{5}$
4. $\tan\theta = \frac{\text{Opposite side}}{\text{Adjacent side}} = \frac{4}{3}$
5. $\cot\theta = \frac{\text{Adjacent side}}{\text{Opposite side}} = \frac{3}{4}$
6. $\cot A = \frac{3}{4} = \frac{adj}{opp}$, $\text{hyp} = \sqrt{opp^2 + adj^2} = 5$

   $\sin A = \frac{opp}{\text{hyp}} = \frac{4}{5}, \cos A = \frac{\text{adj}}{\text{hyp}} = \frac{3}{5}$ ;

   $\therefore \frac{\sin A + \cos A}{\sin A - \cos A} = \frac{\frac{4}{5} + \frac{3}{5}}{\frac{4}{5} - \frac{3}{5}} = 7$
14. $\sin 30° \times \cos 30° + \sin 45° \times \cos 45°$

    $\frac{1}{2} \times \frac{\sqrt{3}}{2} + \frac{1}{\sqrt{2}} \times \frac{1}{\sqrt{2}} = \frac{\sqrt{3}}{4} + \frac{1}{2} = \frac{\sqrt{3}+2}{4} = \frac{2+\sqrt{3}}{4}$
15. $\text{Cosec } 45° - \text{Sec } 45° = \sqrt{2} - \sqrt{2} = 0$
16. $\sin 30° + \cos 60° = \frac{1}{2} + \frac{1}{2} = 1$
17. $4\cos^2\theta = 1 \Rightarrow \cos^2\theta = \frac{1}{4} \Rightarrow \cos\theta = \frac{1}{2} \Rightarrow \theta = 60°$

    $\therefore \text{cosec } 60° = \frac{2}{\sqrt{3}}$

## Worksheet – 2

## Level – II

1) B 2) B 3) D 4) A 5) A 6) A 7) C 8)B
9) B 10) D 11) B 12) C 13) C

7. $y = 2\left(\frac{1}{2}\right) + 3 = 4$
8. $y = 3(-2) + 13 = -5$
9. $y = \frac{1}{2}(2) - 3 = -2$
10. $\because$ Differential coefficient of a constant is zero.
11. $\because \frac{d}{dx}(x^3) = 3x^{3-1} = 3x^2 \left(\because \frac{d}{dx}(x^n) = nx^{n-1}\right)$
12. $\because \frac{d}{dx}(\sqrt{x}) = \frac{d}{dx}\left(x^{\frac{1}{2}}\right) = \frac{1}{2}x^{\frac{1}{2}-1} = \frac{1}{2}x^{\frac{-1}{2}} = \frac{1}{2\sqrt{x}}$
13. $\because \frac{d}{dx}\left(\frac{1}{\sqrt{x}}\right) = \frac{d}{dx}\left(x^{\frac{-1}{2}}\right) = \frac{-1}{2}x^{\frac{-1}{2}-1} = \frac{-1}{2}x^{\frac{-1}{2}-1} = \frac{-1}{2}x^{\frac{-3}{2}} = \frac{-1}{2\sqrt{x^3}}$

# Worksheet – 3
# Level – I

| | | | | |
|---|---|---|---|---|
| 1) C | 2) B | 3) A | 4) A | 5) D |
| 6) A | 7) B | 8) B | 9) B | 10) C |
| 11) A | 12) A | 13) A | | |

4. $\int x^3 dx$ will give a length not time.

5. Integration is the inverse of differentiation.

$$-\frac{dV}{dr} = E \Rightarrow dV = -Edr \Rightarrow V = \int dV = -\int Edr \ \therefore V = -\int Edr$$

7. $\because \int_0^{\pi/2} \cos\, xdx = [\sin\, x]_0^{\frac{\pi}{2}} \Rightarrow = \sin\frac{\pi}{2} - \sin\, 0 = 1 - 0 = 1$

8. $\because \int_0^r (2x - x^3)dx = \left[2\frac{x^2}{2} - \frac{x^4}{4}\right]_0^r = r^2 - \frac{r^4}{4}$

9. $\int 6\sin\, xdx = -6\cos\, x$

10. $\because \int_0^{\mathrm{Q}} \frac{\mathrm{q}}{\mathrm{c}}\mathrm{dq} = \frac{1}{\mathrm{c}}\int_0^{\mathrm{Q}} \mathrm{qdq}(\because \mathrm{c\ is\ constan\ t}) = \frac{1}{\mathrm{c}}\left[\frac{\mathrm{q}^2}{2}\right]_0^{\mathrm{Q}} = \frac{1}{\mathrm{c}}\left[\frac{\mathrm{Q}^2}{2} - 0\right] = \frac{\mathrm{Q}^2}{2\mathrm{c}}$

11 $\because \int_{\mathrm{R}}^{\alpha} \frac{\mathrm{GMm}}{\mathrm{x}^2}\mathrm{dx} = \mathrm{GMm}\int_{\mathrm{R}}^{\alpha} \frac{1}{\mathrm{x}^2}\mathrm{dx}$ ($\because$ G, M, m are constants )

$$= \mathrm{GMm}\left[\frac{\mathrm{x}^{-2+1}}{-2+1}\right]_{\mathrm{R}}^{\alpha} = \mathrm{GMm}\left[-\frac{1}{\mathrm{x}}\right]_{\mathrm{R}}^{\alpha} = -\mathrm{GMm}\left[\frac{1}{\propto} - \frac{1}{\mathrm{R}}\right] = \frac{\mathrm{GMm}}{\mathrm{R}}$$

12. $\int_{-\frac{1}{2}}^{+\frac{1}{2}} \frac{M}{I}x^2 dx = \frac{M}{I}\int_{\frac{-1}{2}}^{+\frac{1}{2}} x^2 dx(\because M, I \text{ are constants }) = \frac{M}{I}\left[\frac{x^3}{3}\right]_{-\frac{1}{2}}^{+\frac{1}{2}}$

$$= \frac{M}{31}\left[\frac{I^3}{8} - \left(-\frac{I^3}{8}\right)\right] = \frac{M}{3l}\left[\frac{I^3}{8} + \frac{I^3}{8}\right] = \frac{M}{3l} \times \frac{I^3}{4} = \frac{MI^2}{12}$$

13. $\because \int_0^{\mathrm{t}} \mathrm{A}\sin\, \omega\mathrm{tdt} = \mathrm{A}\left[\frac{-\cos\, \omega\mathrm{t}}{\omega}\right]_0^{\mathrm{t}} = \frac{\mathrm{A}}{\omega}(1 - \cos\, \omega\mathrm{t})$

# 2. VECTORS

## Worksheet – 1

### Level – I

| | | | |
|---|---|---|---|
| 1. B | 4. B | 7. D | 10.D |
| 2. D | 5. D | 8. A | 11.D |
| 3. C | 6. C | 9. D | 12.A |

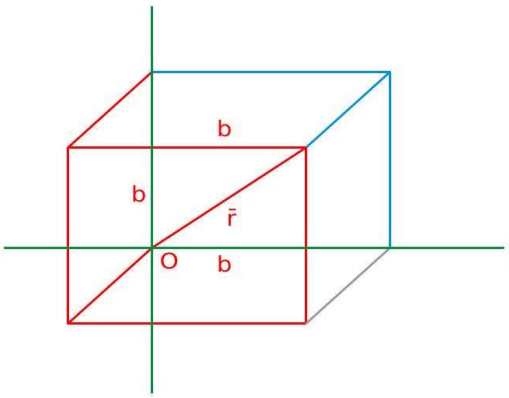
### Level – II

1. D 2. B 3. A 4. D 5. B 6. C 7.B
8. B 9. D 10.B 11. D 12. C 13. A 14. C
15. B 16. C 17. C 18. D 19. A 20. B 21. A
22. C, D 23. $\hat{A} = \frac{\vec{A}}{|\vec{A}|}$ 24. a-q; b-p,q; c-s; d-r,s

2. Coordinate of head of the vector $r_1(x_1, y_1, z_1) = (2,1,0)$
coordinate of tail of the vector $r_2(x_2, y_2, z_2) = (-4,2,-3)$
Then magnitude of vector $|\bar{r}| = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$
$|\bar{r}| = \sqrt{(-4-2)^2 + (2-1)^2 + (-3-0)^2}$
$|\bar{r}| = \sqrt{(-6)^2 + (1)^2 + (-3)^2}$
$|\bar{r}| = \sqrt{36 + 1 + 9}$;
$|\bar{r}| = \sqrt{46}$

3. $\bar{A} = 5\hat{i} + \hat{j} - 2\hat{k}$
$|\bar{A}| = \sqrt{5^2 + 1^2 + (-2)^2}$
$|\bar{A}| = \sqrt{25 + 1 + 4}$
$|\bar{A}| = \sqrt{30}$

4. Three vectors with magnitude $b$ units along the coordinate axes represent the adjacent sides of cube. Let $\bar{r}$ be the vector along diagonal from origin




$\bar{r} = b\hat{i} + b\hat{j} + b\hat{k}$
$\bar{r} = b(\hat{i} + \hat{j} + \hat{k})$
$|\bar{r}| = \sqrt{3}b$
∴ unit vector along the diagonal
$\hat{r} = \frac{\bar{r}}{|\bar{r}|} = \frac{b(\hat{i} + \hat{j} + \hat{k})}{\sqrt{3}b}$
$\therefore \hat{r} = \frac{\hat{i} + \hat{j} + \hat{k}}{\sqrt{3}}$

5. Position of particle in coordinate system $A(3,2,5)$ then position vector from origin is $\overline{OA} = 3\hat{i} + 2\hat{j} + 5\hat{k}$

6. $\overline{A} = 3\hat{i} + 3\hat{j} + 9\hat{k}\ \overline{B} = \hat{i} + aj + 3\hat{k}$
if $a_1\hat{i} + a_2\hat{j} + a_3\hat{k}$ and $b_1\hat{i} + b_2\hat{j} + b_3\hat{k}$ are parallel vectors
$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \frac{a_3}{b_3} \Rightarrow \frac{3}{1} = \frac{3}{a} = \frac{9}{3}$
$\frac{3}{a} = \frac{9}{3} \Rightarrow 9a = 9$
$\therefore a = 1$

8. Magnitude of required vector $|\overline{B}| = 10$
required vector is opposite to $\bar{A} = 6\hat{i} - 8\hat{j}$
$\therefore$ required vector $\overline{B} = |\overline{B}| \times -\hat{A}$
$\therefore \overline{B} = 10 \times -\left(\frac{\overline{A}}{|\overline{A}|}\right)$
$\bar{B} = 10 \times -\frac{(6\hat{i} - 8\hat{j})}{\sqrt{6^2 + (-8)^2}}\ \bar{B} = 10 \times \frac{(-6\hat{i} + 8\hat{j})}{10}$
$\therefore \overline{B} = -6\hat{i} + 8\hat{j}$

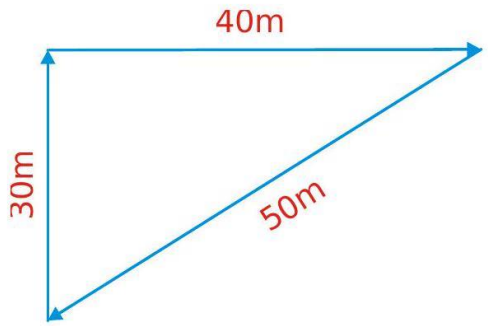
9. Magnitude of position vector $r = 4$
Slope $\tan\theta = \frac{1}{\sqrt{3}} \Rightarrow \theta = 30°$
$\therefore$ The position vector $\bar{r} = r\cos\theta\hat{i} + r\sin\theta\hat{j}$
$\bar{r} = 4\cos 30°\hat{i} + 4\sin 30°\hat{j}$
$\bar{r} = 4 \times \frac{\sqrt{3}}{2}\hat{i} + 4 \times \frac{1}{2}\hat{j}$
$\bar{r} = 2\sqrt{3}\hat{i} + 2\hat{j}$
$\therefore \bar{r} = 2[\sqrt{3}\hat{i} + \hat{j}]$

10. $\overline{A} = 0.4\hat{i} + 0.3\hat{j} + c\hat{k}$ is a unit vector i.e. $|\bar{A}| = 1$
$0.16 + 0.09 + c^2 = 1$
$c^2 = 1 - 0.25$
$c^2 = 0.75 \Rightarrow c = \sqrt{0.75} = 0.866$

11. $(x_1, y_1, z_1) = (1,2,3)$
$(x_2, y_2, z_2) = (5,4,2)$
$\therefore$ displacement vector $\bar{r} = (x_2 - x_1)\hat{i} + (y_2 - y_1)\hat{j} + (z_2 - z_1)\hat{k}$
$\Rightarrow \bar{r} = (5-1)\hat{i} + (4-2)\hat{j} + (2-3)\hat{k}$
$\bar{r} = (4\hat{i} + 2\hat{j} - \hat{k})m$

12. Direction of car is along $\bar{r} = -4\hat{i} + 3\hat{j}$
speed $v = 10$ m/s
velocity of car $\bar{V} = v \cdot \frac{\bar{r}}{|r|}$
$\bar{v} = 10\frac{(-4\hat{i} + 3\hat{j})}{\sqrt{(-4)^2 + 3^2}}$
$\bar{v} = 10\frac{(-4\hat{i} + 3\hat{j})}{3} = (-8\hat{i} + 6\hat{j})$m/s

13. Displacement from original position $A$ is zero




14. If $\bar{r} = 0.2\hat{\imath} + 0.3\hat{\jmath} + Z\hat{k}$ is a unit vector then its magnitude $|\bar{r}| = 1$
$|\bar{r}| = \sqrt{(0.2)^2 + (0.3)^2 + Z^2} = 1$
$0.04 + 0.09 + Z^2 = 1$
$z^2 = 1 - 0.13 = 0.87$
so the answer is none

15. $\bar{P} = \hat{\imath} + 3\hat{\jmath} - 7\hat{k}Q = 5\hat{\imath} - 2\hat{\jmath} + 4\hat{k}$
$\overline{PQ} = \bar{Q} - \bar{P} = (5-1)\hat{\imath} + (-2-3)\hat{\jmath} + (4+7)\hat{k}$
$\overline{PQ} = 4\hat{\imath} - 5\hat{\jmath} + 11\hat{k}$
$\therefore |\overline{PQ}| = \sqrt{4^2 + (-5)^2 + 11^2}$
$|\overline{PQ}| = \sqrt{16 + 25 + 121}$
$|\overline{PQ}| = \sqrt{162}$

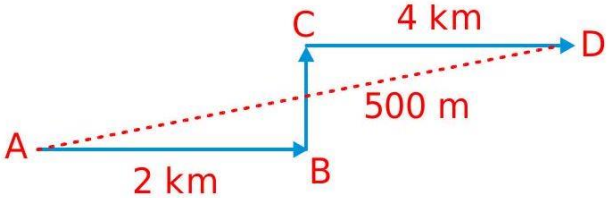
16. $2\hat{\imath} - y\hat{\jmath} + 3\hat{k}$ is parallel to $2\hat{\imath} + 3\hat{\jmath} + 12\hat{k}$ then $\frac{-y}{3} = \frac{3}{12} \Rightarrow y = \frac{-3}{4}$

17. $\bar{A} = 3\hat{\imath} - 4\hat{\jmath}$ parallel vector $\bar{B}$ should be of the form $\bar{B} = \lambda\bar{A}\,\lambda$ should be positive value
$\overline{\mathrm{B}} = \lambda(3\hat{\imath} + 4\hat{\mathbf{j}})$
it $\lambda = \frac{1}{5}$
$$\bar{B} = \frac{1}{5}(3\hat{\imath} - 4\hat{\jmath})$$
$$\therefore \bar{B} = \frac{3}{5}\hat{\imath} - \frac{4}{5}\hat{\jmath}$$

18. $\bar{A} = 5\hat{\imath} - 10\hat{\jmath}$ and $\bar{B} = 2m\hat{\imath} + \frac{1}{2}\hat{\jmath}$
if $\bar{A}$ is parallel to $\bar{B}$ then
$$\frac{5}{2m} = \frac{-10}{\frac{1}{2}}$$
$$5 \times \frac{1}{2} = -10 \times 2 \text{ m}$$
$$\therefore \text{m} = \frac{-1}{8}$$

19. Magnitude of displacement of the mosquito is $\left(\sqrt{7^2 + 4^2 + 3^2}\right)$
$\therefore$ displacement $= \sqrt{74}$ft
If $A$ is the starting point

20. Car moved from A to B 2 km towards east
$\therefore \overline{\mathrm{AB}} = 2\hat{\imath}$km
After taking - perpendicular left turn reached $C$ (north)
then $\therefore \overline{\mathrm{BC}} = 500\ \mathrm{m}\hat{\jmath} \Rightarrow \overline{\mathrm{BC}} = (0.5\ \mathrm{km})\hat{\jmath}$
from $C$ to $D$ it made perpendicular right turn travelling 4.00 km then $\therefore \overline{CD} = 4\hat{\imath}$



displacement of car is $\overline{AD}$

$\overline{AD} = \overline{AB} + \overline{BC} + \overline{CD}$

$\overline{AD} = 2\hat{\imath} + 0.5\hat{\jmath} + 4\hat{\imath}$

$\overline{AD} = 6\hat{\imath} + 0.5\hat{\jmath}$

$|\overline{AD}| = \sqrt{6^2 + (0.5)^2} = \sqrt{36 + 0.25} = \sqrt{36.25} = 6.02$ km

21. $\bar{A} = \frac{3}{5}\hat{\imath} - \frac{2}{5}\hat{\jmath} + \frac{2\sqrt{3}}{5}\hat{k}$

$$|\bar{A}| = \sqrt{\left(\frac{3}{5}\right)^2 + \left(-\frac{2}{5}\right)^2 + \left(\frac{2\sqrt{3}}{5}\right)^2}$$

$$|\bar{A}| = \sqrt{\frac{9}{25} + \frac{4}{25} + \frac{12}{25}} = \sqrt{\frac{25}{25}} = 1$$

$\therefore |\bar{A}| = 1$

# Level – III

25. C 26. D 27. A 28. A,B,D
29. a-r,s; b-p,q;c-r,s;d-r,s 30. D 31. A 32. C

25. The vector of magnitude 6 along negative $y$-axis is $-6\hat{\jmath}$

26. The vector of magnitude $10\sqrt{2}$ in $xy$-plane is $10\hat{\imath} + 10\hat{\jmath}$

27. Initial point $P(x_1, y_1, z_1)$ and terminal point $Q(x_2, y_2, z_2)$ then the vector $\overline{PQ} = (x_2 - x_1)\hat{\imath} + (y_2 - y_1)\hat{\jmath} + (z_2 - z_1)\hat{k}$

$$|\overline{PQ}| = [(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2]^{\frac{1}{2}}$$

30. Initial point $P(1,2,3)$ final point $Q(5,4,2)m$ then displacement vector is

$$\overline{PQ} = (5-1)\hat{\imath} + (4-2)\hat{\jmath} + (2-3)\hat{k}$$

$$\overline{PQ} = 4\hat{\imath} + 2\hat{\jmath} - \hat{k}$$

31. $\bar{r}_1 = (2\hat{\imath} + 3\hat{\jmath} + 5\hat{k})m\ \bar{r}_2 = (\hat{\imath} - 2\hat{\jmath} + \hat{k})$

$\bar{r} = \bar{r}_2 - \bar{r}_1 = (1-2)\hat{\imath}(-2-3)\hat{\jmath} + (1-5)\hat{k}$

$\bar{r} = -\hat{\imath} - 5\hat{\jmath} - 4\hat{k}$

32. $\vec{F}_1 = \hat{\imath} + \hat{k}, \vec{F}_2 = 2\hat{\jmath} + 3\hat{k}, \vec{F}_3 = 3\hat{\imath}, \vec{F}_4 = 3\hat{\jmath} - 4\hat{\imath}$

$$\vec{F}_{\text{net}} = \vec{F}_1 + \vec{F}_2 + \vec{F}_3 + \vec{F}_4$$

$$\vec{F}_{\text{net}} = (\hat{\imath} + \hat{k}) + (2\hat{\jmath} + 3\hat{k}) + (3\hat{\imath}) + (3\hat{\jmath} - 4\hat{\imath})$$

$$\vec{F}_{\text{net}} = 5\hat{\jmath} + 4\hat{k}$$

The body will move in yz plane

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# Worksheet – 2

# Level – I

1. B 3. C 5. B 7. A 9. A
2. C 4. C 6. C 8. A

1. $\ell^2 + m^2 + n^2 = \frac{x^2}{r^2} + \frac{y^2}{r^2} + \frac{z^2}{r^2}$

$\ell^2 + m^2 + n^2 = \dfrac{x^2 + y^2 + z^2}{r^2}$

$\therefore r^2 = x^2 + y^2 + z^2$

$\ell^2 + m^2 + n^2 = 1$

2. $\sin^2\ \alpha + \sin^2\ \beta + \sin^2\ \gamma = 1 - \cos^2\ \alpha + 1 - \cos^2\ \beta + 1 - \cos^2\ \gamma$

$= 3 - (\cos^2\ \alpha + \cos^2\ \beta + \cos^2\ \gamma)$

$= 3 - 1$

$\therefore \sin^2\ \alpha + \sin^2\ \beta + \sin^2\ \gamma = 2$

3. Component along east $= p$

component + along north $= p$

$f = \sqrt{P^2 + P^2} = \sqrt{2}P$

4. Horizontal component of weight mgcos 90°

$W_x = 0$

5. The component of a vector $r$ along $x$-axis will have maximum value it $r$ is along positive $x$-axis

6. It depends on frame of reference

7. $\bar{A} = 3\hat{\imath} - 2\hat{\jmath} + \hat{k}\ \bar{B} = \hat{\imath} - 3\hat{\jmath} + 5\hat{k}$ and $\bar{C} = 2\hat{\imath} + \hat{\jmath} - 4\hat{k}$

$|\bar{A}| = \sqrt{14}$

$|\bar{B}| = \sqrt{35}$

$|\bar{C}| = 21$

Here $|\bar{A}|^2 + |C|^2 = |B|^2$ (pythagorous theorem)

$\therefore$ The given vector from right angled triangle.

# Level – II

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1. B | 2. A | 3. D | 4. C | 5. D | 6. A | 7. C | 8. B |
| 9. B | 10. C | 11. A | 12. D | 13. A | 14. B | 15. D | 16. B |
| 17. 3 | 18. A | 19. D | | | | | |

1. $x$ component of vector = 25 units

y component of vector = 40 units

vector $\bar{r} = x\hat{\imath} + y\hat{\jmath}$

$\bar{r} = -25\hat{\imath} + 40\hat{\jmath}$

magnitude $|\bar{r}| = \sqrt{(25)^2 + (40)^2} = \sqrt{1600 + 625} = \sqrt{2225} = 5\sqrt{89}$ units

Direction of this vector $\theta = \tan^{-1}\left(\frac{x}{y}\right) = \tan^{-1}\left(\frac{-25}{40}\right)$

$\theta = \cos^{-1}\left(\frac{-25}{\sqrt{89}}\right) = \cos^{-1}\left(\frac{-5}{\sqrt{89}}\right)$ with $x$-axis

$= \cos^{-1}\left[\dfrac{-25}{5\sqrt{89}}\right]$

2. $P = \sqrt{F_x + F_y}$

$P^2 = F_x^2 + \left(\dfrac{3P}{5}\right)^2$

$F_x^2 = \dfrac{16P^2}{25}$

$F_x = \dfrac{4P}{5}$

3. $x$ component at angle 30°

$F_x = F\cos\ \theta$

$F_x = 100 \times \dfrac{\sqrt{3}}{2} = 50\sqrt{3}N$

y-component $F_y = F\sin\ \theta$

$F_y = 100 \times \frac{1}{2} = 50N$

4. $P(2,3,5)Q(3,4,5)$
$\overline{PQ}(3,-2)\hat{\imath} + (4-3)\hat{\jmath} + (5-5)\hat{k}$
$\overline{\mathrm{PQ}} = \hat{\imath} + \hat{\jmath}$

5. $\bar{A} = 3\hat{\imath} + 4\hat{\jmath} \bar{B} = 7\hat{\imath} + 24\hat{\jmath}$
$|\overline{\mathrm{A}}| = 5|\overline{\mathrm{B}}| = 25$
$\overline{\mathrm{A}} = \frac{\overline{\mathrm{A}}}{|\overline{\mathrm{A}}|} = \frac{1}{5}(3\hat{\imath} + 4\hat{\jmath})$
The vector with magnitude same as $\bar{B}$ and parallel to $\bar{A}$ is $\bar{C} = |\bar{B}|\hat{A}$
$$\overline{\mathrm{C}} = 25 \times \frac{1}{5}(3\hat{\imath} + 4\hat{\jmath})$$
$$\overline{\mathrm{C}} = 15\hat{\imath} + 20\hat{\jmath}$$

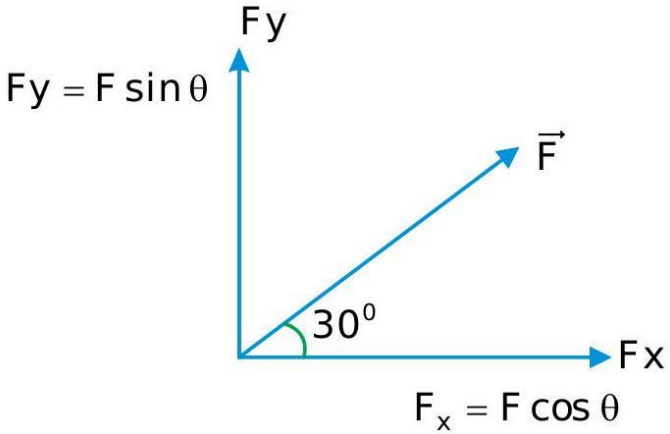
6. The direction ratios of $3\hat{\imath} + 4\hat{\jmath} - 6\hat{k}$ are $3,4,-6$
the direction cosines of $\left(\frac{x}{r}, \frac{y}{r}, \frac{z}{r}\right)$ where are =
$$r = \sqrt{x^2 + y^2 + z^2} = \sqrt{9 + 16 + 36} = \sqrt{61}\ \left(\frac{3}{\sqrt{51}}, \frac{4}{\sqrt{51}}, \frac{-6}{\sqrt{51}}\right)$$

7. $\bar{A} = 2\hat{\imath} + 3\hat{\jmath} - \hat{k}\ |\bar{A}| = \sqrt{14} \Rightarrow -A_y = 3\cos\theta = \frac{A_y}{|\bar{A}|} \Rightarrow \theta = \cos^{-1}\left(\frac{3}{\sqrt{14}}\right)$
$$|\vec{A}| = \sqrt{14}\, Ay = 3$$

8. When $|\bar{A}| = |\bar{B}|$ the components of $A$ and $B$ need not be the same. They may or may not be equal.

9. $\overline{\mathrm{A}} = 2\hat{\imath} - 3\hat{\jmath} + 4\hat{k}$
component in yz-plane $= \sqrt{A_y^2 + A_z^2} = \sqrt{(-3)^2 + 4^2} = 5$
component in zx-plane $= \sqrt{A_x^2 + A_z^2} = \sqrt{2^2 + 4^2} = \sqrt{20} = 2\sqrt{5}$

10. $\mathrm{F_x} = \mathrm{F}\cos\theta; \mathrm{F_y} = \mathrm{F}\sin\theta$




$\mathrm{F} = 10\ \mathrm{N}\theta = 30°$
$\mathrm{F_x} = 10\cos 30°\ \mathrm{F_y} = 10\sin 30°$
$$\mathrm{F_x} = 10 \times \frac{\sqrt{3}}{2}\ \mathrm{F_y} = 10 \times \frac{1}{2}$$
$$\mathrm{F_x} = 5\sqrt{3}\ \mathrm{N},\ \ \mathrm{F_y} = 5\ \mathrm{N}$$

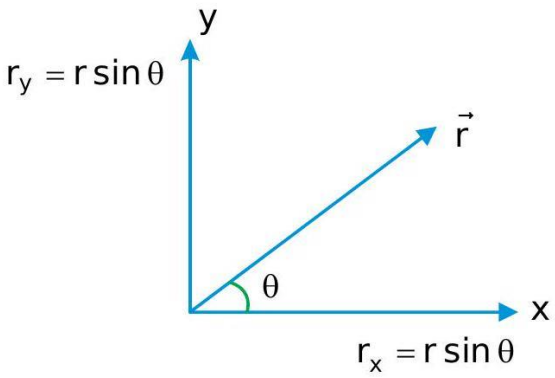
11. $\bar{r} = x\hat{\imath} + y\hat{\jmath}$ and $x = \sqrt{3}y$
$$\tan\theta = \frac{y}{x} \Rightarrow \tan\theta = \frac{y}{\sqrt{3}}$$
$$\theta = \tan^{-1}\left(\frac{1}{\sqrt{3}}\right) = 30°$$

12. $\mathrm{F}^2 = \mathrm{x}^2 + \mathrm{y}^2$
$$y = \sqrt{F^2 - x^2}$$

13. $\bar{F}_1 = (10\hat{\imath} + 14\hat{\jmath})N\bar{F}_2 = 6\hat{\imath} + 10\hat{\jmath}m = 8kg$

$$\bar{F} = \bar{F}_1 + \bar{F}_2 = 16\hat{i} + 24\hat{j}$$
$$a = \frac{\bar{F}}{m} = \frac{16\hat{i} + 24\hat{j}}{8}$$
$$\therefore \vec{a}(2\hat{i} + 3\hat{j})m/s^2$$

14. Let the vector be $\bar{r}$ and its magnitude ' $r$ ' vertical component is $r\sin\theta$ given $r\sin\theta = y$




15. Vertical component $r\sin\theta = H$

$$r = \frac{H}{\sin\theta}$$

horizontal component $= r\cos\theta = \frac{H}{\sin\theta} \times \cos\theta$

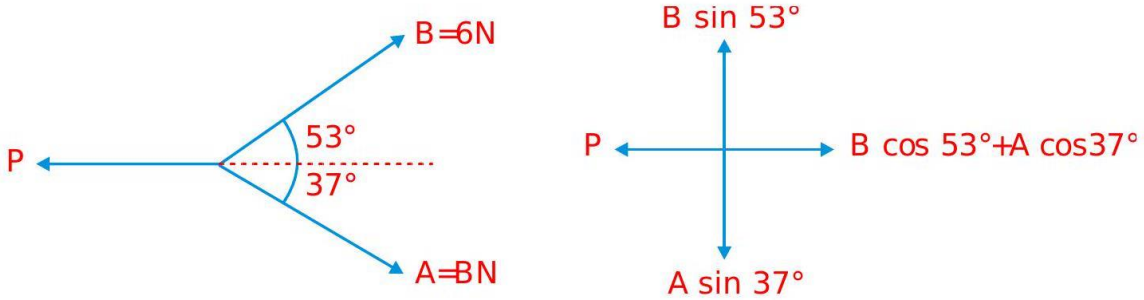
$\therefore$ horizontal component $= H\cot\theta$

16. $x$ and $y$ components of $\bar{A} + \bar{B}$ are 10 and 9
$x$ and $y$ components of $\bar{A}$ are 4 and 6
then $x$ and $y$ components of $\bar{B}$ are (10-4) and (9-6)
i.e $x$ component of $\bar{B} = 6$
y component of $\bar{B} = 3$
angle made by $\bar{B}$ with y axis

$$\tan\theta = \frac{y - \text{component}}{x - \text{component}}$$
$$\tan\theta = \frac{3}{6} \Rightarrow \theta = \tan^{-1}\left(\frac{1}{2}\right)$$

17. $P = B\cos 53° + A\cos 37°$

$$P = 6 \times \frac{3}{5} + 8 \times \frac{4}{5}$$
$$P = \frac{18 + 32}{5} = \frac{50}{5}$$
$$P = 10\ N$$




18. Angle made by the velocity vector of first ball with horizontal

$$\tan\theta_1 = \frac{y - \text{component}}{x - \text{component}} = \frac{u_y}{v_x}$$
$$\tan\theta_1 = \frac{\sqrt{3}}{1} \Rightarrow \theta_1 = \tan^{-1}(\sqrt{3}) = 60°$$

Angle made by the velocity vector of second ball with the horizontal $\tan\theta_2 = \frac{\text{y-component}}{\text{x-component}}$

$\tan\theta_2 = \frac{2}{2}\theta_2 = \tan^{-1}(1) = 45^\circ$

$\therefore$ angle between their directiion $\theta = \theta_1 - \theta_2 = 60^\circ - 45^\circ$

$\theta = 15^0$

19. $\frac{1}{-8}(2\hat{\imath} + 3\hat{\jmath} + 7\hat{k}) = -\frac{1}{4}\hat{\imath} - \frac{3}{8}\hat{\jmath} + x\hat{k}$

$-\frac{1}{4}\hat{\imath} - \frac{3}{8}\hat{\jmath} - \frac{7}{8}\hat{k} = -\frac{1}{4}\hat{\imath} - \frac{3}{8}\hat{\jmath} + x\hat{k}$

$\therefore x = -\frac{7}{8}$

# Level – III

20. B,C,D 21. a-q; b-p; c-r; d-s,t 22. A 23. A 24. 2 25. 9 26. C
27. C 28. A 29. B 30. D 31. D 32. C

22. Magnitude of $\hat{\imath} + \hat{k} = \sqrt{2}$

$\alpha = \cos^{-1}\left(\frac{A_x}{A}\right)\beta = \cos^{-1}\left(\frac{A_y}{A}\right)\gamma = \cos^{-1}\left(\frac{A_z}{A}\right)$

$\cos\alpha = \frac{1}{\sqrt{2}}\cos\beta = 0\cos\gamma = \frac{1}{\sqrt{2}}$

direction cosines of $\hat{\imath} + \hat{k}$ is $\frac{1}{\sqrt{2}}, 0, \frac{1}{\sqrt{2}}$

23. Angle made by $\bar{A}$ with x-axis is $45^\circ$

24. $\bar{A} = \hat{\imath} + \hat{\jmath} + \sqrt{2}k\ |\bar{A}| = 2$

angle made by $\bar{A}$ with $z$-axis $\gamma = \cos^{-1}\left(\frac{A_z}{|\bar{A}|}\right)$

$\gamma = \cos^{-1}\left(\frac{\sqrt{2}}{2}\right)$

$\gamma = \cos^{-1}\left(\frac{1}{\sqrt{2}}\right)$

$\gamma = 45^\circ$

25. $y$-component of $6\hat{\imath} + 9\hat{\jmath}$ is 9

26 conceptual

27. conceptual

28. conceptual

29. $\bar{A} = 6\hat{\imath} + 3\hat{\jmath}$

angle made with $y$-axis is $\tan\theta = \frac{A_x}{A_y}$

$\tan\theta = \frac{6}{3} \Rightarrow \theta = \tan^{-1}(2)$

30. $\bar{A} = A\cos\theta_1\hat{\imath} + A\sin\theta_1\hat{\jmath}\ \bar{B} = B\cos\theta_2\hat{\imath} + B\sin\theta_2\hat{\jmath}$

$A = 2\text{ cm}\theta_1 = 30^\circ$

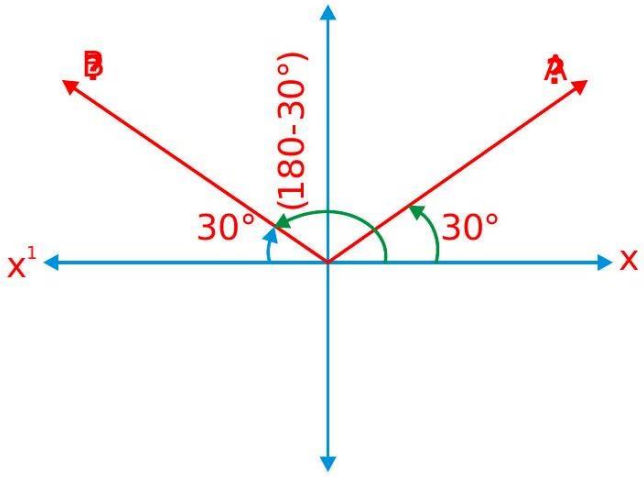
$B = 2\text{ cm}\theta_2 = 180^\circ - 30^\circ = 150^\circ$

$\bar{A} = (2\cos 30^\circ\hat{\imath} + 2\sin 30^\circ\hat{\jmath})\text{cm}$

$\bar{B} = 2\cos 150^\circ\hat{\imath} + 2\sin 150\hat{\jmath}$

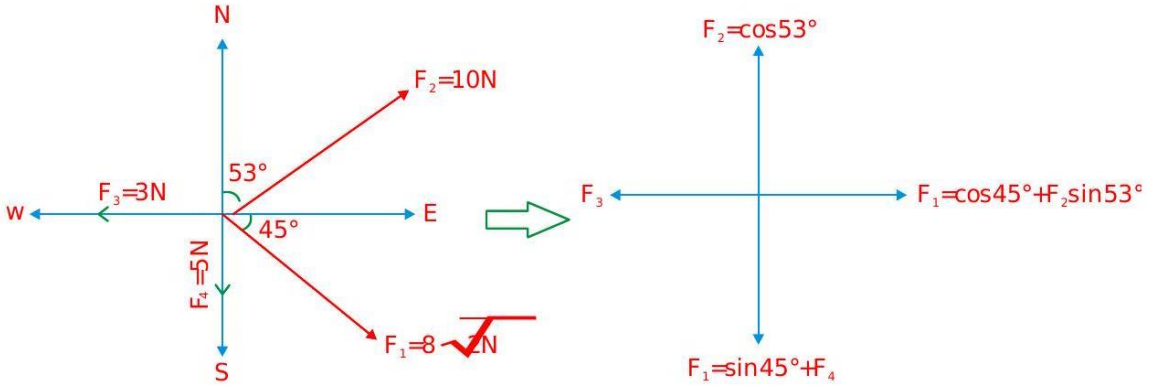
$\bar{A} = (\sqrt{3}\hat{\imath} + \hat{\mathbf{j}})\text{cm}\bar{B} = (-\sqrt{3}\hat{\mathbf{i}} + \hat{\mathbf{j}})\text{cm}$

$\bar{A} + \bar{B} = (\sqrt{3}\hat{\imath} + \hat{\jmath}) + (-\sqrt{3}\hat{\imath} + \hat{\jmath})$




$\bar{A} + \bar{B} = 2\hat{j}$

31. $\sum F_x = F_1 \cos 45° + F_2 \sin 53° - F_3$




$$= \frac{F_1}{\sqrt{2}} + \frac{4}{5}F_2 - F_3$$

$$= \frac{8\sqrt{2}}{\sqrt{2}} + \frac{4}{5} \times 10 - 3$$

$$= 8 + 8 - 3$$

$$\sum F_x = 13N$$

$$\sum F_y = F_2 \cos 53^0 - (F_1 \sin 45^0 + F_4)$$

$$= 10 \times \frac{3}{5} - \left(8\sqrt{2} \times \frac{1}{\sqrt{2}} + 5\right)$$

$$= 6 - (8 + 5)$$

$$= 6 - 13$$

$$\sum f_y = -7 \text{ N}$$

$$\bar{F}_{\text{Resultant}} = \left(\sum f_x\right)\hat{i} + \left(\sum f_y\right)\hat{j}$$

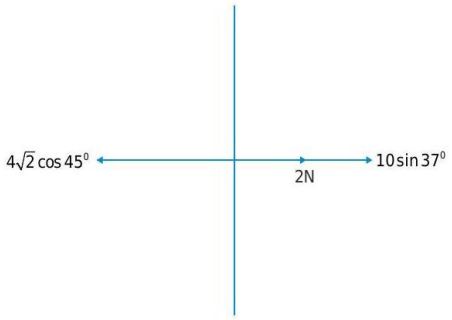
$$\bar{F}_R = (13\hat{i} - 7\hat{j})\text{N}$$

magnitude of resultant force $= \sqrt{13^2 + (7)^2} = \sqrt{218}$

the directiion of $\bar{f}_R = \alpha = \tan^{-1}\left(\frac{-7}{13}\right)$

then $\alpha = \tan^{-1}\left(\frac{7}{13}\right)$ south of east

32.




$F_x = \left(10\sin\ 37° + 2 - 4\sqrt{2}\cos\ 45°\right)i$
$F_x = 4i$
Similarly $F_y = 2j$
$\vec{F} = 4i + 2j\ \therefore \tan\ \theta = \frac{2}{4}$
$\theta = \tan^{-1}\ \frac{1}{2}$ north of east

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# Worksheet – 3

## Level – I

1. A
2. A
3. B
4. C
5. C
6. A
7. A
8. D
9. C
10. C
11. A

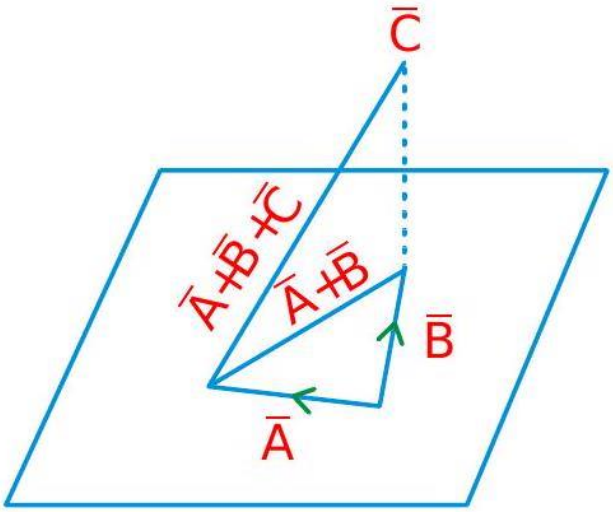
1. $|\bar{A} + \bar{B}| = |\bar{A}| + |\bar{B}|$
$\sqrt{A^2 + B^2 + 2AB\cos\ \theta} = A + B$
$A^2 + B^2 + 2AB\cos\ \theta (A + B)^2$
$2AB\cos\ \theta = 2AB$
$\cos\ \theta = 1$
$\theta = 0°$

3. $|\bar{a} - \bar{b}| = |\bar{a}| + |\bar{b}|$
$a^2 + b^2 - 2ab\cos\ \theta = (a + b)^2$
$a^2 + b^2 - 2ab\cos\ \theta = a^2 + b^2 + 2ab$
$\cos\ \theta = -1$
$\therefore \theta = 180°$

4. $\bar{P} + \bar{Q} = \bar{R}; \bar{P} - \bar{Q} = \bar{S}$
$R^2 = P^2 + Q^2 + 2PQ\cos\ \theta \quad S^2 = P^2 + Q^2 - 2PQ\cos\ \theta$
$R^2 + S^2 = 2(P^2 + Q^2)$

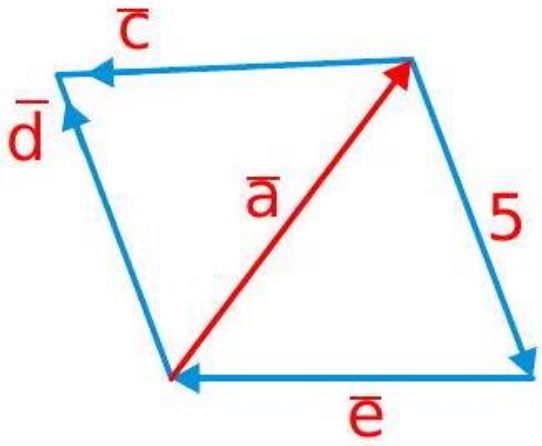
5. $|\bar{A} - \bar{B}| = \sqrt{A^2 + B^2 - 2AB\cos\ \theta}$
The maximum value is obtained when $\cos\ \theta = -1$ then $\pi$ or 180°
$\therefore$ maximum value is $\sqrt{A^2 + B^2 + 2AB} = A + B$

6. $\bar{A} + \bar{B} + \bar{C} \neq 0$ and it lies neither in the plane of $\bar{A}$ or $\bar{B}$ or $\bar{C}$




a. $\bar{a} + \bar{b} + \bar{e} = \bar{o}$

2. $\bar{a} + \bar{c} = \bar{d}$

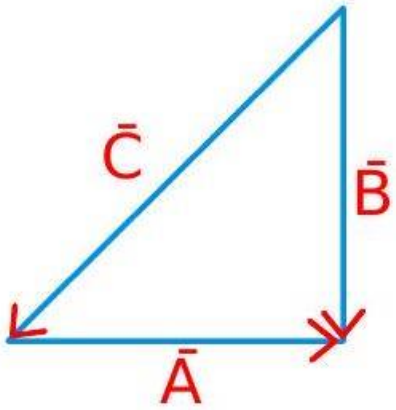



$\bar{c} - \bar{d} = \bar{a}$

3. $\bar{b} + \bar{e} + \bar{d} = \bar{c}$

$\Rightarrow \bar{b} + \bar{e} - \bar{c} = -\bar{d}$

9. $\bar{C} + \bar{A} = \bar{B}$ (Triangle law)




## Level – II

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1. C | 2. B | 3. B | 4. B | 5. B | 6. B | 7. A | 8. A |
| 9. A | 10. A | 11.C | 12. A | 13. A | 14. C | 15. B | 16. C |
| 17. B | 18. D | 19. D | | | | | |

1. $\mathrm{A}+\mathrm{B}=\mathrm{C}$ and $\overline{\mathrm{A}}+\overline{\mathrm{B}}=\overline{\mathrm{C}}$

$C=\sqrt{A^2+B^2+2AB\cos\,\theta}\ \because A+B=C$

$A+B=\sqrt{A^2+B^2+2AB\cos\,\theta}$

$(A+B)^2=A^2+B^2+2AB\cos\,\theta$

$A^2+B^2+2AB=A^2+B^2+2AB\cos\,\theta$

$2AB=2AB\cos\,\theta$

$\cos\,\theta=1$

$\theta=0^0$

2. $\bar{P}=2\hat{\imath}+4\hat{\jmath}+14\hat{k}$

$\bar{Q}=4\hat{\imath}+4\hat{\jmath}+10\hat{k}$

$\bar{P}+\bar{Q}=(2\hat{\imath}+4\hat{\jmath}+14\hat{k})+(4\hat{\imath}+4\hat{\jmath}+10\hat{k})$

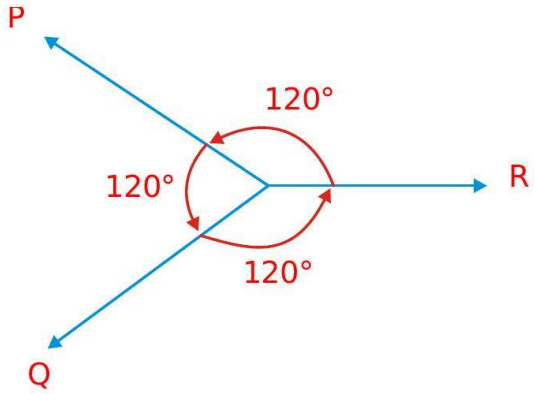
$\bar{P}+Q=6\hat{\imath}+8\hat{\jmath}+24\hat{k}$

$|\overline{\mathrm{P}}+\overline{\mathrm{Q}}|=\sqrt{6^2+8^2+(24)^2}=\sqrt{676}=26$

3. $\mathrm{P}+\mathrm{Q}=\mathrm{R}$

$\mathrm{R}^2=\mathrm{P}^2+\mathrm{Q}^2+2\mathrm{PQ}\cos\,\theta$

If $P+Q=R$ angle between them is 120°




$P^2=2P^2+2P^2\cos\,\theta$

$-P^2=2P^2\cos\,\theta$

$\cos\,\theta=-\frac{1}{2}\ \theta=120°$

4. $\overline{\mathrm{A}}=\overline{\mathrm{B}}+\overline{\mathrm{C}}\ \mathrm{A}=5\ \ \mathrm{B}=4\mathrm{C}=3$

$\overline{\mathrm{C}}=\overline{\mathrm{A}}-\overline{\mathrm{B}}$

$C=\sqrt{A^2+B^2-2AB}\cos\,\theta$ where $\theta$ is the angle between $A$ and $B$

$3^2=5^2+4^2-2\times5\times4\cos\,\theta$

$9=25+16-40\cos\,\theta$

$40\cos\,\theta=32$

$\cos\,\theta=\frac{32}{40}=\frac{4}{5}$

$\theta=\cos^{-1}\left(\frac{4}{5}\right)$

5. $\bar{A}=\hat{\imath}+2\hat{\jmath}+4\hat{k}\ \bar{B}=5\hat{\imath}$ then the third side

$\overline{\mathrm{C}}=\overline{\mathrm{A}}+\overline{\mathrm{B}}$

$\bar{C}=6\hat{\imath}+2\hat{\jmath}+4\hat{k}$

$|\overline{\mathrm{C}}|=\sqrt{36+4+16}=\sqrt{56}$

6. $\overline{\mathrm{A}}+\overline{\mathrm{B}}=\overline{\mathrm{A}}-\overline{\mathrm{B}}$

$2\bar{B}=0$

$\overline{\mathrm{B}}=0$

8. $\mathrm{P}=\mathrm{Q}=5\ \theta=60°$

$|\bar{P}+\bar{Q}| = \sqrt{P^2+Q^2+2P\theta\cos\,\theta} = \sqrt{5^2+5^2+2\times5\times5\times\cos\,60^\circ}$

$= \sqrt{5^2+5^2+5^2}$

$|\bar{P}+\bar{Q}| = 5\sqrt{3}$

$|\bar{P}-\bar{Q}| = \sqrt{P^2+Q^2-2PQ\cos\,\theta} = \sqrt{5^2+5^2-2\times5\times5\times\frac{1}{2}}$

$|\bar{P}-\bar{Q}| = 5$

9. The maximum and minimum resultants of two vectors are in the ratio 4: 3

$$\frac{P+Q}{P-Q} = \frac{4}{3}$$

$3P+3Q = 4P-4Q$

$7Q = P$

$\therefore P: Q = 7:1$

10. $R_1 = 141.4$ N when $\theta = 90^\circ$ and P = Q

$141.4 = \sqrt{P^2+P^2+2P^2\cos\,90^\circ}$

$141.4 = \sqrt{2}P$

$141.4 = 141.4 \times P$

$\therefore P = 100$

when $\theta = 120^\circ$ $R_2 = \sqrt{P^2+P^2+2P^2\cos\,120^\circ}$

$$R_2 = \sqrt{P^2+P^2+2P^2\times-\frac{1}{2}}$$

$R_2 = P = 100$ N

11. A = 2P

$B = \sqrt{2}P$

$R = \sqrt{10}P$ $\theta$ =?

$R^2 = A^2+B^2+2AB\cos\,\theta$

$(\sqrt{10}P)^2 = (2P)^2+(\sqrt{2}P)^2+2\times2P\times\sqrt{2}P\cos\,\theta$

$10P^2 = 4P^2+2P^2+4\sqrt{2}P^2\cos\,\theta$

$4P^2 = 4\sqrt{2}P^2\cos\,\theta \cos\,\theta = \frac{1}{\sqrt{2}} \Rightarrow \theta = 45^\circ$

12. $\bar{R} = \bar{P}-\bar{Q}$ and P − Q = R

$R = \sqrt{P^2+Q^2-2PQ\cos\,\theta}$

$P-Q = \sqrt{P^2+Q^2-2PQ\cos\,\theta}$

$P^2+Q^2-2PQ = P^2+Q^2-2PQ\cos\,\theta$

$\cos\,\theta = 1$

$\theta = 0^\circ$

14. $\bar{v}_1-\bar{v}_2 = \sqrt{v_1^2+v_2^2-2v_1v_2\cos\,\theta}$

15. |A| = |B| = 1

$|\bar{A}-\bar{B}| = \sqrt{A^2+B^2-2AB\cos\,\theta}$

$|\bar{A}-\bar{B}| = \sqrt{1+1-2\times1\times1\cos\,\theta}$

$|\bar{A}-\bar{B}| = \sqrt{2(1-\cos\,\theta)}$

$$|\bar{A}-\bar{B}| = \sqrt{2\times2\sin^2\,\frac{\theta}{2}}$$

$$(A-B) = 2\sin\,\frac{\theta}{2}$$

16. $\bar{A} = 3\hat{\imath}+5\hat{\jmath}-2\hat{k}$ $\bar{B} = -3\hat{\jmath}+6\hat{k}$

2 A + 7 B + 4C = 0

$2(3\hat{\imath} + 5\hat{\jmath} - 2\hat{k}) + 7(-3\hat{\jmath} + 6\hat{k}) + 4c = 0$
$(6\hat{\imath} + 10\hat{\jmath} - 4\hat{k}) + (-21\hat{\jmath} + 42\hat{k}) + 4c = 0$
$6\hat{\imath} - 11\hat{\jmath} + 38\hat{k} + 4c = 0$
$4c = -6\hat{\imath} + 11\hat{\jmath} - 38\hat{k}$
$c = \frac{1}{4}(-6\hat{\imath} + 11\hat{\jmath} - 38\hat{k})$
$C = -1.5\hat{\imath} + 2.75\hat{\jmath} - 9.5\hat{k}$

17. Two forces $\mathrm{P} = 4\ \mathrm{N} Q = 3\ \mathrm{N}$
resultant $R_1$ when they have an angle ' $\theta$ ' between them is
$\mathrm{R}_1 = \sqrt{\mathrm{P}^2 + \mathrm{Q}^2 + 2\mathrm{PQ}\cos\ \theta}$
square of resultant is $\mathrm{R}_1^2 = \mathrm{P}^2 + \mathrm{Q}^2 + 2\mathrm{PQ}\cos\ \theta$
square of resultant of right angles $R_2^2 = P^2 + Q^2$
given $R_1$ exceeds $R_2$ by
$\mathrm{R}_1 = \mathrm{R}_2 + 12$
$\mathrm{P}^2 + \mathrm{Q}^2 + 2\mathrm{PQ}\cos\ \theta = \mathrm{P}^2 + \mathrm{Q}^2 + 12$
$2\mathrm{PQ}\cos\ \theta = 12$
$2 \times 4 \times 3\cos\ \theta = 12$
$\cos\ \theta = \frac{1}{2}$
$\theta = 60°$

18. R is the resultant of $\overline{\mathrm{P}}$ and $\overline{\mathrm{Q}}$ if the magnitude of $\overline{\mathrm{Q}}$ is doubled new resultant $\mathrm{R}_1$ is perpendicular to $\overline{\mathrm{P}}$

$$R = \sqrt{P^2 + Q^2 + 2PQ\cos\ \theta}$$

$\tan\ \alpha = \frac{2Q\sin\ \theta}{P + 2Q\cos\ \theta}$
here $\alpha = 90°$
$\frac{1}{0} = \frac{2Q\sin\ \theta}{P + 2Q\cos\ \theta}$
$P + 2Q\cos\ \theta = 0$
$P = -2Q\cos\ \theta$
$P^2 = 4Q^2\cos^2\ \theta$
$R = \sqrt{P^2 + Q^2 + 2PQ\cos\ \theta}$
$R = \sqrt{4P^2\cos^2\ \theta + Q^2 + 2 \times -2P\cos\ \theta \times Q \times \cos\ \theta}$
$R = \sqrt{4Q^2\cos^2\ \theta + Q^2 - 4Q^2\cos^2\ \theta}$

19. Using phythogoras theorm
$\sin\ \theta = \frac{\mathrm{AB}}{\mathrm{AC}} = 2\sqrt{2}$
$\sin\ \theta = \frac{\sqrt{3}}{2}$
$\theta = 60°$
$60° + 90° = 120°$

# Level – III

20. B 21.a-r; b-p,t; c-s; d-q 22. A 23. D
24. A 25. A,B,D 26. A 27. D 28. C 29. C 30. D 31. C

22. $5\ \mathrm{N} + 3\ \mathrm{N} = 8\ \mathrm{N}$
23. $4\ \mathrm{N} - 3\ \mathrm{N} = 1\ \mathrm{N}$
24. Resultant = 39 N
$\vec{P} = 20$
$\vec{Q} = 25$

$\cos\ \theta = \frac{1}{2}$

$\theta = 60°$

26. $a = 4i + 3j$

$= \sqrt{16 + 9}$

$a = \sqrt{25}$

$a = 5$

27. $a = 4i + 3j$

$b = 3i + 4j$

$|a + b| = 7\sqrt{2}$

$= \sqrt{1 + 1}$

$= \sqrt{2}$

29. $a^2 + b^2 = 25ab = 12$

$(a + b)^2 = 25 + 2(12) = 49, a + b = 7, ab = 12$

$\sqrt{\mathrm{f}_1^2 + \mathrm{f}_2^2} = 5\ \mathrm{N}$

$f_1^2 + f_2^2 = 25$

$\sqrt{\mathrm{f}_1^2 + \mathrm{f}_2^2 + 2\mathrm{f}_1\mathrm{f}_2\cos\ 120^0} = \sqrt{13}\ \mathrm{N}$

$f_1^2 + f_2^2 - f_1 f_2 = 13$

from (1)

$25 - f_1 f_2 = 13$

$\mathrm{f}_1\mathrm{f}_2 = 12$

$\therefore\ \mathrm{f}_2 = \frac{12}{\mathrm{f}_1}$

substrative $f_2$ value in .....(1)

$f_1^2 + \left(\frac{12}{f_1}\right)^2 = 25$

$\mathrm{f}_1^2 + \frac{144}{\mathrm{f}_1^2} = 25$

$f_1^4 - 25f_1^2 + 144 = 0$

$(f_1^2)^2 - 9f_1^2 - 16f_1^2 + 144 = 0$

$f_1^2(f_1^2 - 9) - 16(f_1^2 - 9) = 0$

$(f_1^2 - 16)(f_1^2 - 9) = 0$

$f_1^2 = 16; f_1^2 = 9$

$\mathrm{f}_1 = 4$ or $\mathrm{f}_1 = 3$

when $f_1 = 4N\ f_2 = \frac{12}{f_1} \Rightarrow f_2 = \frac{12}{4} = 3N$

when $\mathrm{f}_1 = 3\ \mathrm{N}_2 = \frac{12}{3} \Rightarrow 4\ \mathrm{N}$

$\therefore$ the forces are 3 N and 4 N

30. Three forces of equal magnitude with equal angle between them make a resultant zero

31. $\mathrm{P} + \mathrm{Q} = 18$

$P^2 + Q^2 + 2PQ\cos\ \theta = 144$

$\tan\ 90° = \frac{P\sin\ \theta}{P + Q\cos\ \theta}$

$\mathrm{P} + \mathrm{Q}\cos\ \theta = 0$

$\mathrm{Q} = \cos\ \theta = -\mathrm{P}$

$P^2 + Q^2 + 2P \times -P = 144$

$Q^2 - P^2 = 144$

$(\mathrm{Q} + \mathrm{P})(\mathrm{Q} - \mathrm{P}) = 144$

$18(\mathrm{Q} - \mathrm{P}) = 144$

$Q - P = 8$
Solving (1) and (2)
we get $P = 5Q = 13$
$\therefore$ The magnitude of forces are $5\ \text{N}, 13\ \text{N}$

# Worksheet – 4

## Level – I

1. D 2. A 3. A 4. D

## Level – II

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1] A | 2] A | 3] A | 4] D | 5] D | 6] D | 7] D | 8] B |
| 9] B | 10] C | 11] C | 12] A | 13] B | 14] A | 15] A | 16] D |
| 17] A | 18] B | | | | | | |

1. $\bar{A} = \hat{\imath} + 2\hat{\jmath} - 3\hat{k}; \bar{B} = 3\hat{\imath} + m\hat{\jmath} - 9\hat{k}$
When $\bar{A} \perp \bar{B}$ then $\bar{A} \cdot \bar{B} = 0$

$$(\hat{\imath} + 2\hat{\jmath} - 3\hat{k}) \cdot (3\hat{\imath} + m\hat{\jmath} - 9\hat{k}) = 0$$
$$(1 \times 3) + (2 \times m) + (-3 \times -9) = 0$$
$$3 + 2m + 27 = 0$$
$$2m = -30$$
$$\therefore m = -15$$

2. $(\bar{a} + \bar{b}) \cdot (\bar{a} - \bar{b}) = 0$

$$\bar{a} \cdot (\bar{a} - \bar{b}) + \bar{b} \cdot (\bar{a} - \bar{b}) = 0$$
$$\bar{a} \cdot \bar{a} - \bar{a} \cdot \bar{b} + \bar{b} \cdot \bar{a} - \bar{b} \cdot \bar{b} = 0$$
$$|\bar{a}|^2 - \bar{a} \cdot \bar{b} + \bar{a} \cdot \bar{b} - |\bar{b}|^2 = 0 \left[\because \bar{a} \cdot \bar{b} = \bar{b} \cdot \bar{a}\right]\left[\bar{a} \cdot \bar{a} = |\bar{a}|^2 : \bar{b} \cdot \bar{b} = |\bar{b}|^2\right]$$
$$|\bar{a}|^2 - |\bar{b}|^2 = 0$$
$$\frac{|\bar{a}|^2}{|\bar{b}|^2} = 1 \ \therefore |\bar{a}| : |\bar{b}| = 1:1$$

4. resultant of $\bar{A} + \bar{B}$ and $\bar{A} - \bar{B}$ is $2\bar{A}$
The angle between $2\bar{A}$ and $\bar{A}$ is zero.
5. The maximum value of dot product will be at 0°
6. $\theta = \cos^{-1}\left(\frac{\bar{A} \cdot \bar{B}}{AB}\right)$

$$\bar{A} \cdot \bar{B} = (\hat{\imath} + 2\hat{\jmath} - \hat{k}) \cdot (-\hat{\imath} + \hat{\jmath} - 2\hat{k}) = 3$$
$$A = \sqrt{1^2 + 2^2 + (-1)^2} = \sqrt{6}$$
$$B = \sqrt{(-1)^2 + 1^2 + (-2)^2}$$
$$B = \sqrt{1 + 1 + 4} = \sqrt{6}$$
$$\theta = \cos^{-1}\left(\frac{3}{\sqrt{6} \times \sqrt{6}}\right)$$
$$\theta = \cos^{-1}\left(\frac{3}{6}\right) = \cos^{-1}\left(\frac{1}{2}\right)$$
$$\therefore \theta = 60°$$

7. clearly the dot product of $\hat{\imath} + \hat{\jmath} + \hat{k}$ and $3\hat{\imath} + 2\hat{\jmath} - 5\hat{k}$ is zero
8. $|\bar{A}| = |\bar{B}| = 1$

$$\frac{1 + \bar{A} \cdot \bar{B}}{1 - \bar{A} \cdot \bar{B}} = \frac{1 + |\bar{A}||\bar{B}|\cos\theta}{1 - |\bar{A}||\bar{B}|\cos\theta} = \frac{1 + \cos\theta}{1 - \cos\theta}$$

9. conceptual

10. $|\vec{a}| = 4$ and $|\vec{b}| = 3$ angle between $\vec{a}$ and $\vec{b}$ is 90° then $\vec{a} \cdot \vec{b} = 0$

11. $\bar{F} = (-3\hat{\imath} - \hat{\jmath} + 2\hat{k})N$
$\bar{r}_1 = 4\hat{\imath} - 3\hat{\jmath} - 5\hat{k};$
$\bar{r}_2 = -\hat{\imath} + 4\hat{\jmath} + 3\hat{k}$
$\bar{S} = \bar{r}_2 - \bar{r}_1 = -5\hat{\imath} + 7\hat{\jmath} + 8\hat{k}$
$W = \bar{F} \cdot \bar{S} = (-3\hat{\imath} - \hat{\jmath} + 2\hat{k}) \cdot (-5\hat{\imath} + 7\hat{\jmath} + 8\hat{k})$
$= 15 - 7 + 16 = 24J$

12. $\bar{F} = (5\hat{\imath} - 3\hat{\jmath} + 2\hat{k})$
$\bar{r}_1 = 2\hat{\imath} + 7\hat{\jmath} + 4\hat{k};$
$\bar{r}_2 = 5\hat{\imath} + 2\hat{\jmath} + 3\hat{k}$
$\bar{S} = \bar{r}_2 - \bar{r}_1 = (5\hat{\imath} + 2\hat{\jmath} + 3\hat{k}) - (2\hat{\imath} + 7\hat{\jmath} + 4\hat{k})$
$\bar{S} = 3\hat{\imath} - 5\hat{\jmath} - \hat{k}$
$W = \bar{F} \cdot \bar{S} = (5\hat{\imath} - 3\hat{\jmath} + 2\hat{k}) \cdot (3\hat{\imath} - 5\hat{\jmath} - \hat{k})$
$W = 15 + 15 - 2; W = 28$ units

13. $\bar{F} = (8\hat{\imath} + 4\hat{\jmath})N; \bar{S} = (3\hat{\imath} - 3\hat{\jmath})N; P = 0.6$ W

$$P = \frac{W}{t} \Rightarrow P = \frac{\bar{F} \cdot \bar{S}}{t} \quad \therefore t = \frac{\bar{F} \cdot \bar{S}}{P}$$

$$\mathrm{t} = \frac{(8\hat{\imath} + 4\hat{\jmath}) \cdot (3\hat{\imath} - 3\hat{\jmath})}{0.6}; \ \mathrm{t} = \frac{(8 \times 3) + (4 \times -3)}{0.6}$$

$$\mathrm{t} = \frac{24 - 12}{0.6}; \ \mathrm{t} = \frac{12 \times 10}{6} = 20 \text{ s}$$

$\therefore$ The time of action of force is 20 s

14. $A = 2m, B = 3m,$

$$\theta = 60°$$

$\bar{\mathrm{A}} \cdot \bar{\mathrm{B}} = \mathrm{AB}\cos\theta$
Scalar Product $\bar{\mathrm{A}} \cdot \bar{\mathrm{B}} = 2 \text{ m} \times 3 \text{ m} \times \frac{1}{2}$
$\bar{\mathrm{A}} \cdot \bar{\mathrm{B}} = 3 \text{ m}^2$

15. $\bar{a} = 2\hat{\imath} + 6\hat{\jmath} + m\hat{k}; \ \bar{b} = n\hat{\imath} + 18\hat{\jmath} + 3\hat{k}$
$\bar{\mathrm{a}}$ and $\bar{\mathrm{b}}$ are parallel then $\mathrm{a} = \lambda\bar{\mathrm{b}}$
$2\hat{\imath} + 6\hat{\jmath} + m\hat{k} = \lambda(n\hat{\imath} + 18\hat{\jmath} + 3\hat{k})$
$2\hat{\imath} + 6\hat{\jmath} + m\hat{k} = 3\lambda\left(\frac{n}{3}\hat{\imath} + 6\hat{\jmath} + \hat{k}\right)$
On comparision
$\frac{\mathrm{n}}{3} = 2; \ \mathrm{m} = 1; \mathrm{n} = 6; \ \mathrm{m} = 1$

16. $\bar{F}_1 = 2\hat{\imath} + 2\hat{\jmath} + 5\hat{k};$
$\bar{F}_2 = 3\hat{\imath} - 5\hat{\jmath} - 4\hat{k};$
$\bar{S} = 18\hat{\imath} + 17\hat{\jmath} - 7\hat{k}$
Resultant force $\bar{\mathrm{F}} = \bar{\mathrm{F}}_1 + \bar{\mathrm{F}}_2$

$$\bar{F} = (2\hat{\imath} + 2\hat{\jmath} + 5\hat{k}) + (3\hat{\imath} - 5\hat{\jmath} - 4\hat{k})$$
$$\bar{F} = (2 + 3)\hat{\imath} + (2 - 5)\hat{\jmath} + (5 - 4)\hat{k}$$
$$\bar{F} = 5\hat{\imath} - 3\hat{\jmath} + \hat{k}$$

Work done by the two forces $\bar{F}_1$ and $\bar{F}_2$ is same as work done by the resultant force $\bar{F}$

$$\therefore \mathrm{W} = \bar{\mathrm{F}} \cdot \bar{\mathrm{S}}$$

$W = (5\hat{\imath} - 3\hat{\jmath} + \hat{k}) \cdot (18\hat{\imath} + 17\hat{\jmath} - 7\hat{k})$
$\mathrm{W} = (5 \times 18) + (-3 \times 17) + (1 \times 7)$

$W = 90 - 51 - 7$

$W = 32$ J

17. Initial velocity $\bar{u} = (2\hat{i} - 3\hat{j} + 11\hat{k})ms^{-1}$ and acceleration $\bar{a} = (10\hat{i} + 10\hat{j} + 10\hat{k})ms^{-1}$

final velocity $\bar{v}$ after time $t = 0.25$ s

$\bar{a} = \dfrac{\bar{v} - \bar{u}}{t} \Rightarrow \bar{v} = \bar{u} + \bar{a}$

$\bar{v} = (2\hat{i} - 3\hat{j} + 11\hat{k}) + 0.25(10\hat{i} + 10\hat{j} + 10\hat{k})$

$\bar{v} = (2\hat{i} - 3\hat{j} + 11\hat{k}) + \left(\dfrac{5}{2}\hat{i} + \dfrac{5}{2}\hat{j} + \dfrac{5}{2}\hat{k}\right)$

$\bar{v} = \left(2 + \dfrac{5}{2}\right)\hat{i} + \left(\dfrac{5}{2} - 3\right)\hat{j} + \left(11 + \dfrac{5}{2}\right)\hat{k}$

$\bar{v} = \dfrac{9}{2}\hat{i} - \dfrac{1}{2}\hat{j} + \dfrac{27}{2}\hat{k}$

$\bar{v} = \dfrac{1}{2}(9\hat{i} - \hat{j} + 27\hat{k})$

$\therefore \bar{v} = \dfrac{1}{2}\sqrt{(9)^2 - (-1)^2 + (27)^2}$

$\bar{v} = \dfrac{1}{2}\sqrt{81 + 1 + 729}$

$\bar{v} = \dfrac{1}{2}\sqrt{811}\ ms^{-1}$

18. Component of $\bar{A}$ along $\bar{B} = A\cos\theta$

Component of $\bar{B}$ along $\bar{A} = B\cos\theta$

$A\cos\theta = \sqrt{3} \times B\cos\theta$

$\dfrac{A}{B} = \dfrac{\sqrt{3}}{1} \Rightarrow A : B = \sqrt{3} : 1$

# Level – III

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 19] B,D | 20]a-r,t;b-s,t;c-p,t;d-q | | 21] C | 22] B | 23) D | | |
| 24] 6 | 25] A | 26] B,C,D | 27] B | 28]A,B,C | 29] B,D | 30] A | 31] D |

19. $(\bar{V}_1 + \bar{V}_2) \cdot (\bar{V}_1 - \bar{V}_2) = 0$

$|\bar{V}_1|^2 - \bar{V}_1\bar{V}_2 + \bar{V}_2\bar{V}_1 - |\bar{V}_2|^2 = 0$

$|\bar{V}_1|^2 = |\bar{V}_2|^2 \Rightarrow |\bar{V}_1| = |\bar{V}_2|$

Here the angle between $\bar{V}_1$ and $\bar{V}_2$ can be any value.

20. $W = \bar{F} \cdot \bar{S}; P = \bar{F} \cdot \bar{V};\ \bar{A} \cdot \bar{B} = \dfrac{AB}{2} = AB\cos 60°$; acceleration is a vector

21. $P = \bar{F} \cdot \bar{V}$

$P = (5\hat{i} - 10\hat{j} + 6\hat{k}) \cdot (4\hat{i} - 2\hat{j} + \hat{k})$

$P = 20 + 20 + 6 = 46w$

22. $W = F\cos\theta = 100 \times 5 \times \dfrac{4}{5} = 400$ J

23. $\vec{A} = 6\hat{i} + 5\hat{j},\ \vec{B} = 4\hat{i} - 3\hat{j}$

$\vec{A} + \vec{B} = 10\hat{i} + 2\hat{j}$

$\vec{A} - \vec{B} = 2\hat{i} + 8\hat{j}$

$(\vec{A} + \vec{B}) \cdot (\vec{A} - \vec{B}) = (10\hat{i} + 2\hat{j}) \cdot (2\hat{i} + 8\hat{j})$

$= 20 + 16$

$\therefore (\vec{A} + \vec{B}) \cdot (\vec{A} - \vec{B}) = 36$ units

24. $F = 1200$ N; $S = 10$ m; $\theta = 60°$
Work done (W) $= \bar{F} \cdot \bar{S} = FS\cos\theta$

$$W = 1200 \times 10 \times \frac{1}{2};\ W = 6000 \text{ J}$$

$$W = 6 \times 10^3 \text{ J};\ \text{Ans} : 6$$

25. vector $\bar{A} = \hat{\imath} + \hat{\jmath} + \sqrt{2}\hat{k}$ $A = 2$
angle made by $\bar{A}$ with x-axis
take $\bar{A} = \hat{\imath} + \hat{\jmath} + \sqrt{2}\hat{k}$; $\bar{B} = \hat{\imath}$

$$\theta_x = \cos^{-1}\left(\frac{\bar{A} \cdot \bar{B}}{AB}\right) = \cos^{-1}\left(\frac{(\hat{\imath} + \hat{\jmath} + \sqrt{2}\hat{k}) \cdot \hat{\imath}}{2 \times 1}\right)$$

$$\theta_x = \cos^{-1}\left(\frac{1}{2}\right) = 60°$$

Similarly $\theta_y = 60°$ and $\theta_z = 45°$

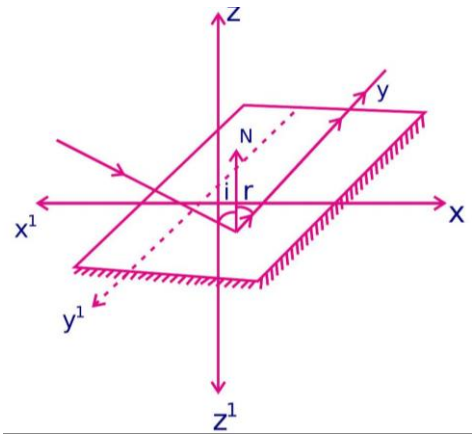
26. conceptual

27. Force applied on the body $\bar{F} = 3\hat{\imath} - 2\hat{\jmath} + 4\hat{k}$ at intial point $A(8, -2, -3)$ to the point $B(-2,0,6)$.
The position vector to $A$ is $O\bar{A} = 8\hat{\imath} - 2\hat{\jmath} - 3\hat{k}$ and position vector to $B$ is $\overline{OB} = -2\hat{\imath} + 6\hat{k}$ drawn from origin.
displacement $\bar{S} = \overline{OB} - \overline{OA}$

$$\bar{S} = \overline{OB} - \overline{OA}$$

$\bar{S} = (-2\hat{\imath} + 6\hat{k}) - (8\hat{\imath} - 2\hat{\mathbf{j}} - 3\hat{k})$
$\bar{S} = -10\hat{\imath} + 2\hat{\jmath} + 9\hat{k}$
work done $W = \bar{F} \cdot \bar{S}$
$W = (3\hat{\imath} - 2\hat{\jmath} + 4\hat{k}) \cdot (-10\hat{\imath} + 2\hat{\jmath} + 9\hat{k})$
$W = -30 - 4 + 36;$
$W = 2$ units

28. Light ray incident along the vector $\bar{A} = \frac{1}{2}\hat{\imath} - \frac{1}{\sqrt{2}}\hat{\jmath} + \hat{k}$ on a plane mirror placed in xy-plane.




from the figure
Here the normal on incident point will be along z-axis
Angle made by $\bar{A}$ with z-axis gives angle of incidance

$$\theta = \cos^{-1}\left(\frac{\left(\frac{1}{2}\hat{\imath} - \frac{1}{2}\hat{\jmath} + \hat{k}\right) \cdot \hat{k}}{\frac{\sqrt{7}}{2} \times 1}\right) = \cos^{-1}\left(\frac{2}{\sqrt{7}}\right)$$

29. $\bar{A} = 6\hat{\imath} + 3\hat{\jmath} + 2\hat{k};$

$\bar{B} = 2\hat{\imath} - 2\hat{\jmath} - \hat{k}$

$A = 7;\ B = 3$

$$\cos\theta = \frac{\bar{A}\cdot\bar{B}}{AB} = \frac{(6\hat{\imath} + 3\hat{\jmath} + 2\hat{k})\cdot(2\hat{\imath} - 2\hat{\jmath} - \hat{k})}{7\times 3}$$

$$\cos\theta = \frac{12 - 6 - 2}{21} = \frac{4}{21};\ \theta = \cos^{-1}\left(\frac{4}{21}\right)$$

30. $\phi = \vec{B}\cdot\vec{A} = 15\times 10^{-2} - 15\times 10^{-2} = 0$

31. $P = \frac{w}{t}$

$$P = \frac{\vec{F}\cdot\vec{S}}{t}$$

$$P = \frac{(2\hat{\imath} + 3\hat{\jmath} + 4\hat{k})\cdot(3\hat{\imath} + 4\hat{\jmath} + 5\hat{k})}{4}$$

$$= \frac{38}{4} = 9.5\text{w}$$

# Worksheet – 5

## Level – I

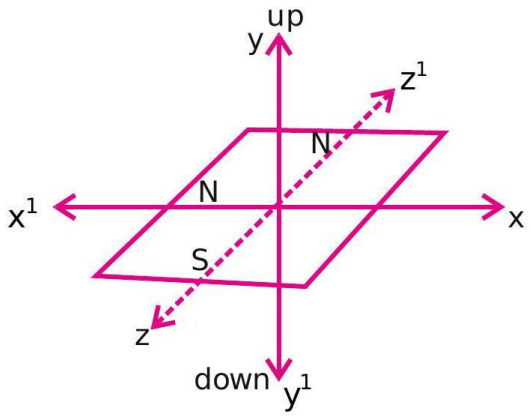
1] C 2] B 3] D 4]B 5]A 6]B 7]A 8]C
9]A 10]C 11]A

1. Cork screw right hand rule
If a Cork screw is placed in a direction perpendicular to the plane that contains vectors $\bar{A}$ and $\bar{B}$ and its handle is turned in the direction from first to second vector in the product. The direction of the cross product is given by the progression of the Cork screw

2. $(\bar{A}\times\bar{B}) = -(\bar{B}\times\bar{A})$ the angle between $(\bar{A}\times\bar{B})$ and $(\bar{B}\times\bar{A})$ is 180°

3. $|\bar{a}\times\bar{b}| = |\bar{a}\cdot\bar{b}|$
$ab\sin\theta = ab\cos\theta$
$\therefore \sin\theta = \cos\theta$
Here $\theta = 45°$

4. Let the vector $\bar{A} = \hat{\jmath}$ as it is pointing vertically upword $\bar{B} = \hat{k}$ as it is pointng towards south then $\bar{A}\times\bar{B} = \hat{\jmath}\times\hat{k} = \hat{\imath}$




Here $\bar{A}\times\bar{B} = \hat{\imath}$ points towards east

5. When $\bar{P}\perp\bar{Q}$

$$\bar{P}\perp\bar{Q} = PQ\sin\theta = PQ\sin 90°$$

$$|\bar{P}\times\bar{Q}| = PQ$$

6. If $\bar{A}\times\bar{B} = 0$ and $\bar{B}\times\bar{C} = 0$ means $\bar{A}||\bar{B}$ and

$\bar{B}||\bar{C}$ then $\bar{A}||\bar{C}$ i.e., $\bar{A} \times \bar{C} = 0$

7. $\hat{j} \times \hat{k} = \hat{i}$ is true
   $i \cdot i = 1$
   $\hat{\mathbf{j}} \times \hat{\mathbf{j}} = 0$
   $k \cdot i = 0$
8. Given $|\bar{A}| > 1; |\bar{B}| > 1$
   $|\bar{A} \times \bar{B}| = |\bar{A}||\bar{B}| \sin\theta$
   Since the values of sin $\theta$ range from -1 to +1
   the value of $|\bar{A} \times \bar{B}|$ always lie between or equal to $-AB \leq |\bar{A} \times \bar{B}| \leq AB$
   $|\bar{A} \times \bar{B}|$ will never be more than $AB$
9. $\bar{A} \times \bar{B} = \bar{B} \times \bar{A}$
   $\bar{A} \times \bar{B} = -\bar{A} \times \bar{B}$
   $2(\bar{A} \times \bar{B}) = 0$
   $2(AB \sin\theta \hat{n}) = 0$
   for $A \neq 0 \& B \neq 0$
   $\sin\theta = 0$
   then $\theta = n\pi$ where n = 0,1,2,3 ...
   Here $\theta = \pi$
10. $(\bar{a} \times \bar{b}) + (\bar{b} \times \bar{a}) = ab \sin\theta \hat{n} - ab \sin\theta \hat{n} = \bar{0}$
    the resultant of $\bar{a} \times \bar{b}$ and $\bar{b} \times \bar{a}$ is null or zero vector.
11. $\bar{A} \times \bar{B} = |A||B| \sin\theta \hat{n}$

# Level – II

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1] B | 2] C | 3] A | 4] A | 5] D | 6] D | 7] A | 8] C |
| 9] D | 10] D | 11] A | 12] A | 13] C | 14] D | 15] B | 16] A |
| 17] D | 18] A | 19]C | | | | | |

1. $\bar{V} = \bar{\omega} \times \bar{r}$
   $\bar{\omega} = 4\hat{\imath} + \hat{\jmath} - 2\hat{k}$
   $\bar{r} = 2\hat{\imath} + 3\hat{\jmath} - 3\hat{k}$
   $\bar{v} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 4 & 1 & -2 \\ 2 & 3 & -3 \end{vmatrix} = 3\hat{\imath} + 8\hat{\jmath} + 10\hat{k}$
2. $\bar{A}, \bar{B}$ and $\bar{A} + \bar{B}$ are coplaner $\bar{A} \times \bar{B}$ will be perpendicular to the plane containing $\bar{A}, \bar{B}$ and $\bar{A} + \bar{B}$
   $\therefore$ the angle between $\bar{A} + \bar{B}$ and $\bar{A} \times \bar{B}$ is $\frac{\pi}{2}$
3. $A = 2m, B = 3m, \theta = 60°$
   $|\bar{A} \times \bar{B}| = AB \sin\theta$
   $|\bar{A} \times \bar{B}| = 2\ m \times 3\ m \times \sin 60°$
   $|\bar{A} \times \bar{B}| = 6m^2 \times \frac{\sqrt{3}}{2}$
   $|\bar{A} \times \bar{B}| = 3\sqrt{3}\ m^2$
4. $\hat{\imath} \times (\hat{\imath} - \hat{\jmath}) + \hat{\jmath} \times (\hat{\imath} - \hat{\jmath})$
   $= -(\hat{k}) + (-\hat{k}) = \widehat{-k} - \hat{k} = -2\hat{k}$
5. $\bar{A} = 2\hat{\imath} - 3\hat{\jmath} + \hat{k};\ \bar{B} = -\hat{\imath} + 3\hat{\jmath} + \hat{k}$
   $\bar{A} \times \bar{B} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 & -3 & 1 \\ -1 & 3 & 1 \end{vmatrix} = -6\hat{\imath} - 3\hat{\jmath} + 3\hat{k}$

6. $\bar{F} = -3\hat{\imath} + 2\hat{\jmath} + \hat{k}; \bar{r} = 8\hat{\imath} + 2\hat{\jmath} + 3\hat{k}$

$$\bar{\tau} = \bar{r} \times \bar{f} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 8 & 2 & 3 \\ -3 & 2 & 1 \end{vmatrix} = -4\hat{\imath} - 17\hat{\jmath} + 22\hat{k}$$

7. $\bar{\mathrm{F}} = 2\hat{\imath} - \hat{\jmath} - \hat{\mathrm{k}};\ \bar{\mathrm{r}}_1 = -5\hat{\imath} + 2\hat{\jmath} + \hat{\mathrm{k}};\ \bar{\mathrm{r}}_2 = 0\hat{\imath} + 0\hat{\jmath} + 0\hat{\mathrm{k}}$

$$\bar{r} = \bar{r}_1 - \bar{r}_2 = -5\hat{\imath} + 2\hat{\jmath} + \hat{k}$$

$$\bar{\tau} = \bar{r} \times \bar{f} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ -5 & 2 & 1 \\ 2 & -1 & -1 \end{vmatrix} = -\hat{\imath} - 3\hat{\jmath} + \hat{k}$$

$$\bar{\tau} = -\hat{\imath} - 3\hat{\jmath} + \hat{k};\ |\bar{\tau}| = \sqrt{(-1)^2 + (-3)^2 + (1)^2} = \sqrt{11}$$

$$\therefore |\bar{\tau}| = \sqrt{11}$$

8. As $\overline{\mathrm{F}_1}$ is acting along positive $x$-axis. Let $\overline{\mathrm{F}_1} = \mathrm{a}\hat{\imath}$.
The vector product of $\bar{F}_1$ with $\overline{F_2}$ is zero
i.e., $\bar{F}_1 \times \overline{F_2} = 0 \Rightarrow F_1F_2 \sin\,\theta = 0$
$\bar{F}_1$ and $\overline{F_2}$ should have an angle of 0 or $\pi$
between them $\therefore \overline{F_2}$ can take $\overline{F_2} = \pm\mathrm{bi}$
Here $-5\hat{\imath}$ is the suitable optioin.

9. Here $\bar{A} \cdot \bar{B} = |\bar{A} \times \bar{B}|$, then angle between them $\theta = 45°$

$$\therefore \bar{C} = \bar{A} + \bar{B}$$

$$|\bar{C}| = \sqrt{A^2 + B^2 + 2AB\cos\,\theta}$$

$$|\bar{C}| = \sqrt{A^2 + B^2 + 2AB \times \frac{1}{\sqrt{2}}}$$

$$= \sqrt{A^2 + B^2 + \sqrt{2}AB}$$

10. $|\overline{\mathrm{A}} \cdot \overline{\mathrm{B}}|^2 + |\overline{\mathrm{A}} \times \overline{\mathrm{B}}|^2 = (\mathrm{AB}\cos\,\theta)^2 + (\mathrm{AB}\sin\,\theta)^2$

$$= A^2B^2(\cos^2\,\theta + \sin^2\,\theta) = A^2B^2$$

11. $\bar{A} = 2\hat{\imath} + p\hat{\jmath} + q\hat{k};$

$$\bar{B} = 5\hat{\imath} + 7\hat{\jmath} + 3\hat{k}$$

If $\overline{\mathrm{A}} || \overline{\mathrm{B}}\ \overline{\mathrm{A}} \times \overline{\mathrm{B}} = \overline{\mathrm{O}}$

$$\bar{A} \times \bar{B} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 & p & q \\ 5 & 7 & 3 \end{vmatrix} = (3p - 7q)\hat{\imath} - (6 - 5q)\hat{\jmath} + (14 - 5q)\hat{k}$$

$$4 - 5p = 0 \Rightarrow p = \frac{14}{5}$$

$$6 - 5q = 0 \Rightarrow q = \frac{6}{5}$$

12. $\bar{p} = 3\hat{\imath} + 4\hat{\jmath} - 2\hat{k}; \bar{r} = \hat{\imath} + 2\hat{\jmath} - \hat{k}$
Angular momentum $\bar{\mathrm{L}} = \bar{\mathrm{r}} \times \bar{\mathrm{p}}$

$$\bar{L} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 1 & 2 & -1 \\ 3 & 4 & -2 \end{vmatrix} = -\hat{\jmath} - 2\hat{k}$$

X - component of $\bar{L}$ is zero
$\bar{\mathrm{L}}$ is perpendicular to x-axis

13. $\bar{\mathrm{F}} = 4\hat{\imath} - 5\hat{\jmath} + 3\hat{\mathrm{k}};\ \overline{r_1} = \hat{\imath} - 2\hat{\mathbf{j}} + 3\hat{\mathrm{k}};\ \overline{r_2} = 3\hat{\imath} - 2\hat{\mathbf{j}} - 3\hat{\mathrm{k}}$

$$\bar{r} = \overline{r_1} - \overline{r_2}$$

$$\bar{r} = (\hat{\imath} - 2\hat{\jmath} + 3\hat{k}) - (3\hat{\imath} - 2\hat{\jmath} - 3\hat{k})$$

$$\bar{r} = -2\hat{\imath} + O\hat{\jmath} + 6\hat{k}$$

$\bar{\tau} = \bar{r} \times \bar{f}$

$\bar{\tau} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ -2 & 0 & 6 \\ 4 & -5 & 3 \end{vmatrix} = 30\hat{\imath} + 30\hat{\jmath} + 10\hat{k}$

14. $\bar{A} = 3\hat{\imath} + \hat{\jmath} + 2\hat{k}; \bar{B} = 2\hat{\imath} - 2\hat{\jmath} + 4\hat{k}$

$\bar{A} \times \bar{B} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 3 & 1 & 2 \\ 2 & -2 & 4 \end{vmatrix} = 8\hat{\imath} - 8\hat{\jmath} - 8\hat{k}$

$|\bar{A} \times \bar{B}| = 8\sqrt{3}$

unit vector perpendicular to the plane contaning $\bar{A}$ and $\bar{B}$ is $\hat{n} = \frac{\bar{A} \times \bar{B}}{|\bar{A} \times \bar{B}|}$

$\hat{n} = \frac{8\hat{\imath} - 8\hat{\jmath} - 8\hat{k}}{8\sqrt{3}}; \therefore \hat{n} = \frac{\hat{\imath} - \hat{\jmath} - \hat{k}}{\sqrt{3}}$

15. If $\bar{P} \times \bar{Q} = \bar{R}$ then $\bar{R}$ is perpendicular to both $\bar{P}\&\bar{Q}$.
Therfore $\bar{P} \cdot \bar{R} = \bar{Q} \cdot \bar{R} = 0$

16. Angle between the vectors is 90°.

17. Velocity $\bar{V} = v\hat{\imath}$
Lineor momentum $\bar{p} = m\bar{v} = mv\hat{\imath}$
$\bar{r} = y\hat{\imath}$
$\bar{L} = \bar{r} \times \bar{p} = y\hat{\imath} \times mv\hat{\jmath}$

$\bar{L} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 0 & y & 0 \\ mv & 0 & 0 \end{vmatrix} = mvy\hat{k}$

18. $\bar{A} = 4\hat{\imath} + 6\hat{\jmath};\ \bar{B} = 2\hat{\imath} + 3\hat{\jmath}$
$\bar{A} = 2(2\hat{\imath} + 3\hat{\jmath})$
$\bar{A} = 2\bar{B}$
$\therefore \bar{A} = \lambda\bar{B}$ and $\bar{A} || \bar{B}$
$\therefore \bar{A} \times \bar{B} = 0$

19. $A \perp B$ and $A \perp C$
$\bar{B} \perp \bar{C} = BC\sin\frac{\pi}{6}\hat{n}$
$\bar{B} \times \bar{C} = \frac{1}{2}BC\hat{n}$
$\bar{A} = 2(\bar{B} \times \bar{C})$ [$\because A = B = C = 1$]

# Level – III

| | | | | |
|---|---|---|---|---|
| 20] A,B,D | 21] D | 22]C | 23] A | 24]a-r;b-s;c-p;d-s |
| 25] D | 26] A,C | 27]a-r;b-s;c-q;d-p | | 28] a-r;b-p;c-s;d-q |
| 29] B | 30] C | 31] D | | |

20. If $|\bar{A} \times \bar{B}| = 0$
the possibilities are
1. $|\bar{A}|$ or $|\bar{B}|$ is zero
2. Angle between $\bar{A}\&\bar{B}$ is 0°
3. Angle between $\bar{A}\&\bar{B}$ is 180°

21. $|\bar{A}| = 2\hat{\imath} - 3\hat{\jmath} + \hat{k}$ and $|\bar{B}| = -\hat{\imath} + 3\hat{\jmath} + \hat{k}$

$$\bar{A} \times \bar{B} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 & -3 & 1 \\ -1 & 3 & 1 \end{vmatrix} = -6\hat{\imath} - 3\hat{\jmath} + 3\hat{k}$$

22. $|\overline{\mathrm{A}} \times \overline{\mathrm{B}}|^2 + |\overline{\mathrm{A}} \cdot \overline{\mathrm{B}}|^2 = \mathrm{A}^2\ \mathrm{B}^2$

23. $\bar{F} = (4\hat{\imath} - 5\hat{\jmath} + 3\hat{k})N;$

$$\overline{r_1} = (\hat{\imath} + 2\hat{\jmath} + 3\hat{k})m;\ \overline{r_2} = (3\hat{\imath} - 2\hat{\jmath} - 3\hat{k})m$$

$\bar{r} = \overline{r_1} - \overline{r_2} = (\hat{\imath} + 2\hat{\jmath} + 3\hat{k}) - (3\hat{\imath} - 2\hat{\jmath} - 3\hat{k})$

$\bar{r} = -2\hat{\imath} + 4\hat{\jmath} + 6\hat{k}$

$$\bar{\tau} = \bar{r} \times \bar{f} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ -2 & 4 & 6 \\ 4 & -5 & 3 \end{vmatrix} = (42\hat{\imath} + 30\hat{\jmath} - 6\hat{k})\mathrm{Nm}$$

24. a) $\hat{\imath} = \hat{\imath}(\hat{\jmath} \times \hat{k}) + \hat{\jmath} \cdot (\hat{k} \times \hat{\imath}) + \hat{k}(\hat{\imath} \times \hat{\jmath})$

$= \hat{\imath} \cdot \hat{\imath} + \hat{\jmath} \cdot \hat{\jmath} + \hat{k} \cdot \hat{k}$

$= 1 + 1 + 1 = 3$

b) $(\hat{\imath} \cdot \hat{\imath}) + (\hat{\imath} \cdot \hat{\imath}) = 1 + 1 = 2$

c) $(\overline{\mathrm{A}} \cdot \overline{\mathrm{B}})(\overline{\mathrm{A}} \times \overline{\mathrm{B}}) + (\overline{\mathrm{B}} \cdot \overline{\mathrm{A}})(\overline{\mathrm{B}} \times \overline{\mathrm{A}})$

$= (\overline{\mathrm{A}} \cdot \overline{\mathrm{B}})[(\overline{\mathrm{A}} \times \overline{\mathrm{B}}) + (\overline{\mathrm{B}} \times \overline{\mathrm{A}})]$

$= (\overline{\mathrm{A}} \cdot \overline{\mathrm{B}})[(\overline{\mathrm{A}} \times \overline{\mathrm{B}}) - (\overline{\mathrm{A}} \times \overline{\mathrm{B}})] = 0$

d) $= (\hat{\imath} + \hat{\jmath}) \cdot (2\hat{\jmath} + 2\hat{k})$

$= 1 \times 0 + 1 \times 2 + 0 \times 2 = 2$

25. $r = 4m;\ F = 20N;\ \theta = 30°$

$\bar{\tau} = \bar{r} \times \bar{F} = r\bar{F}\sin\ \theta\hat{n}$

$|\bar{\tau}| = rF\sin\ \theta = 4 \times 20 \times \sin\ 30° = 40\ \mathrm{N.m} = 4 \times 10\ \mathrm{N.m}$

a. $\bar{A} = 3\hat{\imath} + 4\hat{\jmath};\ A = \sqrt{3^2 + 4^2} = \sqrt{25} = 5$

2. $\bar{F} = (3\hat{\imath} + 4\hat{\jmath}); \bar{S} = 6\hat{\jmath}; W = \bar{F} \cdot \bar{S} = (3\hat{\imath} + 4\hat{\jmath}) \cdot 6\hat{\jmath} = 24J$

so (2) is not correct

3. Area of parallelogram with $\overline{\mathrm{A}}$&$\overline{\mathrm{B}}$ as adjacent sides is given by $|\overline{\mathrm{A}} \times \overline{\mathrm{B}}|$

4. $\mathrm{F} = 20\ \mathrm{N};\ \theta = 60°; \mathrm{f}\cos\ \theta = 20 \times \cos\ 60°$

So (4) is not correct

$$= 20 \times \frac{1}{2} = 10\ \mathrm{N}$$

27. $a - r; b - s; c - q; d - p$

28. a) $|\overline{\mathrm{A}} + \overline{\mathrm{B}}| = |\overline{\mathrm{A}} - \overline{\mathrm{B}}|$

$\sqrt{A^2 + B^2 + 2AB\cos\ \theta} = \sqrt{A^2 + B^2 - 2AB\cos\ \theta}$

$2AB\cos\ \theta = -2AB\cos\ \theta$

$4AB\cos\ \theta = 0$

$\cos\ \theta = 0$

$\theta = 90°$

b) $|\overline{\mathrm{A}} \times \overline{\mathrm{B}}| = \overline{\mathrm{A}} \cdot \overline{\mathrm{B}}$

$AB\sin\ \theta = AB\cos\ \theta$

$\frac{\sin\ \theta}{\cos\ \theta} = 1$

$\tan\ \theta = 1$

$\theta = 45°$

C) $\bar{A} \cdot \bar{B} = \frac{AB}{2}$

$AB\cos\ \theta = \frac{AB}{2}$

$\cos\ \theta = \frac{1}{2};$

$\theta = 60°$

D) $|\bar{A} \times \bar{B}| = \frac{AB}{2}; AB\sin\theta = \frac{AB}{2}$

$\sin\theta = \frac{1}{2}$

$\theta = 30°$

29. $|\vec{c}| = |\vec{a} \times \vec{b}|$

30. $\bar{A} = 2\hat{\imath} + 3\hat{\jmath} + 6\hat{k}; \bar{B} = 3\hat{\imath} - 6\hat{\jmath} + 2\hat{k}$

Vector perpendicular to $\bar{A}$ and $\bar{B}$ is $\bar{A} \times \bar{B}$

$$\bar{A} \times \bar{B} = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 & 3 & 6 \\ 3 & -6 & 2 \end{vmatrix} = 42\hat{\imath} + 14\hat{\jmath} - 21\hat{k}$$

given $\bar{A} \times \bar{B} = k(6\hat{\imath} + 2\hat{\jmath} - 3\hat{k})$

$42\hat{\imath} + 14\hat{\jmath} - 21\hat{k} = k(6\hat{\imath} + 2\hat{\jmath} - 3\hat{k})$

$= 7(6\hat{\imath} + 2\hat{\jmath} - 3\hat{k}) = k(6\hat{\imath} + 2\hat{\jmath} - 3\hat{k})$

$$K = \frac{7(6\hat{\imath} + 2\hat{\jmath} - 3\hat{k})}{6\hat{\imath} + 2\hat{\jmath} - 3\hat{k}}$$

k = 7.

31. $\bar{F} = -f\hat{k}$

$r = (1,1) = \hat{\imath} + \hat{\jmath}$

$\bar{\tau} = \bar{r} \times \bar{F}$

$$\begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 1 & 1 & 0 \\ 0 & 0 & -f \end{vmatrix}$$

$-f\hat{\imath} + f\hat{\jmath} + 0$

$= -f(\hat{\imath} - \hat{\jmath})$

# Worksheet – 6

## Level – I

1. D 2. D 3. B 4. C

## Level – II

1. D
2. A
3. A
4. A
5. B
6. A
7. B
8. C
9. A
10. D
11. B
12. D
13. D
14. B
15. D
16. B
17. B
18. B
19. A
20. A, C, D
21. A
22. A
23. B
24. a-q; b-p; c-S; d-t
25. A,B
26. D
27. D
28. C

1. $\bar{A} = (\hat{\imath} + \hat{\jmath} + \hat{k}), \bar{B} = (2\hat{\imath} - \hat{\jmath} + 3\hat{k})$ and $\bar{C}$ are acting on a body to keep equilibrium. Then the resultant of $\bar{A}, \bar{B}$ and $\bar{C}$ is zero

i.e., $\bar{A} + \bar{B} + \bar{C} = 0$

$(\hat{\imath}+\hat{\jmath}+\hat{k})+(2\hat{\imath}-\hat{\jmath}+3\hat{k})+\bar{C}=0$

$(3\hat{\imath}+4\hat{k})+\bar{C}=0;$

$\bar{C}=-(3\hat{\imath}+4\hat{k})$

2. $\overline{F_1}=a(\hat{\imath}-\hat{\jmath}+\hat{k});\overline{F_2}=2\hat{\imath}-3\hat{\jmath}+4\hat{k}$ and $\overline{F_3}=8\hat{\imath}-7\hat{\jmath}+6\hat{k}$

$\overline{F_1}+\overline{F_2}+\overline{F_3}=\bar{O}$

$a(\hat{\imath}-\hat{\jmath}+\hat{k})+(2\hat{\imath}-3\hat{\jmath}+4\hat{k})+(8\hat{\imath}-7\hat{\jmath}+6\hat{k})=\bar{O}$

$(a+2+8)\hat{\imath}+(-a-3-7)\hat{\jmath}+(a+4+6)\hat{k}=0$

$\therefore a+10=0\Rightarrow a=-10$

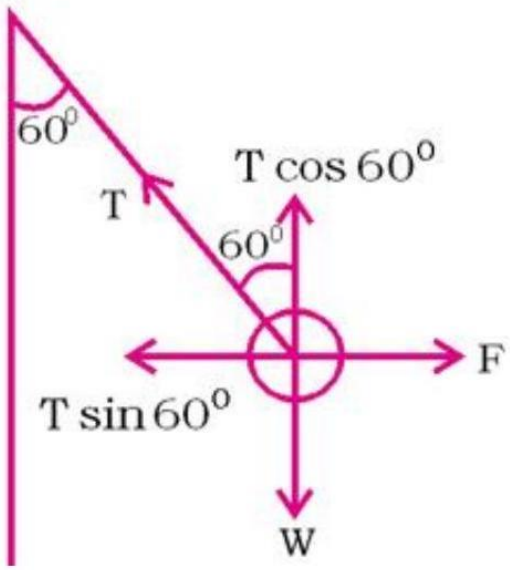
3. Given number of forces is 11

required number of forces $=\frac{360^\circ}{\theta}=\frac{360^\circ}{30^\circ}=12$

So one force of equal magnitude i.e., 5 N is required

∴ The resultant of all eleven forces is 5 N.

4. W = 10 kg wt




$W=T\cos 60^\circ; F=T\sin 60^\circ; T\cos 60^\circ=10\ kg\cdot wt$

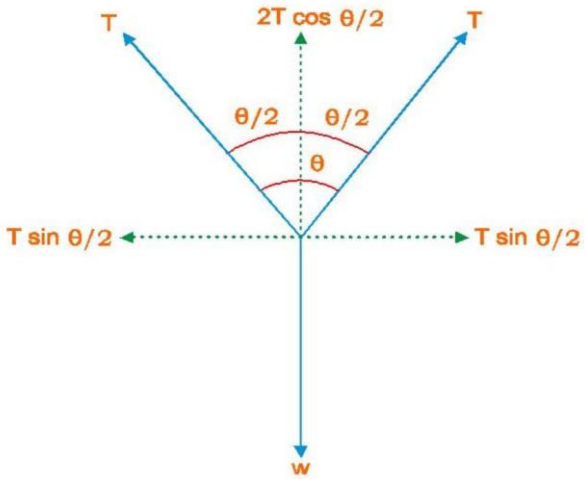
$T\times\frac{1}{2}=10;\ T=20\ N$

$F=T\sin 60^\circ;\ F=20\times\frac{\sqrt{3}}{2};\ F=10\sqrt{3}kgwt$

6. Number of identical forces = N

Required number of forces of maintain equilibrium

∴ Their resultant is zero



$W = 2\ T\cos\frac{\theta}{2}$

Maximum value at, $T = mg$

$2\ T\cos\frac{\theta}{2} = mg$

$\cos\frac{\theta}{2} = \frac{1}{2}$

$\frac{\theta}{2} = 60°$

$\theta = 120°$

2. $\bar{a} = -2\hat{\imath} + \hat{\jmath} - 3\hat{k}\ \vec{b} = 5\hat{\imath} + 3\hat{\jmath} - 2\hat{k}$

$3\bar{a} + 2\bar{b} - \bar{c} = 0 \Rightarrow \vec{c} = 3\bar{a} + 2\bar{b}$

$\bar{c} = 3(-2\hat{\mathbf{i}} + \mathbf{j} - 3\hat{k}) + 2(5\mathbf{i} + 3\mathbf{j} - 2\hat{k})$

$\bar{C} = -6\hat{\imath} + 3\hat{\jmath} - 9\hat{k} + 10\hat{\imath} + 6\hat{\jmath} - 4\hat{k}$

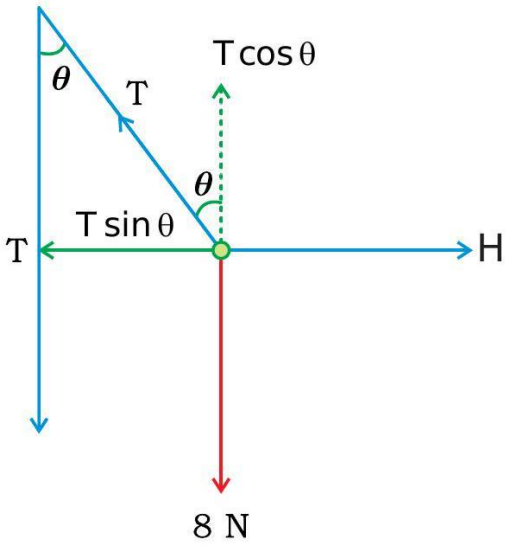
$c = 4\hat{\imath} + 9\hat{\jmath} - 13\hat{k}$

3. $H = T\sin\theta$

$w = T\cos\theta$

$\frac{H}{W} = \tan\theta$

$H = w\tan\theta$




$w = mg \Rightarrow H = mg \times \tan\theta$

$H = 3 \times 9.8 \times \frac{1}{\sqrt{3}} = 9.8\sqrt{3} = 16.97 \approx 17\ N$

4. $\vec{F}_1 = (2\hat{\imath} + \hat{\jmath} - \hat{k})N\ \vec{F}_2 = (2\hat{\imath} + 3\hat{\jmath} - 3\hat{k})N$

$F_3 = a(\hat{\imath} + \hat{\jmath} - \hat{k})$

$\vec{F}_1 + \vec{F}_2 + \vec{F}_3 = \bar{0}$

$(2\hat{\imath} + \hat{\jmath} - \hat{k}) + (2\hat{\imath} + 3\hat{\jmath} - 3\hat{k}) + a(\hat{\imath} + \hat{\jmath} - \hat{k}) = 0$

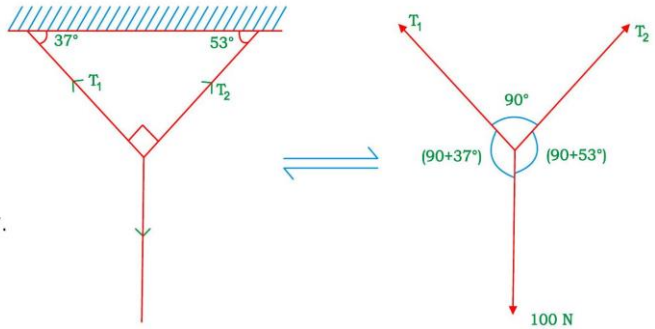
$(2 + 2 + a)\hat{\imath} + (1 + 3 + a)\hat{\jmath} + (-1 - 3 - a)\hat{k} = 0$

$a + 4 = 0 \Rightarrow a = 4$

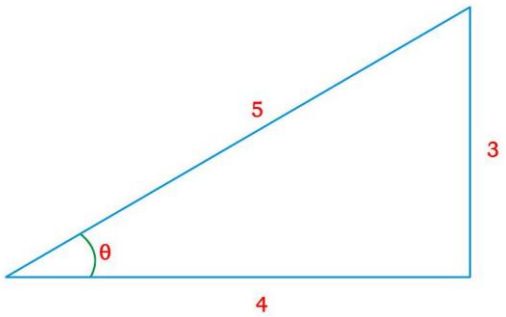
5. $\frac{P}{AB} = \frac{Q}{BC} = \frac{R}{CA} = K$

6. Lamis Theorm $\frac{P}{\sin\alpha} = \frac{Q}{\sin\beta} = \frac{R}{\sin\gamma}$

7.




Applying lamis theorem




$$\frac{T_1}{\sin(90+53°)} = \frac{T_2}{\sin(90+37°)} = \frac{100}{\sin 90°}$$

$$\frac{T_1}{\cos 53} = \frac{T_2}{\cos 37°} = 100$$

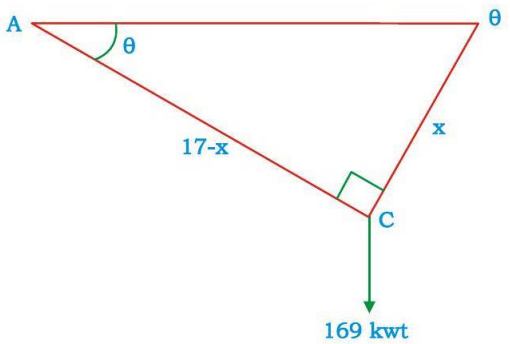
$T_1 = 100\cos 53°$ $T_2 = 100\cos 37°$

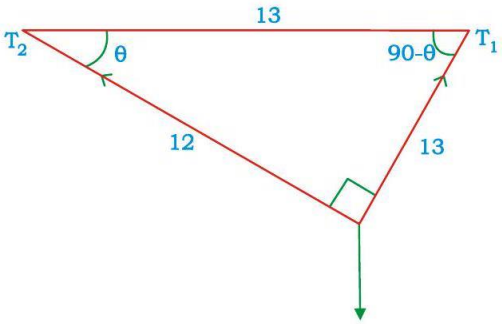
Given $\text{Tan } 37° = \frac{3}{4} \Rightarrow \sin 37° = \cos 53° = \frac{3}{5}$ and

$\sin 53° = \cos 37° = \frac{4}{5}$

$\therefore T_1 = 100 \times \frac{3}{5}$ $T_2 = 100 \times \frac{4}{5}$

$T_1 = 60$ N $T_2 = 80$ N




In $\triangle ABC, \angle C = 90°$

$x^2 + (17 - x)^2 = 13^2$

$x^2 - 17x + 60 = 0$

$(x - 5)(x - 12) = 0$

$x = 5,12$

$$\frac{T_1}{\sin(90+0)} = \frac{T_2}{\sin(180-0)\sin} = \frac{169}{90°}$$

$$\frac{T_1}{\cos\theta} = \frac{T_2}{\sin\theta} = 169$$

$T_1 = 169\cos\theta\ T_2 = 169\sin\theta$

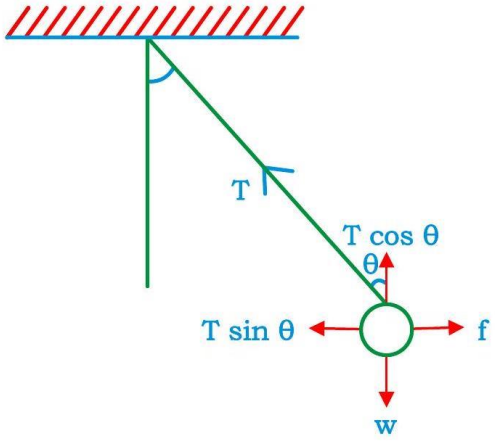
$T_1 = 169 \times \frac{12}{13}\ \ T_2 = 169 \times \frac{5}{13}$

$T_1 = 156\ N\ \ T_2 = 65\ N$

9. $F = 20N$ each angle between each force is same i.e. $\frac{360}{5} = 72°$. Then the resultant is zero.

10. $F = W\tan\theta \Rightarrow 20 \times \tan 60° = 20\sqrt{3}$kgwt

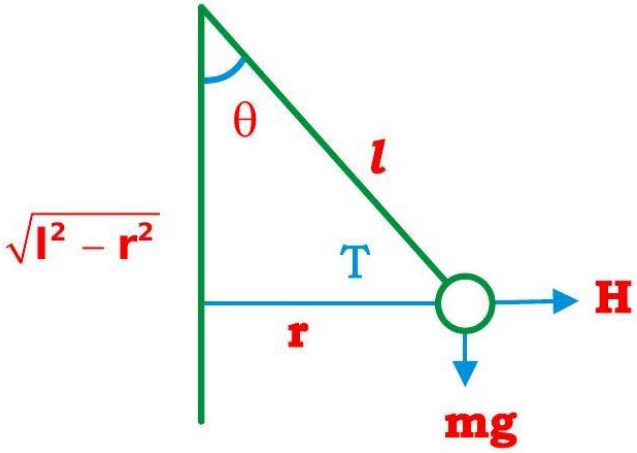
11. $F = T\sin\theta$




$\frac{F}{T} = \sin\theta = \sin 60° = \frac{\sqrt{3}}{2}$.

$\therefore F:T = \sqrt{3}:2$

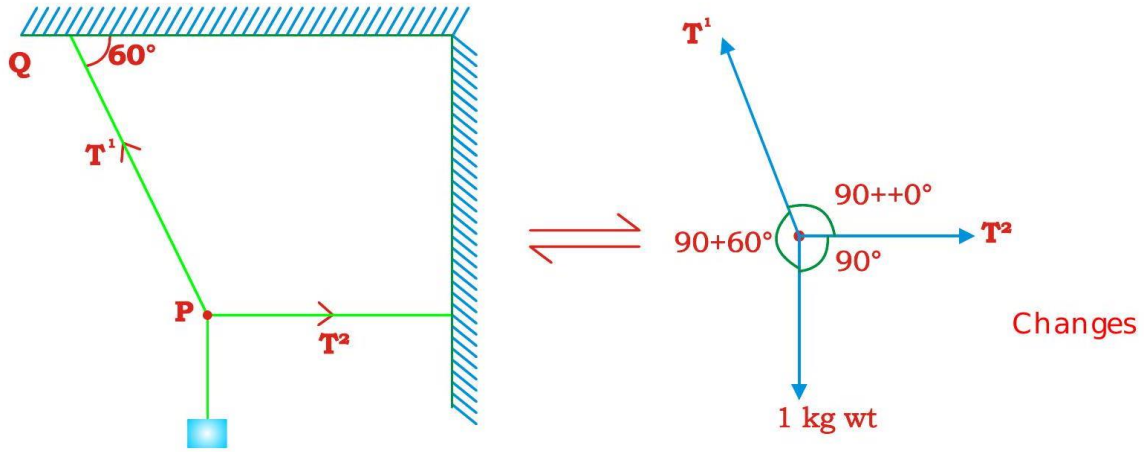
12. $H = mg + \tan\theta$




$$H = \frac{mgr}{\sqrt{l^2 - r^2}}$$

13. $\angle AP_2P_1 = 120°(W_1 = W_2 = T)$

14. $\frac{T_1}{\sin 90°} = \frac{T_2}{\sin(90+60°)} = \frac{1\text{kgwt}}{\sin(90+30°)}$




1 kg

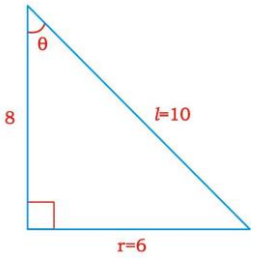
$$\frac{T_1}{1} = \frac{1}{\cos 30^\circ} \Rightarrow T_1 = \frac{2}{\sqrt{3}}$$

$$T_1 = \frac{2 \times \sqrt{3}}{\sqrt{3} \times \sqrt{3}} = \frac{2}{3}\sqrt{3}\text{kgwt}$$

15. $m_1 g\sin\theta = m_2 g\sin\alpha$

$$\frac{m_2}{m_1} = \frac{\sin\theta}{\sin\alpha} = \sqrt{3}:1$$

16. $w = 20\text{kgwt} L = 10$ cm r = 6 cm




$H = mg\tan\theta$

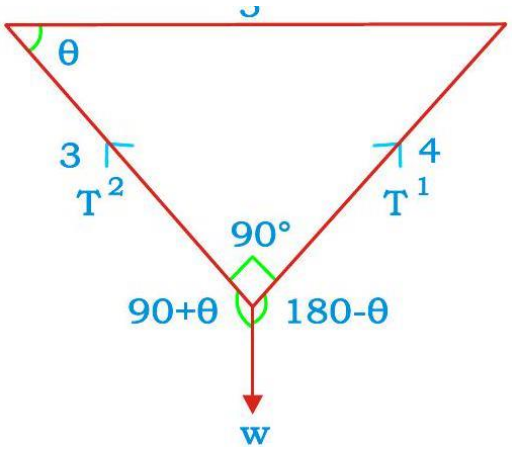
$\Rightarrow H = 20 \times \frac{3}{4}$

$H = 5 \times 3 = 15\text{kgwt}$

17. $T\cos\theta = 1 \Rightarrow T = \frac{1}{\cos\theta}$

18. $\frac{T_1}{\sin(90+\theta)} = \frac{T_2}{\sin(180-\theta)} = \frac{w}{\sin 90^\circ}$

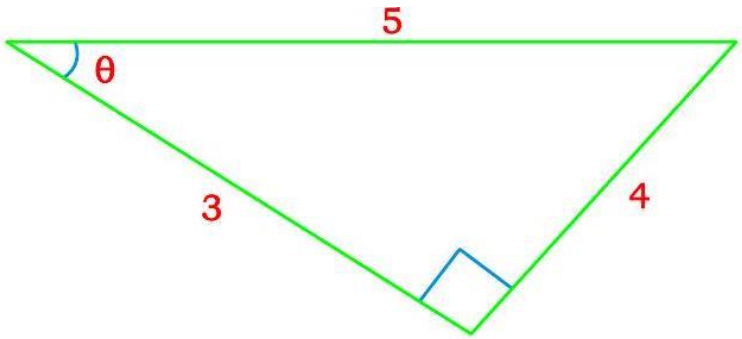
5




$$\frac{T_1}{\cos\theta} = \frac{T_2}{\sin\theta} = w$$

$T_1 = w\cos\theta\ T_2 = w\sin\theta$

5

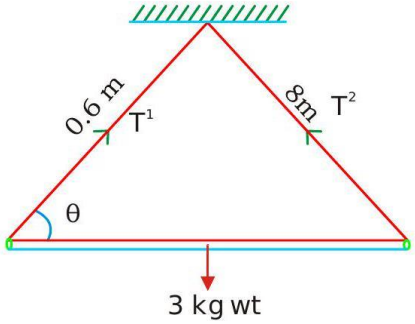


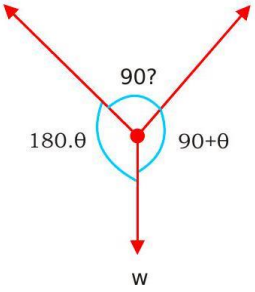

$T_1 : T_2 = 3:4$

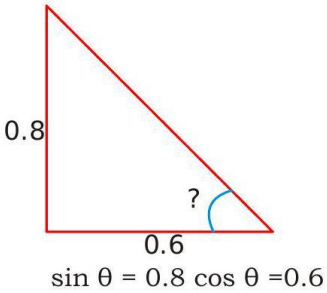
$\cos\theta = \frac{3}{5}\ \sin\theta = \frac{4}{5}$

$T_1 : T_2 = 3:4$

19. $\frac{T_1}{\sin(180-\theta)} = \frac{T_2}{\sin(90+\theta)} = \frac{w}{\sin 90°}$



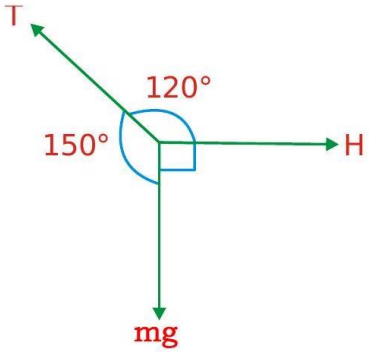




$T_1 = w\sin\theta\ T_2 = w\cos\theta$

$T_1 = 3 \times 0.8\ \ T_2 = 3 \times 0.6$

$T_1 = 2.4\text{kgwt} T_2 = 1.8\text{kgwt}$

22,23,24




$\frac{T}{\sin 90°} = \frac{mg}{\sin 120°} = \frac{H}{\sin 150°}$

$T = \frac{mg}{\sqrt{3/2}} = \frac{H}{1/2}$

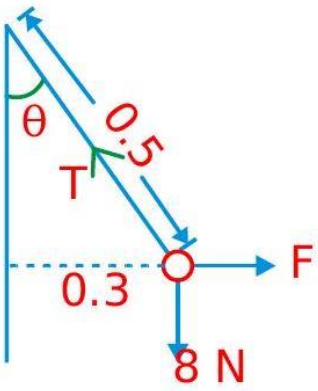
$T = \frac{2}{\sqrt{3}} \times 0.05 \times 98$

$T = 0.56\ N$

$H = \frac{T}{2} = \frac{0.56}{2} = 0.28\ N$

$H : T = 0.28 : 0.56 \Rightarrow 1:2$ (or) 0.5

25. $F = w\tan\theta$



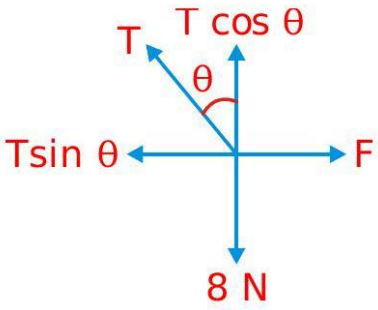

$F =$

$T\cos\theta = g$

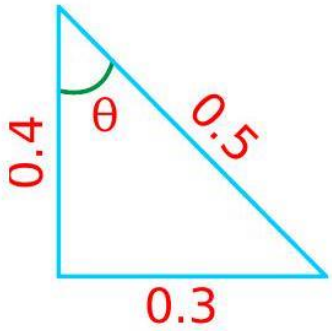
$T \times \frac{4}{5} = 8$

$T = 8 \times \frac{5}{4}$

$T = 10\ N$




Tension × horizontal force $= 10 \times 6 = 60$



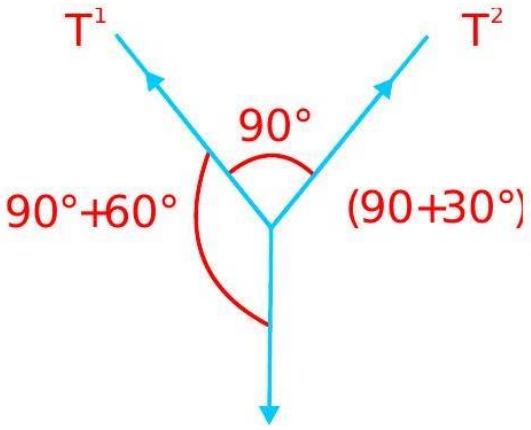

$\text{Tan}\ \theta = \frac{3}{4}$

$\cos\theta = \frac{4}{5}$

$\sin\theta = \frac{3}{5}$

$$\frac{\text{Tension}}{\text{horizontal force}} = \frac{10}{6}$$

26. Applying Lami's theorem




80 kg wt

$$\frac{80}{\sin 90^\circ} = \frac{T_1}{\sin(90+30)} = \frac{T_2}{\sin(90+60^\circ)}$$

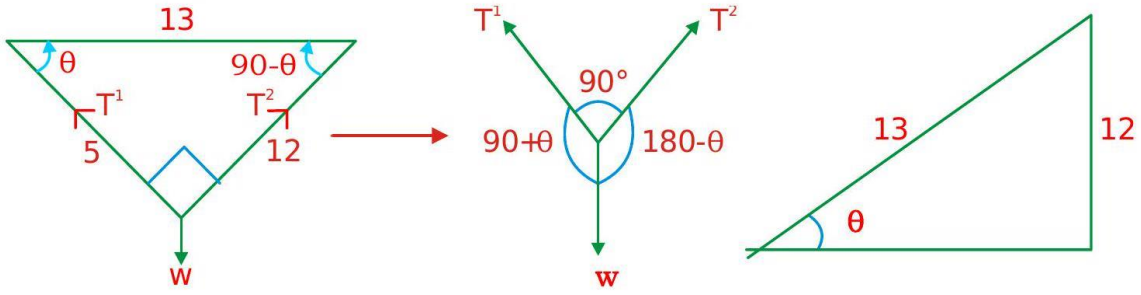
$$80 = \frac{T_1}{\cos 30^\circ} = \frac{T_2}{\sin 60^\circ}$$

$$T_1 = 80\cos 30^\circ \; T_2 = 80\sin 60^\circ$$

$$T_1 = 80 \times \frac{\sqrt{3}}{2} \quad T_2 = 80 \times \frac{1}{2}$$

$$T_1 = 40\sqrt{3}\text{kgwt}\; T_2 = 40\text{kgwt}$$

28. Applyiing Lamis theorem




$$\frac{T_1}{\sin(180-\theta)} = \frac{T_2}{\sin(90+\theta)} = \frac{w}{\sin 90^\circ}$$

$$\frac{T_1}{\sin\theta} = \frac{T_2}{\cos\theta} = w$$

$$T_1 = \frac{12}{13}w \; T_2 = \frac{5w}{13}$$

## Worksheet – 7

# Level – I

| | | | |
|---|---|---|---|
| 1. C | 4. A | 7. C | 10. B |
| 2. D | 5. D | 8. C | 11. D |
| 3. A | 6. C | 9. A | |

# Level – II

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1. A | 2. A | 3. A | 4. B | 5. C | 6. C | 7. B | 8. D |
| 9. D | 10. A | 11.C | 12. A | 13. B | 14. C | 15. B | 16. A |
| 17. A | | | | | | | |

---

1. $[\hat{\imath}\hat{\jmath}\hat{k}] = \hat{\imath} \cdot (\hat{\jmath} \times \hat{k}) = \hat{\imath}.\hat{\imath} = 1$

2. $[\bar{A}\bar{B}\bar{C}] = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 3 & 3 & 3 \\ 5 & 5 & 5 \end{vmatrix} = 0$

3. $\hat{\imath} \cdot (\hat{\imath} \times \hat{\jmath}) = \hat{\imath} \cdot \hat{k} = 0$

4. $\hat{\imath}.(\hat{\jmath} \times \hat{k}) = \hat{\imath}.\hat{\imath} = 1$

5. $\hat{\jmath} \cdot (\hat{\imath} \times \hat{k}) = \hat{\imath}.-\hat{\jmath} = -1$

6. Volume of paralleopiped $(2\hat{\imath} - 3\hat{\jmath}) \cdot \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 1 & 1 & -1 \\ 3 & 0 & -1 \end{vmatrix} = (2\hat{\imath} - 3\hat{\jmath}) \cdot (-\hat{\imath} - 2\hat{\jmath} - 3\hat{k}) = 4$ cubic units

7. $[\bar{a} \quad \bar{b} \quad \bar{a} \times \bar{b}] = (\bar{a} \times \bar{b}) \cdot (\bar{a} \times \bar{b}) = |(\bar{a} \times \bar{b})|^2 = 2^2 = 4$

8. Conceptual

9. $[\bar{a} \quad \bar{b} \quad \bar{c}] = \bar{a} \cdot \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 & 4 & -1 \\ 1 & 1 & 3 \end{vmatrix} = (\hat{\imath} + \hat{\jmath} + \hat{k}) \cdot (13\hat{\imath} - 7\hat{\jmath} - 2\hat{k}) = 4$

10. Conceptual

11. $(\hat{\imath} - 2\hat{\jmath} - 3\hat{k}) \cdot \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 2 & 1 & -1 \\ 1 & 3 & -2 \end{vmatrix} = (\hat{\imath} - 2\hat{\jmath} - 3\hat{k}) \cdot (\hat{\imath} + 3\hat{\jmath} + 5\hat{k}) = -20$

12. $(\hat{\imath} - \hat{\jmath}) \times (\hat{\jmath} - \hat{k}) \cdot (\hat{k} - \hat{\imath}) = \begin{vmatrix} \hat{\imath} & \hat{\jmath} & \hat{k} \\ 1 & -1 & 0 \\ 0 & 1 & -1 \end{vmatrix} \cdot (\hat{k} - \hat{\imath}) = \hat{\imath} + \hat{\jmath} + \hat{k} \cdot (\vec{k} - \vec{\imath}) = 0$

13. $(\bar{a} - \lambda\bar{b}) \cdot (\bar{b} - 2\bar{c}) \times (\bar{c} - 2\bar{a}) = (\bar{a} - \lambda\bar{b}) \cdot \begin{vmatrix} \bar{a} & \bar{b} & \bar{c} \\ 0 & 1 & -2 \\ -2 & 0 & 1 \end{vmatrix} = (\bar{a} - \lambda\bar{b}) \cdot (\bar{a} - 4\bar{b} + 2\bar{c})$

$1 + 4\lambda = 0$

$\lambda = -\frac{1}{4}$

14,15,16,17 : CONCEPTUAL

# 3. KINEMATICS

## Worksheet – 1

## Level – I

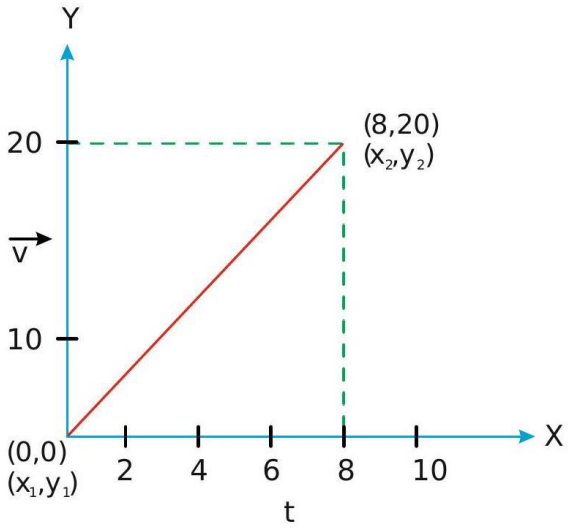
| | | | | |
|---|---|---|---|---|
| 1. B | 4. A | 7. A | 10. C | 13. B |
| 2. C | 5. D | 8. D | 11. A | 14. B |
| 3. C | 6. A | 9. A | 12. B | 15. B |

## Level – II

| | | | | |
|---|---|---|---|---|
| 1. A | 6. D | 11. D | 16. D | 21. A |
| 2. B | 7. D | 12. A | 17. A, B, C | 22. A |
| 3. B | 8. B | 13. A | 18. D | |
| 4. A | 9. B | 14. B | 19. D | |
| 5. A | 10. D | 15. B | 20. D | |

23. D
24. a-p,b-q,c-r,d-s
25. A, B, C
26. B
27. D
28. B

1. From 0 to 2sec, it is parallel to $x$-axis. So the body moving with uniform velocity.
2. Body has positive acceleration. Since slope increases.
3. From fig $\frac{V_A}{V_B} = \frac{\text{Tan } 60^\circ}{\text{Tan } 45^\circ} = \frac{\sqrt{3}}{1} = \sqrt{3}:1$
4. From $U = 10$ m/s
   $V = 30$ m/s
   $S = 4$ km $= 4 \times 1000$ m $= 4000$ m
   From $V^2 - U^2 = 2as$
   $(30)^2 - (10)^2 = 2(a)4000$
   $900 - 100 = 8000 \times a$
   $800 = 8000 \times a$
   $a = \frac{800}{8000} = \frac{1}{10}$ m/s$^2$
   And also $a = \frac{V-U}{t}$
   $t = \frac{V-U}{a} = \frac{30-10}{1/10} = 20 \times 10 = 200$sec
5. Given $U = 0$
   $V = 20$ m/s
   $t = 4$ s.
   From $a = \frac{V-U}{t}$
   $a = \frac{20-0}{4}$
   $a = 5$ m/s$^2$
   And also $V^2 - U^2 = 2aS$
   $(20)^2 - 0^2 = 2 \times 5 \times S$
   $400 = 10 \times S$
   $S = \frac{400}{10}$
   $S = 40$ m
6.




Area under v-t graph is distance

$\therefore$ distance $= \frac{1}{2} \times 8 \times 20 = 80$ m

Slope of v-t graph gives acceleration

Slope $= \frac{y_2 - y_1}{x_2 - x_1} = \frac{20-0}{8-0}$

Slope $= \frac{20}{8} = 2.5$ m/s$^2$

i.e acceleration $= 2.5$ m/s$^2$

7. From figure upto time to the particle moves constant velocity and then stops.

8. Given $U = 100$ m/s

$g = 10$ m/s$^2$

$T_f = \frac{2U}{g} = \frac{2 \times 100}{10} = 20$sec

9. We know that $h \alpha t^2 [\because U = 0]$

Hence $\frac{t_1}{t_2} = \sqrt{\frac{h_1}{h_2}}$

$\therefore t_1 : t_2 = \sqrt{h_1} : \sqrt{h_2}$

10. x distance in the first 2sec

$$\therefore x = \frac{1}{2} g(2)^2 [\because U = 0]$$

$$x = 2g ----(1)$$

And $y$ distance in next 2sec

i.e $x + y = \frac{1}{2} g(4)^2$

$2g + y = 8g$ [$\because$ fromm (1)]

$y = 6g$

$y = 3 \times 2g$

$y = 3x$ [$\because$ from (1)]

11. Area under v-t graph gives distance

distance $= A_1 + A_2 + A_3 + A_4 + A_5$

$= \frac{1}{2} \times 1 \times 20 + 1 \times 20 + \frac{1}{2} \times 1 \times 10 + 1 \times 10 + 1 \times 10$

$= 10 + 20 + 5 + 10 + 10 = 55$ m

12. Area under v-t graph gives distance

distance $= A_1 + A_2 + A_3 + A_4 + A_5 + A_6$

$= \frac{1}{2} \times 1 \times 4 + 2 \times 4 + \frac{1}{2} \times 1 \times 4 + \frac{1}{2} \times 1 \times 2 + 2 \times 2 + \frac{1}{2} \times 1 \times 2$

$= 2 + 8 + 2 + 1 + 4 + 1$

distance $= 18$ m upto 8 seconds

13. The time taken to reach the water surface is $t = \sqrt{\frac{2\,h}{g}} = \sqrt{\frac{2\times44.1}{9.8}} = 3\text{sec}$

After 1sec second stone thrown i.e 2sec

$\therefore h = ut + \frac{1}{2}gt^2$

$44.1 = U(2) + \frac{1}{2}\times 9.8 \times 2^2$

Solve above equation, we get $U = 12.25$ m/s

14. $\frac{\text{slopeof graph first '2'seconds}}{\text{slopeof graph in next'4'seconds}}$

$= \frac{\frac{S}{2}}{\frac{S}{4}} = \frac{S}{2}\times\frac{4}{S} = 2:1$

15. Area under $v$-t graph gives displacement

$\therefore$ displacement $= A_1 - A_2 + A_3$

$= \frac{1}{2}\times 3\times 2 - \left[\frac{1}{2}\times 1\times 2\right] + 1\times 1$

$= 3 - 1 + 1 = 3$ m

16. Given, $u = SU, a = 6\ m/s^2$ $S_{6^{th}} = 41$

$S_{n^{th}} = U + \frac{a}{2}(2n-1)$

$S_{6^{th}} = 8U + \frac{6}{2}(2\times 6 - 1)$

$41 = 8U + 3(12-1)$

$8U = 41 - 33 = 8$

$U = \frac{8}{8} = 1$ m/s

17. From the graph maximum acceleration in the region is BC. Slop of BC graph is

Slope $= \frac{y_2 - y_1}{x_2 - x_1}$

$= \frac{60-20}{1-0.75}$

$= \frac{40}{0.25}$

$= \frac{40}{25}\times 100$

slope $= 160$ km/h$^2$

$\therefore a_{max} = 160$ km/h$^2$

18. Conceptual

19. $t_i = 0, x_i = 60$

$t_f = 10, x_f = 40$

$v_{avg} = \frac{|x_f - x_i|}{t_f - t_i} = \frac{|40-60|}{10-0} = \frac{20}{10} = 2$ m/s

20. Area under v-t graph gives distance for first 40 seconds distance

$= \frac{1}{2}\times 20\times 5 + \frac{1}{2}\times 20\times 5$

distance $= 100$ m

displacement $= \frac{1}{2}\times 20\times 5 - \frac{1}{2}\times 20\times 5$

$= 0$

Avg velocity $= \frac{\text{total displacement}}{\text{total time}}$

Avgvelocity $= \frac{0}{40} = 0$

21,22,23

Body moving with uniform velocity so there is No change in velocity up to 10 min.

25. Slope of v-t graph gives acceleration
$\therefore$ Slope $= \frac{y_2-y_1}{x_2-x_1}$
acceleration $= \frac{8-2}{10-0} = \frac{6}{10}$
$\therefore$ a = 0.6 m/s$^2$
distance travelled in 0 to 10sec
→ Area under v-t graph gives distance
distance = 50 m
→ displacement = 50mb/w0 to 10sec.
0 to 10sec.

26. From figure
$H_{max} = \frac{u^2}{2g}$
$H_{max} = \frac{(40)^2}{2\times10} = \frac{40\times40}{2\times10} = 80$ m.

27. Net displacement of the body
Area under v-t graph gives displacement
displacement $= \frac{1}{2}\times4\times40 - \frac{1}{2}\times4\times40 = 0$

28. Total distance covered by the body is
Area under v-t graph gives distance
displacement $= \frac{1}{2}\times4\times40 + \frac{1}{2}\times4\times40 = 160$ m

30. Acceleration of car from 0 to $A$
Slope $= \frac{y_2-y_1}{x_2-x_1} = \frac{40-0}{10-0}$
Slope $= \frac{40}{10}$
Acceleration = 4 m/s$^2$

31. Acceleration of car A to B from A to B it moves with uniform velocity. so acceleration Zero.

32. Retardation from $B$ to $C$
Slope $\frac{y_2-y_1}{x_2-x_1} = \frac{40-0}{50-30} = \frac{40}{20} = 2$ m/s$^2$

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# Worksheet – 2

## Level – I

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1) A | 2) C | 3) D | 4) C | 5) C | 6) D | 7) A | 8) C |
| 9) C | 10) B | 11) C | 12) B | 13) D | 14) B | 15) D | |

1. $a = g$
2. goes on decreasing
3. $\theta = 45^\circ \Rightarrow R_{max} = \frac{u^2}{g}$
$H = \frac{u^2\sin^2\theta}{2g} \Rightarrow H = \frac{u^2}{4g} \Rightarrow H = \frac{R}{4}$
4. $\bar{u} = u\cos\theta i + u\sin\theta j$
$\bar{v} = \cos\theta(i) - u\sin\theta(j)$
$\Delta v = -2u\sin\theta$
$\Delta p = -2mu\sin\theta$
5. Projectile reaches its highest point,it looses all of its vertical velocity.
6. maximum range $\theta = 45^\circ \Rightarrow \tan 45^\circ = 1$
7. Slightly higher than the target.

goes on increasing upto 90°.

# Level – II

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1) B | 2) C | 3) B | 4) A | 5) A | 6) D | 7) D | 8) B |
| 9) B | 10) B | 11) B | 12) B | 13) D | 14) A | 15) B | 16) A |
| 17) B | 18) C | 19) B | | | | | |

1. Time (t) $= \frac{\text{usin } \theta}{\text{g}} = \frac{20\sqrt{2}\times\frac{1}{\sqrt{2}}=2}{10} = 2$
2. (t) $= \frac{2\text{usin } \theta}{\text{g}} = 4\text{sec}$
3. $\text{h} = \frac{\text{u}^2\sin^2\theta}{2\text{ g}} = 92$ m
4. $\text{R} = \frac{\text{u}^2\sin 2\theta}{\text{g}} = 80$ m
5. $\text{R}_{\max} = \frac{\text{u}^2}{\text{g}}$

$\theta = 45°$ for max.range $\Rightarrow \frac{u^2}{g} = 16000$

$\Rightarrow u = \sqrt{16000} = 400$ m/s

6. $\bar{u} = u\cos\theta\hat{\imath} + u\sin\theta\hat{\jmath}$

$= 10\frac{\sqrt{3}}{2}\hat{\imath} + 10\frac{1}{2}\hat{\jmath} = 5\sqrt{3}\hat{\imath} + 5\hat{\jmath}$

$s = ut + \frac{1}{2}at^2 \Rightarrow 60 = 5t + \frac{1}{2}(-10)t_2$

$\Rightarrow (t-4)(t+3) = 0 \Rightarrow t = 4\text{sec}$

7. $\text{R} = \text{H} \Rightarrow \frac{\text{u}^2\sin 2\theta}{\text{g}} = \frac{\text{u}^2\sin^2\theta}{2\text{ g}} \Rightarrow \tan\theta = 4 \Rightarrow \sec^2 = 1 + \tan^2\theta$

$\Rightarrow \sin^2 = \frac{16}{17}$

$H = \frac{u^2\sin^2\theta}{2g} = \frac{u^2}{2g}\left(\frac{16}{17}\right) = \frac{8u^2}{17g}$

8. $R = 245$ m, u $= 49$ m/s

$\text{R} = \frac{\text{u}^2\sin 2\theta}{\text{g}} \Rightarrow \theta = 45°$

time, $\text{t} = \frac{\text{R}}{\text{ucos }\theta} = \frac{245\times\sqrt{2}}{49} = 5\sqrt{2}\text{sec}$

9. $\text{R}_{\max} = \frac{\text{u}^2}{\text{g}}, \text{u}_\text{n} = \text{ucos }\theta$
10. $\text{V}_\text{y} = \text{u}_\text{y} - \text{gt} = \text{usin }\theta - \text{gt} = -8$ m/s; $\text{V}_\text{y} = 8$ m/s downwards
11. $\text{h} = \text{usin }\theta - \frac{1}{2}\text{ g}^2 \Rightarrow 2 = 14\sin\theta - 5$

$\Rightarrow \sin\theta = \frac{1}{2} \Rightarrow \theta = 30°$

12. $\text{h} = \frac{\text{u}^2}{2\text{ g}}$

ratio of the heights $\left(\frac{h_1}{h_2}\right) = ?$

$\frac{\text{h}_1}{\text{h}_2} = \frac{\frac{\text{u}_1^2}{2\text{ g}}}{\frac{\text{u}_2^2}{2\text{ g}}}$

$\frac{h_1}{h_2} = \frac{u_1^2}{u_2^2}$

w.k.t. $v = u + at$

$v = 0, t = \frac{-u}{a}$

$a$ is some as equal to g and t is same, so u is same ratio of heights = 1:1

13. $T = \frac{2u\sin\theta}{g_p}$

14. $\theta = 30°$ u = 30 m/st = 1.5sec

$V_y = u\sin\theta - gt$

$$V_y = 30 \times \frac{1}{2} - 10 \times 1.5$$
$$V_y = 15 - 15$$
$$V_y = 0$$

15. $V_{y_1} = v\sin 30,\ V_{y_2} = v\sin 60°$

$$u_1 = u_2 = u; \theta_1 = 30°\ \theta_2 = 60°$$
$$\frac{H_1}{H_2} = \frac{\frac{u_1^2\sin^2 30}{2g}}{\frac{u_2^2\sin^2 60}{2g}} = \frac{\sin^2 30}{\sin^2 60} = \frac{\left(\frac{1}{2}\right)^2}{\left(\frac{\sqrt{3}}{2}\right)^2} = \frac{\left(\frac{1}{2}\right)^2}{\left(\frac{\sqrt{3}}{2}\right)^2} = \frac{1}{3}$$

16. $\frac{H}{T^2} = \frac{u^2\sin^2\theta/2g}{4u^2\sin^2\theta/g^2} = \frac{g}{8} = \frac{10}{8} = \frac{5}{4}$

17. $H_{max} = \frac{A^2}{4B} = \frac{16\times16}{4\times5} = 12.8$ m

18. $R = \frac{A}{B} = \frac{10}{5/9} = 18$ m

19. $R = H \Rightarrow \frac{u^2\sin 2\theta}{g} = \frac{u^2\sin^2\theta}{2g}$

$$\Rightarrow 4 = \tan\theta \Rightarrow \theta = \tan^{-1} 4$$

# Level – III

| | | | | |
|---|---|---|---|---|
| 20) A,B,C,D | 21) A | 22) C | 23) A | |
| 24) a-t,b-p,c-r,d-q | 25) A,C,D | 26) C | 27) D | 28) B |
| 29) a-q,b-r,c-s,d-p | 30) A,C | 31) a-r,b-p,c-s,d-q | | 32) A |

21. u = 60 $ms^{-1}$

$\theta = 30°$

Velocity at maximum height $u\cos\theta = 60 \times \cos 30°$

$$= 60 \times \frac{\sqrt{3}}{2}$$
$$= 30\sqrt{3}\ ms^{-1}$$

22. $H_{max}$ = ?

$$H_{max} = \frac{u^2\sin 2\theta}{2g}$$
$$= \frac{(60)^2 \times \sin 30 \times \sin 30}{2g}$$
$$= \frac{\frac{900}{3600} \times \frac{1}{} \times \frac{1}{}}{20} = \frac{90\emptyset}{\emptyset\emptyset} = 45$$

$$H_{max} = 45 \text{ m}$$

23. Range = ?

$R = \frac{u^2 \sin 2\theta}{g} = \frac{(60)^2 \times \sin 60}{10} = \frac{360\emptyset \times \frac{\sqrt{3}}{}}{1\emptyset} = 180\sqrt{3}$ m

25. $H_{max} = \frac{u^2 \sin^2 \theta}{2g}$, $T = \frac{2u \sin \theta}{g}$, $R = \frac{u^2 \sin 2\theta}{g}$

26. $\vec{u} = u\cos\theta\hat{i} + u\sin\theta\hat{j}$
$= 60 \cdot \cos 30\hat{i} + 60 \cdot \sin 30\hat{j}$
$= 60 \times \frac{\sqrt{3}}{2}\hat{i} + 60 \times \frac{1}{2}\hat{j}$
$= \frac{30}{60} \times \frac{\sqrt{3}}{2}\hat{i} + 60 \times \frac{1}{2}\hat{j}$
$= 30\sqrt{3}\hat{i} + 30\hat{j}$

27. Velocity after 3sec
$v = u\cos\theta\hat{\mathbf{i}} + (u\sin\theta)\hat{j}$
$= 60 \times \cos 30\hat{i} + (60 \times \sin 30 - 10 \times 3)\hat{j}$
$= 60 \times \frac{\sqrt{3}}{2}\hat{i} + \left(60 \times \frac{1}{2} - 30\right)\hat{j}$
$= 30\sqrt{3}\hat{i} + (0)\hat{j}$
$= 30\sqrt{3}i$

28. displacement after 2 s is (along x-axis) $= S = ut + \frac{1}{2}at^2$
along x-direction $a_x = 0$
$S_x = u_x \cos\theta \times t + 0$
$S_x = 60 \times \cos 30 \times 2 + 0$
$S_x = 60 \times \frac{\sqrt{3}}{} \times$
$S_x = 60\sqrt{3}$
Along y-direction
$a_y = -g$
$s_y = ut - \frac{1}{2}gt^2$
$= u\sin\theta \times t - \frac{1}{2}gt^2$
$= 60 \times \sin 30 \times (2) - \frac{1}{2} \times 10 \times (2)^2$
$= 60 \times \frac{1}{} \times$
$= 60 - 20 = 40$
$S = S_x\hat{i} + S_y\hat{j}$
$= 60\sqrt{3}\hat{i} + 40\hat{j}$

29. (a) R = H
$\frac{x^2 \sin 2\theta}{\emptyset} = \frac{x^2 \sin^2 \theta}{2\phi}$
$2\sin\theta\cos\theta = \frac{\sin^2 \theta}{2}$
$4\cos\theta = \sin\theta$
$4 = \frac{\sin\theta}{\cos\theta}$
$\tan\theta = 4$
$\theta = \tan^{-1}(4)$
(b) R = 2H
$\frac{x^2 \sin 2\theta}{\emptyset} =$

$2\sin\theta\cos\theta = \sin^2\theta$

$2\cos\theta = \sin\theta$

$2 = \frac{\sin\theta}{\cos\theta}$

$2 = \tan\theta$

$\theta = \tan^{-1}(2)$

(c) R = 3H

$$\frac{x^2\sin 2\theta}{\Phi} = 3 \times \frac{x^2\sin^2\theta}{2\phi}$$

$2\sin\theta\cos\theta = 3\frac{\sin^2\theta}{2}$

$4\cos\theta = 3\sin\theta$

$\frac{4}{3} = \frac{\sin\theta}{\cos\theta}$

$\frac{4}{3} = \tan\theta$

$\theta = \tan^{-1}\left(\frac{4}{3}\right)$

(d) R = 4H

$\frac{x^2\sin 2\theta}{\phi} = {}^{2}4 \times \frac{\varphi^2\sin^2\theta}{2\phi}$

$2\sin\theta\cos\theta = 2\sin^2\theta$

$\cos\theta = \sin\theta$

$1 = \frac{\sin\theta}{\cos\theta}$

$1 = \tan\theta$

$\theta = \tan^{-1}(1)$

30. Given

$y = AX - BX^2$

$y = x\tan\theta - \frac{g}{2u^2\cos^2\theta}x^2$

Compare equation (1) and (2)

$A = \tan\theta$

$B = \frac{g}{2u^2\cos^2\theta}$

Now, Range

$R = \frac{u^2 2\sin\theta\cdot\cos\theta}{g} \times \frac{\cos\theta}{\cos\theta}$

$R = \frac{u^2 \times 2 \times \tan\theta\cdot\cos^2\theta}{g}$

$R = \tan\theta\cdot\frac{2u\cos^2\theta}{g}$

$\tan\theta = A; \frac{2u^2\cos^2\theta}{g} = \frac{1}{B}$

$R = \frac{A}{B}$

31. (a) $u = 20\ ms^{-1}; \theta = 30° \Rightarrow 2 \times 30° \Rightarrow \theta = 60°$

$R = \frac{u^2\sin 2\theta}{g}$

$= \frac{(20)^2\sin 60}{10}$

$$= \frac{20\emptyset\emptyset \times \frac{\sqrt{3}}{20}}{1\emptyset}$$

$R = 20\sqrt{3}$ m

(b) $u = 30\ ms^{-1}, \theta = 60°$

$$R = \frac{u^2 \sin\ 2\theta}{g}$$

$$R = \frac{(30)^2 2\sin\ \theta\cos\ \theta}{g}$$

$$R = \frac{90\phi \times 2 \times \frac{\sqrt{3}}{} \times \frac{1}{}}{1\emptyset} = 45\sqrt{3}$$

$R = 45\sqrt{3}$ m

(c) $u = 30\ ms^{-1}, \theta = 45°$

$$R = \frac{(30)^2 \sin\ 90}{10}$$

$$R = \frac{90\emptyset \times 1}{1\emptyset}$$

$R = 90$ m

(d) $u = 20\ ms^{-1}, \theta = 45°$

$$R = \frac{(20)^2 \sin\ 90}{10}$$

$$R = \frac{40\emptyset}{1\emptyset}$$

$R = 40$ m

32. Given $u = 9.8\ ms^{-1}$

$\theta = 30°$

$g = 9.8\ ms^{-2}$

$$t_f = \frac{2u\sin\ \theta}{g}$$

$$t_f = \frac{2 \times 9.8 \times \sin\ 30°}{9.8}$$

$$t_f = 2 \times \frac{1}{}$$

$t_f = 1sec$

# Worksheet – 3

## Level – I

**CUQ's:**

| | | | | |
|---|---|---|---|---|
| 1. D | 4. D | 7. C | 10. C | 13. D |
| 2. C | 5. B | 8. A | 11. C | 14. C |
| 3. C | 6. B | 9. C | 12. A | 15. C |

## Level – II

| | | | | |
|---|---|---|---|---|
| 1. 3 | 6. 3 | 11. 4 | 16. 1 | 19. 3 |
| 2. 4 | 7. 2 | 12. 2 | 15)2 | |
| 3. 4 | 8. 1 | 13. 4 | 16. 3 | |
| 4. 3 | 9. 2 | 14. 2 | 17. 3 | |
| 5. 3 | 10. 3 | 15. 4 | 18. 4 | |

**JEE MAINS_SOLUTIONS**

1. $v_y = gt \Rightarrow 10 \times 4 \Rightarrow u_x = 30$

$v = \sqrt{vx^2 + vy^2}$

$v = \sqrt{(30)^2 + (40)^2} \Rightarrow v = \sqrt{900 + 1600}$

$\sqrt{2500} = 50$ m/s

2. $S_x = u_x t = 20m$

$S_y = \frac{1}{2} gt^2 = \frac{1}{2} \times 10 \times 4 = 20$

$S = \sqrt{20^2 + 20^2} = 20\sqrt{2}$

3. $R_1 = u_x \sqrt{\frac{2h}{g}} \Rightarrow h = \frac{R^2 g}{2u^2} = \frac{(75)^2 \times 10}{2 \times (12.5)^2} = 180$ m

4. $R = u\sqrt{\frac{2h}{g}} = 10\sqrt{\frac{2 \times 19.6}{9.8}} = 20$ m

5. $\theta = \tan^{-1}(gt/4) = \tan^{-1}\left(\frac{9.8 \times 2}{19.6}\right) = 45°$

6. $v_y = gt = 2v_x \Rightarrow t = \frac{2v_x}{g} = 2\text{sec}$

7. $R = u\sqrt{\frac{2h}{g}} \Rightarrow R = (u_2 - u_1)\sqrt{\frac{2h}{g}} = 20\sqrt{\frac{2 \times 78.4}{9.8}} = 20 \times 4 = 80$ m

8. $\Rightarrow R = u \times t = u\sqrt{\frac{2h}{g}} = g = 9.8$

9. $R = u\sqrt{\frac{2h}{g}} \Rightarrow (u_1 + u_2)$

10. $\text{Tan } \theta = \frac{v_y}{u_x} \Rightarrow \theta = \tan^{-1}(gt/u) = 45°$

11. $R = u\sqrt{\frac{2h}{g}}$

12. $\tan \theta = \frac{V_y}{V_x}$

$= \frac{\frac{1}{2} gt^2}{u}$

$\tan 30° = \frac{\frac{1}{2} gt^2}{u}$

$u = \frac{\frac{1}{2} gt^2}{\frac{1}{\sqrt{3}}} = \frac{\frac{1}{2} \times 10 \times (2)^2}{\frac{1}{\sqrt{3}}} = 2g\sqrt{3}$

13. $R = u\sqrt{\frac{2(h_1 - h_2)}{g}}$

$\Rightarrow u^2 = \frac{R^2 \times g}{2(h_1 - h_2)} \Rightarrow g = 9.8$ m/s$^2$

$\Rightarrow u^2 = 8^2 \Rightarrow u = 8$ m/s

14. $u = R\sqrt{\frac{g}{2h}}$

15. $t = \sqrt{\frac{2h}{g}}, R = u \times t$

16. $v = \sqrt{2gh}, v = \sqrt{vx^2 + vy^2}$

17. $v = gt$

18. $vx = u\cos\theta$

$= 33.947$

$vy = u\sin\ \theta$

$= 19.6$

$= ugt^2 - 19.6t156.8 = 0$

$t^2 - ut - 32 = 0$

$t(t-8) + 4(t-8) = 0$

$(t-4)(t-8) = 0(\text{ or })t = -4(\text{ or })8.$

$BD = u\cos\ \theta \times t = 271.57$

19. $u^2\cos^2\ \theta = \frac{2}{5} \times \frac{u^2\sin^2\ \theta}{2} + u^2\sin^2\ \theta$

$$u^2\cos^2\ \theta - \frac{2}{5}u^2\cos^2\ \theta = \frac{u^2\sin^2\ \theta}{5}$$

$$\frac{3\cos^2\ \theta}{5} = \frac{\sin^2\ \theta}{5}$$

$$\Rightarrow \frac{\sin^2\ \theta}{\cos^2\ \theta} = \frac{1}{3}$$

$$\Rightarrow \tan^2\ \theta = \frac{1}{3}$$

$$\Rightarrow \tan\ \theta = \frac{1}{\sqrt{3}}$$

$$\theta = 60°$$

# Level – III

20) 1 21) 4 22) 3 23) 3 24) a-r,b-q,c-p,d-s,t 25) 1,3,4 26) 1
27) 2 28) a-q,b-r,c-p,d-s 29) 3 30) 1 31) a-s,b-s,c-r,d-p

---

21. $h = u_y t + \frac{1}{2}gt^2 \Rightarrow (0)t + \frac{1}{2}gt^2 \Rightarrow \times \frac{1}{2} \times 10 \times (0.4)^2 \Rightarrow 5 \times 0.16 \Rightarrow 0.8$ m

22. Horizontal distance=horizontal velocity × time.

$$u_x \times t = 4 \times 0.4 \Rightarrow 1.6 \text{ m}$$

23. $v_y = \sqrt{2gh} \Rightarrow \sqrt{2 \times 10 \times 0.8} \Rightarrow 4$ m/s $\Rightarrow v_x = 4$ m/s

$$V = \sqrt{v_x^2 + v_y^2} \Rightarrow \sqrt{4^2 + 4^2} \Rightarrow \sqrt{32} \Rightarrow 4\sqrt{2} \text{ m/s}$$

25. $t = \sqrt{\frac{2h}{g}}$

26. $v_x = -10 \times 4 = -40$

$$v_y = 10i\tan\ \theta = \frac{v_y}{v_x}$$

28. $a - q; b - r; c - p; d - s$

29. $\tan\ \theta = \frac{v_y}{v_x} = 1$

$$v_y = u$$

$$v_y = 0 + 9.8 \times 2 = 19.6 \text{ m/s}$$

$$y = u_y t + \frac{1}{2}ayt^2 = 0 + \frac{1}{2} \times 9.8 \times 4 = 19.6 \text{ m}$$

31. a-s, b-s, c-r, d-p

# Worksheet – 4

## Level – I

| | | | | |
|---|---|---|---|---|
| 1. D | 4. D | 7. A | 10. B | 13. A |
| 2. D | 5. A | 8. B | 11. A | |
| 3. A | 6. A | 9. A | 12. C | |

## Level – II

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1) B | 2) A | 3) B | 4) C | 5) A | 6) A | 7) B | 8) A |
| 9) B | 10) B | 11) B | 12) A | 13) B | 14) A | 15) A | 16) A |
| 17) A | 18) B | 19) B | 20) B | 21) C | 22) A | 23) D | 24) A |

1. elative velocity of Romu w.r.t Srinu $= 3 - (-2) = 5$ m/s.
2. Relative velocity of somu w.r.t Ramu $= 4 - 3 = 1$ m/s.
3. $V_{AB} = V_A - V_B = 4 - 3 = 1$ m/s
4. $V_{BA} = V_B - V_A = 3 - 4 = -1$ m/s

$$N_{BA}| = |-1| = 1 \text{ m/s}$$

11. $a_A = 4$ m/s$^2$

$$a_B = 10 \text{ m/s}^2$$
$$a_B = a_A - a_B = 4 - 10$$
$$|a_{AB}| = |-6 \text{ m/s}^2|$$
$$a_{AB} = 6 \text{ m/s}^2$$

15. $\bar{a}_{AB} = \bar{a}_A - \bar{a}_B = 3\bar{\imath} + 7\bar{\imath} = 10\bar{\imath}$

$$|\bar{a}_{AB}| = 10 \text{ m/s}^2$$

17. $\bar{a}_{BA} = \bar{a}_B - \bar{a}_A$

$$\bar{a}_{AB} = -7\bar{\imath} - 3\mathrm{i} = -10\bar{\imath}$$
$$|\bar{a}_{BA}| = |-10\bar{\imath}|$$
$$a_{BA} = 10 \text{ m/s}^2$$

18. $\bar{a}_{BA} = -10\mathrm{i}$
-ve $x$-axis, west direction.
19. Total distance = 100 m
$V_{AB} = 25$ m/s
$\text{time} = \frac{\text{distance}}{V_{AB}}$
$t = \frac{100}{25}$
$t = 4$sec
20. $\bar{V}_{\text{Ramu,srinu}} = V_{\text{Ramu}} - V_{\text{srinu}}$
$= 5\bar{\imath} + 5\bar{\imath}$
$\bar{V}_{R,S} = 10\bar{\imath}$
$V_{RS}| = 10$ m/s
21. $|\bar{a}_{AB}| = 2 - (-7)$
$\bar{a}_{AB} = 9$ ms/ $^2$

## Level – III

22) A 23) D 24) A 25) B 26) a-r,b-s,c-q,d-p 27) B 28) B
29) B 30) a-q,b-r,c-r,d-r

---

22. $\overset{V_A=10\ m/s}{\longrightarrow}$ $\longleftarrow V_B = 12\ m/s$

$V_{AB} = V_A - V_B = 10 + 12$

$V_{AB} = 22\ m/s$

$V_{AB} = V_B - V_A = -12 - 10$

$V_{BA} = -22 m/s$

23. 4: $V_{boy} = 5\ m/s, V_{man} = 10\ m/s$

$$V_{man\ ,\ boy} = \sqrt{V_{boy}^2 + V^2\ man}$$

$$= \sqrt{5^2 + 10^2}$$

$$= \sqrt{25 + 100} = \sqrt{125}$$

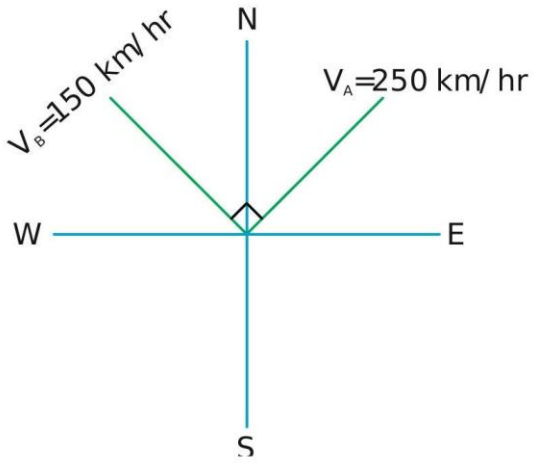
$$= 5\sqrt{5}\ m/s$$

24. $\bar{V}_A = 8\bar{j}$kmph, $\bar{V}_B = -6\bar{\imath}$kmph

$$\bar{V}_{AB} = \sqrt{V_A^2 + V_B^2 - 2V_AV_B\cos\ \theta}\ (\because \theta = 90°)$$

$$\bar{V}_{AB} = \sqrt{64 + 36} = \sqrt{100} = 10\text{kmph}$$

25.




$$V_{AB} = \sqrt{V_A^2 + V_B^2 + 2\ V_A + V_B\cos\ \theta}$$

$$V_{AB} = \sqrt{(250)^2 + (150)^2}\ (\because \theta = 90)$$

$$V_{AB} = \sqrt{85000}$$

$$V_{AB} = 291.54\ km/hr$$

26. a-r; b-s; c-q; d-p.

$$\bar{V}_A = 4\bar{j}m/s$$

$$\bar{V}_B = -9\bar{j}m/s$$

$$\bar{V}_{AB} = \bar{V}_A - \bar{V}_B = 4\bar{\jmath} + 9\bar{\jmath}$$

$|\bar{V}_{AB}| = |13\bar{\jmath}|$

$V_{AB} = 13\ m/s$

28. $\bar{V}_{BA} = \bar{V}_B - \bar{V}_A$

$$= -9\bar{\jmath} - 4\bar{\jmath}$$

$\bar{V}_{BA}| = | - 13\bar{\jmath}\ |$ (along south)

$V_{BA} = 13\ m/s$

29. $\bar{V}_{BA} = -13\bar{j}$
Along south direction.

# Worksheet – 5

## Level – I

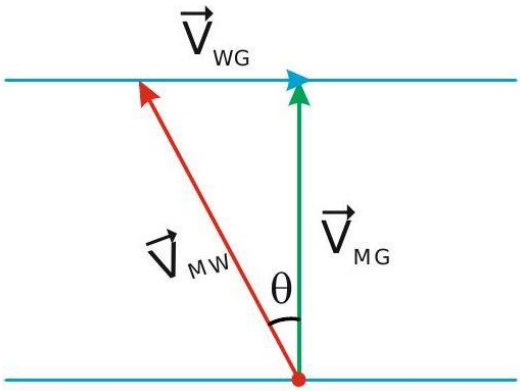
1) B 2) A 3) C 4) C 5) B 6) A 7) C
8) D 9) D

## Level – II

1) D 2) A 3) B 4) C 5) C 6) B 7) A 8) B
9) A 10) B 11) C 12) D 13) B 14) D 15) A 16) D
17) B 18) D 19) A,B,C 20) D 21) D 22) A
23) a- q; b- p; c- r; d-s 24) 25 25) a- q; b- p; c-r; d-s 26) A 27) A 28)A
29) A 30)A 31)B 32) A

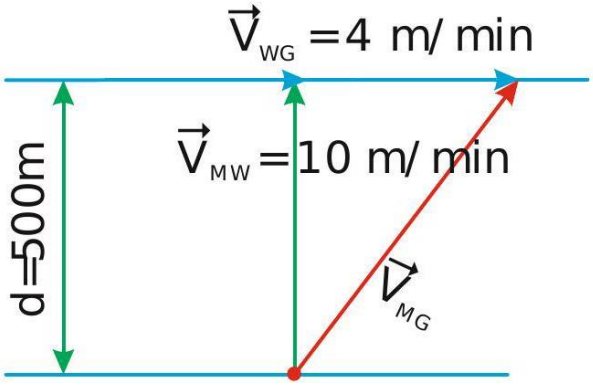
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1. North




$\vec{V}_{WG} = 5$ m/min
$\vec{V}_{MW} = 10$ m/min
from fig $\sin\theta = \frac{V_{WG}}{V_{MW}}$
$\sin\theta = \frac{5}{10}$ $\sin\theta = \frac{1}{2}$
$\sin\theta = \sin 30°$
$\theta = 30°$W of N

3.




Form the figure

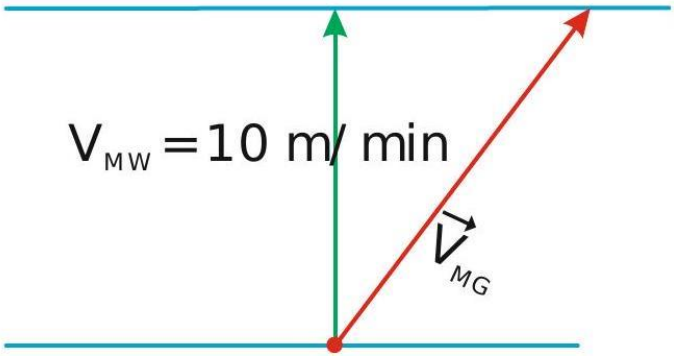
$\text{time} = \frac{\text{width of theriver}}{V_{MW}}$

$t = \frac{500\ m}{10\ m/min}$

$t = 50\ min$

$V_{WG} = 5\text{kmph}$

4.




Form figure

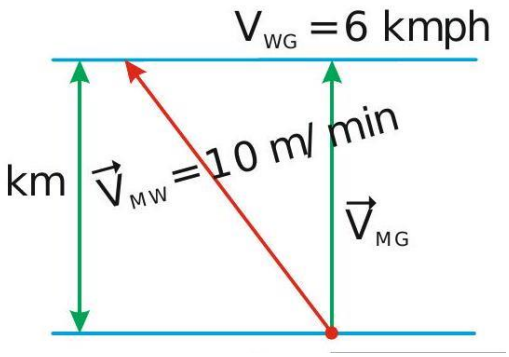
$V_{MG} = \sqrt{V_{MW}^2 + V_{WG}^2}$

$= \sqrt{10^2 + 5^2}$

$= \sqrt{125}$

$V_{MG} = 5\sqrt{5}\text{kmph}$

5.




From figure $\overrightarrow{V_{MG}} = \sqrt{V_{MW}^2 - V_{WG}^2}$

$= \sqrt{10^2 - 6^2}$

$= \sqrt{100 - 36}$

$= \sqrt{64}$

$\overrightarrow{V_{MG}} = 8\text{kmph}$

∴ time taken to cross the river width

1 km is

$t = \frac{\text{width}}{V_{MG}}$

$t = \frac{1}{8}\text{hr}$

6. Given time $t_1 = 12\text{hr}$

$$t_2 = 8\text{hr}$$

Distance between two points in river is same.

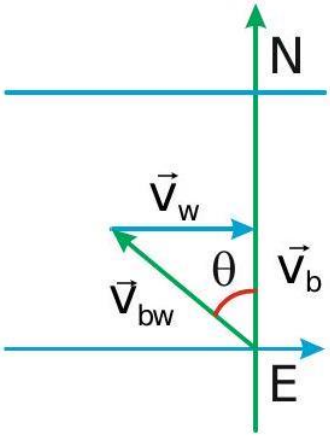
$\therefore t = \dfrac{2t_1t_2}{t_1 + t_2}$

$t = \dfrac{2 \times 12 \times 8}{12 + 8}$

$t = \dfrac{48}{5}$

$t = 9.6\text{hr}$

7. From figure




$$V_b = \sqrt{V_{bw}^2 - V_w^2} = \sqrt{13^2 - 5^2} = 12\text{kmph}$$

$$\tan\theta = \frac{V_w}{V_b} = \frac{5}{12}$$

$$\theta = \tan^{-1}\left(\frac{5}{12}\right) \text{W of N}$$

8. The boat travels 16 km in 2 hours in still water

$\therefore$ Velocity of boat, $V_b = \frac{16\text{ km}}{2\text{hr}}$

$= 8$ km/hr

velocity of river flow, $v_r = 4$ km/hr velocity of boat in down stream

$= V_b + V_r = 8 + 4 = 12$ km/hr

Time taken by the boat to cover a distance 8 km in downstream is to

$$\frac{8\text{ km}}{\frac{12\text{ km}}{\text{hr}}} = \frac{2}{3}\text{hr}$$

velocity of boat in up-

stream $= V_b - V_r = 8 - 4 = 4$ km/hr

Time taken by the boat to cover a distance 8 km in upstream is

$T_u = \dfrac{8\text{ km}}{4} = 2\text{hr}$

$\therefore$ Total time of travel

$= t_D + t_U = \left(\dfrac{2}{3} + 2\right)\text{hr} = 2\text{hr}40\text{ min}$

9. $t_1 = 10 = \frac{d}{v}$

$\Rightarrow d = 10v$

$x = ut_1$

$\Rightarrow 120 = u \times 10$

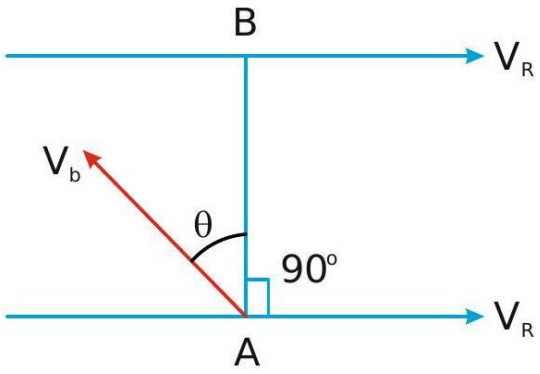
$\Rightarrow 12 mmin^{-1} = u$

$t_2 = 12.5 = \frac{d}{\sqrt{v^2 - u^2}}$

$\Rightarrow 12.5 = \frac{10v}{\sqrt{v^2 - 12^2}}$

$v = 20 \text{ m min}^{-1}$

10.




$\sin\theta = \frac{\vec{V}_{NG}}{\vec{V}_{MW}}$

$\sin\theta = \frac{4}{8}$

$\sin\theta = \frac{1}{2}$

$\theta = 30°$

∴ He should makes an angle

$90 + 30 = 120°$

$\Rightarrow 120°$ to flow of water

11. For down stream $t_1 = \frac{d}{u+v}$

For up stream $t_2 = \frac{d}{u-v}$

from (1) and (2)

$\frac{t_1}{t_2} = \frac{u-v}{u+v} = \frac{5-3}{5+3} = \frac{2}{8} = \frac{1}{4}$

$\therefore t_1 : t_2 = 1:4$

12. Time taken to cross river $= \frac{\text{width}}{V_{BW}}$

$t = \frac{400 \times 10^{-3} \text{ km}}{8 \text{kmph}}$ $t = 50 \times 10^{-3} \text{hr}$

Drift (x) $= V_{WG} \times t$

$= 2 \text{kmph} \times 50 \times 10^{-3} \text{hr}$

$= 100 \times 10^{-3} \text{ km} = 100 \times 10^{-3} \times 1000 \text{ m}$

Drift $(x) = 100$ m

13. $V_{MG} = \sqrt{V_{MW}^2 - V_{WG}^2}$

$V_{MG} = \sqrt{6^2 - 3^2}$

$V_{MG} = \sqrt{36 - 9}$

$V_{MG} = \sqrt{27}$kmph

∴ The time taken to cross the river

$t = \frac{d}{V_{MG}} = \frac{2}{\sqrt{27}}$hr

14. Given d = 0.5 km

d = 0.5 × 1000 m

d = 500 m

$\vec{V}_{MW}$ = 7.2kmph = 2 m/s

time taken to cross the river

$t = \frac{d}{V_{MW}} = \frac{500}{2} = 250$sec

t = 250sec

And also shift = 150 m

∴ Shift = $V_{MG}$ × time

150 = $V_{MG}$ × 250

$V_{MG} = \frac{150}{250}$

$V_{MG}$ = 0.6 m/s

15. Time taken to cross the river is

$t = \frac{d}{V_{MW}}$

$t = \frac{180}{1.65}$

t = 109.09sec

shift = $V_{MG} \times t$

shift = 0.85 × 109.09

shift = 92.7 m

16. $t = \frac{d}{v_r + v_{br}}$

$t = \frac{30}{5 + 10}$

$t = 2hr$

17. Time taken in up stream

$t_1 = \frac{d}{u - v}$

Time taken in down stream

$t_2 = \frac{d}{u + v}$

From euation (1) and equation (2)

$\frac{t_1}{t_2} = \frac{u + v}{u - v}$

$\frac{4}{2} = \frac{u + v}{u - v}$

$4u - 4v = 2u + 2v$

$2u = 6v$

$\frac{u}{v} = \frac{6}{2}$

$u : v = 3 : 1$

18. $W = 2\sqrt{3}$ km

$x = 2$ km

$S = \sqrt{W^2 + x^2}$

$= \sqrt{(2\sqrt{3})^2 + (2)^2}$

$\therefore \theta = \sin^{-1}\left(\frac{v_t}{V_{br}}\right) = \sin^{-1}\left(\frac{2}{4}\right)$

$= \sin^{-1}\left(\frac{1}{2}\right) = 30^\circ$

Hence, to reach the point directly opposite to starting point he should the boat at an angle of $30^\circ$ with $AB$ or $90^\circ + 30^\circ = 120^\circ$ with the river

21. Time taken by the boatman to cross the river

w = width of river = 4 km

$V_{br} = 4$ km/hr and $\theta = 30^\circ$

$t = \frac{4}{4\cos\ 30^\circ} = \frac{2}{\sqrt{3}}$ hr

22. For shortest time $\theta = 0^\circ$ and

$t_{min} = \frac{W}{V_{br}\cos\ 0^0} = \frac{4}{4} = 1$hr

Hence, he should head his boat perpendicular to the river current for crossing the river in shortest time and this shortest time is 1hr.

23. $a - q; b - p; c - r; d - s$

24. 25sec

25. $a - q; b - p; c - r; d - s$

26. Velocity of river $\vec{V}_r = 2$ m/s velocity of boat with respect to river $v_{br} = 4$ m/s

$|\vec{V}_m| = \sqrt{\vec{V}_{br} + \vec{V}_{re}} = \sqrt{16 + 4} = \sqrt{20}$ m/s

$\vec{V}_{be} = 45$ m/s to find direction we have

$\tan\ \theta = \frac{V_r}{V_{br}} = \frac{2}{4} = \frac{1}{2}$

$\Rightarrow \theta = \tan^{-1}\left(\frac{1}{2}\right)$

27. $t = \frac{800}{4} = 200$sec

28. Shift $= 200 \times 2 = 400$ m Hence the boat is drifted by 400 m away from position $A$.

case (i)

$t_1 = 10$ min $= 10 \times 60 = 600$ s

drift $= 120 = V_r \times 600$

$V_r = \frac{1}{5}$ m/s $= 0.2$ m/s $= 0.72$kmph

$V_{br} = \frac{d}{600}$

$d = V_{br}600$

30. case (ii)

$t_2 = 12.5 \text{ min} = 12.5 \times 60 = 750 \text{ s}$

$t_2 = \frac{d}{V_{br}\cos\alpha}$

$V_{br}\cos\alpha = \frac{d}{750}$

$V_r = V_{br}\sin\alpha$

$\sin\alpha = \frac{V_r}{V_{br}} = \frac{1}{5\ V_{br}}$

$V_{br}\cos\alpha = \frac{d}{750}$

$V_{br}\sqrt{1-\sin^2\alpha} = \frac{d}{750}$

Substituting (1) & (2) in equation (3), We get

$V_{br} = \frac{1}{3}$ m/s

$V_{br} = 20$ m/min

31. Width of the river; from equation (1)

$$d = 600 \times \frac{1}{3} = 200 \text{ m}$$

32. $\sin\alpha = \frac{V_r}{V_{br}} = \frac{(1/5)}{(1/3)} = \frac{3}{5}$

$$\alpha = \sin^{-1}\left(\frac{3}{5}\right)$$

# Worksheet – 6

## Level – I

| 1.A | 2.D | 3.B | 4.A | 5.D |
|---|---|---|---|---|

1. $\overrightarrow{V_{rm}} = \overrightarrow{V_r} - \overrightarrow{V_M}$
2. $\vec{V_r} = a\hat{\imath} + b\hat{\jmath}$

$\overrightarrow{V_m} = V_m\hat{\imath}$

$\overrightarrow{V_{rm}} = \overrightarrow{V_r} - \overrightarrow{V_M}$

$= a\hat{\imath} + b\hat{\jmath} - V_m\hat{\imath}$

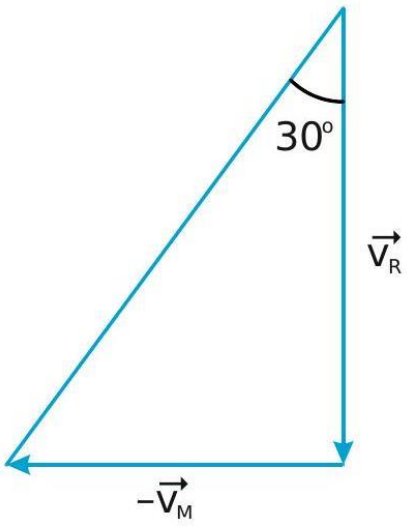
$\overrightarrow{V_{rm}} = (a - V_m)\hat{\imath} + b\hat{\jmath}$

3. Conceptual

## Level – II

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1) A | 2) B | 3) B | 4) A | 5) A | 6) B | 7) A | 8) A |
| 9) C | 10) C | 11) A | 12) C | 13)C | 14) A | 15)D | 16)B |
| 17)B | 18)B | 19) B | 20)D | 21) a-s; b-p; c-q,r; d-r | | 22)B | |
| 24) 4 m/s | 25) B | 26) B | 27) B | 28)B | 29) A,B | | |

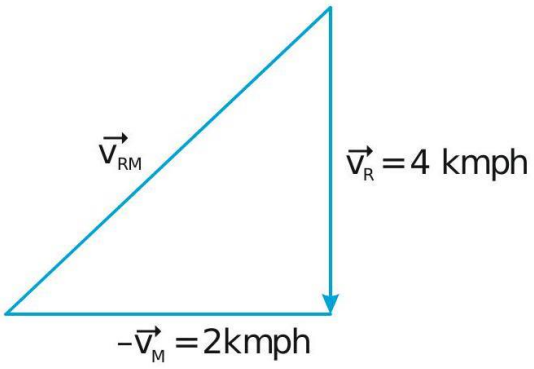
1.




$\frac{\vec{V}_R}{\vec{V}_m} = \cot 30°$

$\frac{\vec{V}_R}{\vec{V}_B} = \sqrt{3}:1$

2.



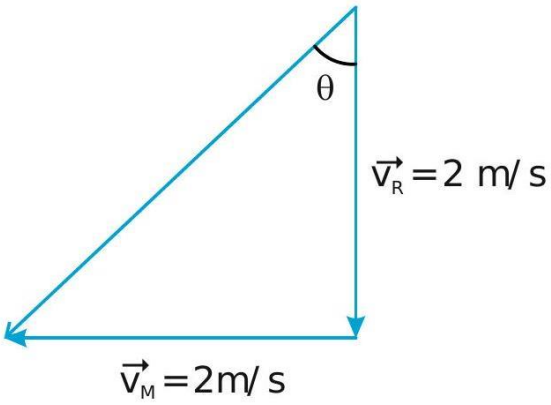

$\vec{V}_{Rm} = \vec{V}_R - \vec{V}_m$

$|V_{Rm}| = \sqrt{V_R^2 + V_m^2}$

$V_{Rm} = \sqrt{4^2 + 2^2} = \sqrt{20}$

$= 2\sqrt{5}$kmph

3.



$\text{Tan } \theta = \frac{V_m}{V_R}$

$\text{Tan } \theta = \frac{2}{2}$

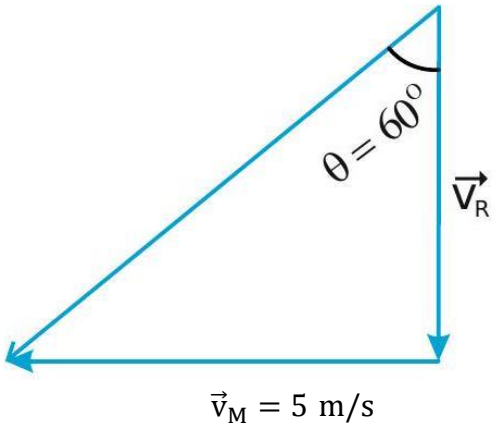
$\text{Tan } \theta = 1$

$\theta = 45°$

4. $\vec{V}_m = 18\text{kmph} = 5 \text{ m/s}$

$\theta = 60°$ to the vertical

$\vec{V}_R = ?$

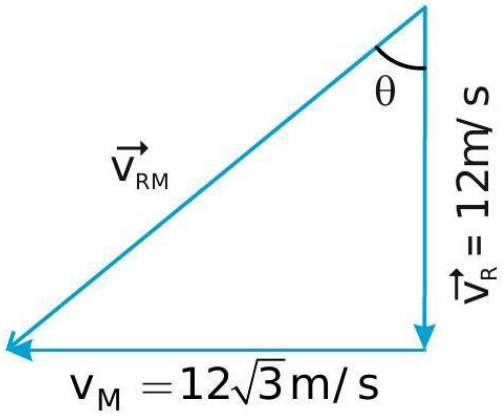



from fig $\text{Tan } 60° = \frac{V_m}{V_R}$

$\sqrt{3} = \frac{5}{V_R}$

$V_R = \frac{5}{\sqrt{3}}$ m/s

5.



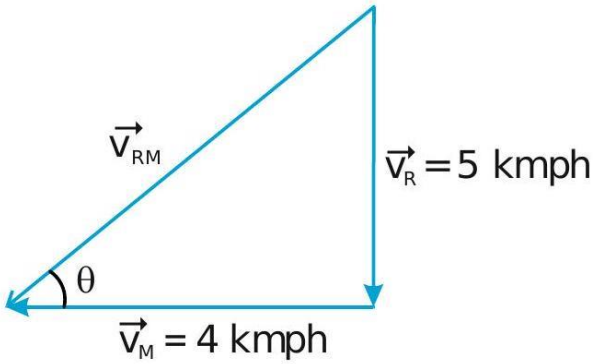

$\text{Tan } \theta = \frac{V_m}{V_R}$

$\text{Tan } \theta = \frac{12\sqrt{3}}{12}$

$\text{Tan } \theta = \sqrt{3}$

$\text{Tan } \theta = \text{Tan } 60° \ \theta = 60°$

6.

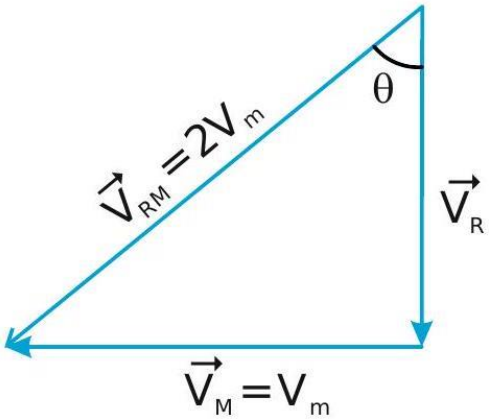



from fig Tan $\theta = \frac{V_R}{V_m}$

$= \frac{5}{4}$

$\theta = \text{Tan}^{-1}\left(\frac{5}{4}\right)$ with horizontal

7.




Given $\vec{V}_{Rm} = 2\vec{V}_m$

from fig

$\sin\theta = \frac{V_m}{V_R}$

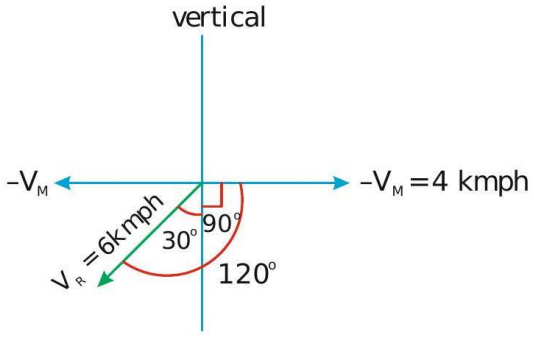
$\sin\theta = \frac{V_m}{2V_m}$ $(\therefore V_{Rm} = 2V_m)$

$\sin\theta = \frac{1}{2}$

$\theta = 30°$

vertical

8.



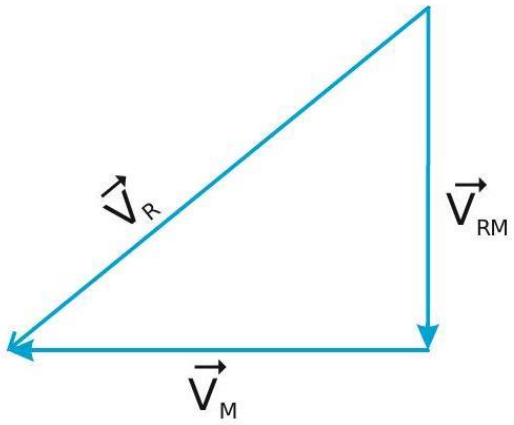
$\vec{V}_{Rm} = \sqrt{V_R^2 + V_m^2 - 2V_R V_m \cos\ \theta}$

$= \sqrt{6^2 + 4^2 - 2 \times 6 \times 4 \times \cos\ 120}$

$\sqrt{36 + 16 - 48 - \left(\frac{1}{2}\right)} = \sqrt{76}$

$V_{Rm} = 8.718$kmph

9.




From fig $|V_r| = \sqrt{V_{rm}{}^2 + V_r{}^2}$

$= \sqrt{5^2 + 5^2} = \sqrt{50}$

$= \sqrt{25 \times 2}$

$V_r = 5\sqrt{2}$kmph

10. $\vec{V}_R = 4$kmph

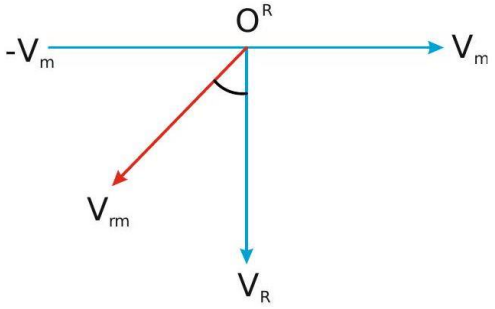
$\vec{V}m = 3$kmph

$\vec{V}_{RM} = \vec{V}_R - \vec{V}_M$

$|\vec{V}_{RM}| = \sqrt{V_R^2 + V_M^2} = \sqrt{4^2 + 3^2}$

$= \sqrt{25} = 5$kmph

11. Given $\vec{V}_{RM} = 1.5vm$




$\sin\ \theta = \frac{v_m}{v_{rm}}$

$\sin\ \theta = \frac{v_m}{1.5v_m}$

$\sin\ \theta = \frac{10}{15}$

$\sin\theta = \frac{10}{15}$

$\sin\theta = \frac{2}{3}$

from figure $\tan\theta = \frac{2}{\sqrt{5}}$

12. The actual direction of the rain is 30° to the vertical.

$\tan\theta = \frac{V_M}{V_{RM}}$

$\tan 30° = \frac{2}{V_{RM}}$

$V_{RM} = 2\sqrt{3}$kmph

$V_R^2 = V_{RM}^2 + V_M^2$

$= (2\sqrt{3})^2 + (2)^2 = 4$kmph

13. Given $\theta = 60°$ velocity of person

$\vec{V}_P = \overrightarrow{OA} = 20$ m/s

This velocity is same as the velocity of person w.r.t ground

$\vec{V}_P = \overrightarrow{OB}$ = velocity of rain w.r.t person Speed of the rain w.r.t road is from

$\triangle OCB, \frac{CB}{OC} = \sin 60°$

$OC = \frac{CB}{\sin 60°} = \frac{20}{\sqrt{3/2}} = \frac{40\sqrt{3}}{3}$ m/s

14. Velocity of rain = 20j

Velocity of man = 5i

Velocity of wind = $15i$

$V_{RM}$ when wind is blowing is

$V_{RM} = 15i - 5i + 20j$

$V_{RM} = 10i + 20j$

∴ Direction $\tan\theta = \frac{20}{10}$

$\tan\theta = 2$ from horizontal

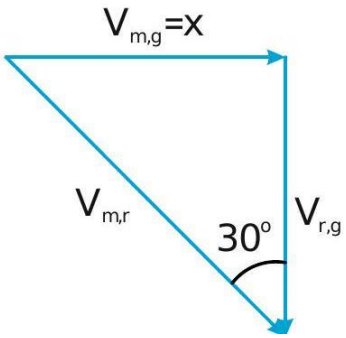
Hence the person should hold umbrella at $\alpha$ (with vertical)

$\tan\theta = \frac{10}{20}$

$\tan\theta = \frac{1}{2}$

$\theta = \tan^{-1}\left(\frac{1}{2}\right)$

15. case-i:

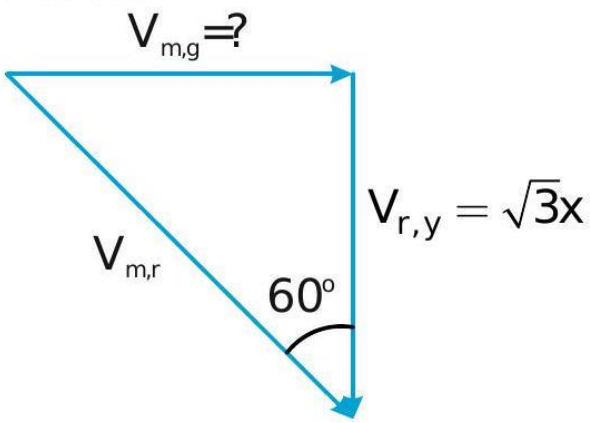



$\tan 30° = \frac{v_m}{v_r}$

$\frac{1}{\sqrt{3}} = \frac{v_m}{v_r}$

$v_m \times \sqrt{3} = v_m$

$\therefore v_r = \sqrt{3} \times v_m$ case-ii:




$\tan 60 = \frac{v_m^1}{v_r}$

$v_m^1 = \tan 60 \times V_r$

$v_m^1 = \sqrt{3} \times \sqrt{3} \times v_m$ [$\therefore$ from (1)]

$v_m^1 = 3v_m$

16. $\tan 45° = \frac{V_{m_2} - V_{m_1}}{V_{RMI}} = \frac{3}{V_{RMI}}$

$V_{RMI} = 3$kmph

$V_R = \sqrt{V_{RMI}^2 + V_{MI}^2}$

$= \sqrt{9+9} = 3\sqrt{2}$kmph

17. $\tan \alpha = \frac{V_2}{V_1}$

$\alpha = \tan^{-1}\left(\frac{V_2}{V_1}\right)$

18. $\cos 53 = \frac{3}{5}$ $\therefore \text{Tan } 53 = \frac{4}{3}$

$\text{Tan } 53 = \frac{V_R}{V_M}$

$\therefore V_R - V_M \text{Tan } 53 = 3 \times \frac{4}{3} = 4$ m/sec

19. $\overrightarrow{v_r} = 5\sqrt{2}$ m/s

$\overrightarrow{v_m} = 5\sqrt{2}$ m/s

$\vec{V}_{rm} = \vec{V}_r - \vec{V}_m$

$|\vec{V}_{rm}| = \sqrt{V_r^2 - V_m^2}$

$= \sqrt{(5\sqrt{2})^2 + (5\sqrt{2})^2} = \sqrt{100}$

$\vec{V}_{rm} = 10$ m/s

ii) $\tan\theta = \frac{v_m}{v_R}$

$\tan\theta = \frac{5\sqrt{2}}{5\sqrt{2}}$

$\tan\theta = 1$

$\theta = 45°$

20. $\frac{v_r}{v_m} = \cot 60°$

$\frac{v_r}{v_m} = \frac{1}{\sqrt{3}}$

$v_r : v_m = 1:\sqrt{3}$

21. $\vec{V}_r = 2$kmph

$\vec{V}_m = 2$kmph

$V_{rm} = \sqrt{V_r^2 + V_m^2}$

$= \sqrt{2^2 + 2^2} = \sqrt{4+4} = \sqrt{8}$

$v_{rm} = 2\sqrt{2}$kmph

22. $V_{RM}^2 = V_R^2 + V_M^2$

$= (2)^2 + (2)^2 = 8$kmph

$V_{RM} = 2\sqrt{2}$kmph

23. a-s; b-p ; c- q,r ; d-r

24. $V_M = 10.8\text{kmph} = 10.8 \times \frac{5}{18} = 3$ m/s $\tan 53° = \frac{V_R}{V_M} \Rightarrow \frac{4}{3} = \frac{V_R}{3}$

$V_R = 4$ m/s

25.26.27

Velocity of rain with respect to woman = $\vec{V}_{rw}$

Velocity of rain with respect to earth = $\vec{V}_{RE}$

Velocity of woman with respect to earth = $\vec{V}_{WE}$

$V_{RW} = V_{RE} - V_{WE} = V_{RE} + (-V_{WE})$

$V_{RW} = \sqrt{V_{RE}^2 + (-V_{WE})^2}$

$= \sqrt{20^2 + 5^2} = 20.6$ m/s

$\tan\theta = \frac{|v_{we}|}{|v_{re}|} = \frac{5}{20} = \frac{1}{4} = \theta = 14°$ from

figured = $1.5\tan\theta$

$d = 1.5 \times \frac{1}{4} = 0.375$ m $= 37.5$ cm

28. The situation is shown in figure. Let $A$ and $B$ be the position of two ships respec tively. The ship $A$ at 9 A.M. is 30 km southwest of $B$, i.e., $AB = 30$ km. $AB$ makes an

angel 45° from east. $A$ travels in the direction $AC$ with velocity 10 km/hr. Let $B$ travels in the direction $BC$ and interception takes place at point $C$ after a time $t$. Then AC = 10 T

and $BC = 15t$

(a) Let BC makes an angel q with BA

In triangle $ABC$, or $\frac{10t}{\sin\theta} = \frac{15t}{\sin 45^\circ}$

$\therefore \sin\theta = \frac{10t \times \sin 45^\circ}{15t}$ or

$\sin\theta = \frac{10}{15} \times \frac{1}{\sqrt{2}} = 0.4717$ or $\theta = 28^\circ$

Now, $\angle bca = 180^\circ - (45^\circ - 28^\circ) = 107^\circ$

(b) From triangle

$ABC, \frac{15t}{\sin 45^\circ} = \frac{30}{\sin 107^\circ}$

$\therefore 15t = \frac{30\sin 45^\circ}{\sin 107^\circ}$

$t = \frac{30}{15} \times \frac{\sin 45^\circ}{\sin 107^\circ} = 2 \times \frac{0.7070}{0.9563} = 1.48hr$

29. case i : The man is at rest $\therefore v_m = 0$,

$\therefore \vec{V}_m = 0\hat{i}$

$\therefore \vec{V}_{rm} = a\hat{i} + b\hat{j}$

The rain makes an angle 30° with the vertical

$\therefore \tan 30^\circ = \left|\frac{a}{b}\right| \Rightarrow |b| = a\sqrt{3}$

Case ii:

$\therefore Vm = 10\hat{i}; \vec{V}_{rm} = (a-10)\hat{i} + b\hat{j}$

Now rain drops hitting him vertically

$\therefore$ horizontal component of $\vec{V}_{rm}$ is zero

$\therefore a - 10 \Rightarrow a = 10$

$\therefore$ From (1)

$|b| = 10\sqrt{3} \Rightarrow b = -10\sqrt{3}$

(since b should be negative)

$$\therefore \vec{V}_{rg} = 10\hat{i} - 10\sqrt{3}\hat{j}$$

$$\therefore \left|\vec{V}_{rg}\right| = \sqrt{10^3 + (10\sqrt{3})^2} = 20kmph$$

$$\left|\vec{V}_{rm}\right| = |10\sqrt{3}\hat{i}| = 10\sqrt{3}kmph$$

# 4. NEWTON'S LAWS OF MOTION

## Worksheet – 1

### Level – I

| | | | |
|---|---|---|---|
| 1. D | 3. D | 5. A | 7. B |
| 2. B | 4. A | 6. C | 8. C |

### Level – II

| | | | | |
|---|---|---|---|---|
| 1. C | 6. C | 11. C | 16. B | 21. A |
| 2. B | 7. C | 12. C | 17. C | 22. D |
| 3. B | 8. D | 13. D | 18. A | |
| 4. A | 9. B | 14. C | 19. A | |
| 5. C | 10. D | 15. C | 20. A | |

23. inertia of rest
24. a-p, b-p, c-s, d-t 26) 2 27) 1
25. C
26. B
27. C
28. C

## Worksheet – 2

### Level – I

| | | | | |
|---|---|---|---|---|
| 1. B | 4. D | 7. B | 10. D | 13. A |
| 2. C | 5. D | 8. B | 11. B | 14. A |
| 3. A | 6. B | 9. B | 12. B | 15. A |

### Level – II

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1) B | 2) C | 3) D | 4) A | 5) B | 6) B | 7) A | 8) C |
| 9) B | 10) D | 11) C | 12) D | 13) C | 14) B | 15) B | 16) A |
| 17) A,B,C | 18) A,B | 19) A,D | 20) a-q, b-t, c-p,r d-s. | | 21) 8cm | 22) A | 23) B |
| 24) D | 25) A | 26) C | | | | | |

1. $m = 25$ kg, $p = 125$ kg m/sv = ?
   p = mv
   $v = \frac{p}{m} = \frac{125}{25} = 5$ m/s
   $v = 5$ m/s
2. $m = 75$ kg, $v = 2$ m/s, p = ?
   p = mv
   $= 75(2) = 150$kgm/s
   p = 150 kg m/s
3. $m = 200$ kg, $u = 20$ m/s, $v = 10$ m/s, t = 25sec
   $\Delta P$ = ?
   $\Delta P = m(v - u)$
   $= 200(10 - 20) = 200 \times (-10)$
   $= -2000$kgm/s
   $\Delta P = -2000$kgm/s
4. m = 150 g = $\frac{150}{1000} \Rightarrow 15 \times 10^{-2}$ kg

v = 20 m/s
t = 0.1sec
$F = \frac{mv}{t} = \frac{15 \times 10^{-2} \times 20}{0.1} = 30$ N

5. m = 0.4 kg, $u = 20$ m/s, $v = 0$ m/s,
t = 10sec, t = ?
$F = ma = \frac{m(v-u)}{t}$
$= \frac{0.4(0-20)}{10} = -0.8$ N

6. m = 0.4 kg, u = 0, t = 10sec, F = ?
After removal of force, the body moves with uniform velocity which is equal to
$\frac{100}{20} = 5$ m/s $= v$
$a = \frac{v-u}{t}$
$a = \frac{5-0}{10} = 0.5$ m/s$^2$
$F = ma = 0.4(0.5) = 0.2$ N

7. Impulse (or) change in momentum = 25NS
Impulse = F × t
$F = \frac{\text{Impulse}}{t} = \frac{25}{0.05} = \frac{25}{5 \times 10^{-2}} = 500$ N

8. Average force = Rate of change in momentum
$F = \frac{mv - (-mv)}{\Delta t} = \frac{2mv}{\Delta t}$
$= \frac{2 \times 0.25 \times 2}{10^{-3}} = 10^{-3}$ N

9. $m(v-u) = F_1 t_1 + F_2 t_2$
$2(v-0) = 4 \times 1 + 3 \times 2$
$v = 5$ m/s

10. Force = rate of change of momentum $F = v\frac{dm}{dt} = 20 \times \frac{5}{60} = \frac{5}{3}N$

11. $F = ma = m\frac{dv}{dt}$
$3x = m\frac{dv}{dt} \Rightarrow 3x = m\frac{dv}{dt} \times \frac{dx}{dt}$
$3x = m\frac{dv}{dt}.v \Rightarrow 3x = 8\frac{dv}{dx}.v \Rightarrow 3xdx = 8vdv$
$\int_2^{10} 3xdx = \int_0^v 8vdv$
$3\left(\frac{x^2}{2}\right)_2^{10} = 8\left(\frac{v^2}{2}\right)_0^v \Rightarrow 3(100-4) = 8v^2$
v = 6 m/s.

12. $F = ma$
We know that
$v^2 - u^2 = 2as \Rightarrow a = \frac{v^2-u^2}{2s}$
Then above equation becomes,
$F = m \times \frac{v^2-u^2}{2s}$
$F = \frac{mv^2}{2s} = v^2 = \frac{2Fs}{m} \Rightarrow v\alpha\frac{1}{\sqrt{m}}$

13. $m_1 = \frac{F}{a_1}$, $m_2 = \frac{F}{a_2}$

$$a = \frac{F}{(m_1 + m_2)} = \frac{F}{\left(\frac{F}{a_1} + \frac{F}{a_2}\right)}$$

$$= \frac{F}{F\left(\frac{a_1 + a_2}{a_1 a_2}\right)} = \frac{a_1 a_2}{a_1 + a_2}$$

$$\therefore a = \frac{a_1 a_2}{a_1 + a_2}$$

14) $\mathrm{m} = 0.03\ \mathrm{kg}, \frac{n}{t} = 200, v = 30\ \mathrm{m/s}$ f = ?

$$F = \frac{nmv}{t} = 200 \times 0.03 \times 30 = 180N$$

15. $m = 40\ \mathrm{g}, = \frac{40}{1000} = 4 \times 10^{-2}\ \mathrm{kg}$

$\mathrm{v} = 1200\ \mathrm{m/s}, \mathrm{F} = 144\ \mathrm{N}, \frac{n}{t} = ?$

$$F = \frac{nmv}{t} = \frac{n}{t} = \frac{F}{mv} = \frac{144}{4 \times 10^{-2} \times 1200} = 3$$

$$\therefore \frac{n}{t} = 3$$

16. $\mathrm{F} = \mathrm{ma}; \mathrm{F} = \frac{\mathrm{mv}}{\mathrm{t}}; \mathrm{v} = \frac{\mathrm{Ft}}{\mathrm{m}}$

$$v = \frac{(0.32\hat{\imath} - 0.4\hat{\jmath})6}{0.2}$$

$$v = \frac{(1.8\hat{\imath} - 2.4\hat{\jmath})}{0.2}$$

$$v = 9\hat{\imath} - 12\hat{\jmath}$$

19. $\mathrm{F} = \frac{\mathrm{mv}}{\mathrm{t}}$

$$= \frac{150 \times 10^{-3} \times 20}{0.1} = 30\ \mathrm{N}$$

$= \mathrm{F} \times \mathrm{t} = 30 \times 0.1 = 3\mathrm{Ns}$

21. $\mathrm{F} = \mathrm{ma}; \mathrm{F} = \frac{\mathrm{mv}}{\mathrm{t}}; \mathrm{v} = \frac{\mathrm{Ft}}{\mathrm{m}}$

$$25x = \frac{100 \times 10}{5}$$

$25x = 200 \Rightarrow x = 8$ cm

22. $\frac{\mathrm{m}}{\mathrm{t}} = 0.5\ \mathrm{kg/sec},$

$\mathrm{v} = 2\ \mathrm{m/s};\ \mathrm{F} = ?$

$$\mathrm{F} = \mathrm{ma}; \mathrm{F} = \frac{\mathrm{mv}}{\mathrm{t}}$$

$\mathrm{F} = 0.5 \times 2 = 1\ \mathrm{N}$

23. Change in momentum of one ball

$= 2\mathrm{mu}$, time = 1sec

$$\mathrm{F_{avg}} = \frac{\text{Total change in momentum}}{\text{timetaken}}$$

$$\mathrm{F_{avg}} = \frac{\mathrm{n(2mu)}}{1}$$

$\mathrm{F_{avg}} = 2\mathrm{nmu}$

24. Change in momentum $= m(v - u)$

$= mv - (-mv) = 2mv$

$= 2\ \mathrm{m}\sqrt{2gh}\ (\therefore v = \sqrt{2gh})$

25. $\mathrm{F_1} = \mathrm{m_1 a_1} = 4 \times 5 = 20\ \mathrm{N}$

$\mathrm{F_2} = \mathrm{m_2 a_2} = 10 \times 4 = 40\ \mathrm{N}$

26. Force = Fkgwt = gFN

mass = mkg

time = tsec

$\mathrm{u} = 0; \mathrm{s} = \mathrm{xm}$

Final velocity = Vm/s

Force = ma

$gF = ma \Rightarrow a = \frac{gF}{m}$

$v^2 - u^2 = 2as$

$v^2 = 2 \times \frac{gF}{m} \times x \Rightarrow x = \frac{v^2 m}{2gF}$

$x = \frac{mv}{2gF}$

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# Worksheet – 3

## Level – I

| | | | | |
|---|---|---|---|---|
| 1. B | 4. D | 7. A | 10. C | 13. C |
| 2. C | 5. C | 8. B | 11. C | 14. B |
| 3. C | 6. A | 9. C | 12. A | 15. B |

## Level – II

| | | | | | | |
|---|---|---|---|---|---|---|
| 1) 1 | 2) 2 | 3) 2 | 4) 3 | 5) 2 | 6) 3 | 7) 1 |
| 8) 4 | 9) 4 | 10) 3 | 11) 3 | 12) 4 | 13) 4 | 14) 3 |
| 15) 3 | 16) 4 | 17) 4 | 18) 4 | 19) 1 | 20) 1 | 21) 1 |
| 22) 2 | 23) 2 | 24) a-t, b-t, c-p, d-q | | 25) 3 | 26) 3 | 27) 2 |

1. $F = \frac{mv}{t}$

$= \frac{m}{t} = \frac{F}{v} \Rightarrow \frac{m}{t} = \frac{210}{300} = 0.7$ kg/s

2. $F = \frac{mv}{t}$

$F = \frac{0.05 \times 400}{1} = 20$ N

3. $a = \frac{\frac{1}{F}}{(m_1+m_2+m_3)}$

4. $T_1 = m_1 a$

$T_1 = m_1 \left(\frac{F}{m_1 + m_2 + m_3}\right)$

$T_1 = \frac{m_1 F}{m_1 + m_2 + m_3}$

5. $T_2 - T_1 = m_2 a$

$T_2 = m_2 a + T_1$

$T_2 = m_2 \left(\frac{F}{m_1 + m_2 + m_3}\right) + \frac{m_1 F}{m_1 + m_2 + m_3}$

$T_2 = \frac{(m_1 + m_2)F}{(m_1 + m_2 + m_3)}$

6. $V = -\left(\frac{m}{M}\right) v$

$V = -\left(\frac{20 \times 10^{-3}}{8}\right) \times 100 = -0.25$ m/s

7. $V = -\left(\frac{m}{M}\right) v$

$V = -\left(\frac{10 \times 10^{-3}}{1}\right) \times 5 = -0.05$ m/s

8. $mg\sin\theta = ma\cos\theta = F\cos\theta$

$\mathbf{F} = mg\tan\theta$

9. $a = \frac{F}{m_1+m_2} = \frac{3}{1+2} = 1\ \text{m/s}^2$

10. $\text{v} = \frac{35\times10^{-3}\times1\times400}{20}$

$= 700 \times 10^{-3} = 0.7\ \text{m/s}$

$\text{F} = \frac{\text{mv}}{\text{t}}; \text{F} = \frac{20 \times 0.7}{1} = 14\ \text{N}$

11. $V = \frac{20\times10^{-3}\times4\times300}{10} = 2.4\ \text{m/s}$

$\text{F} = \frac{\text{mv}}{\text{t}};\ \text{F} = \frac{10 \times 2.4}{1} = 24\ \text{N}$

12. $\text{F} = \text{Nmv}; \text{N} = \frac{\text{F}}{\text{mv}}$

$= \frac{144}{40 \times 10^{-3} \times 1200}$

$= \frac{144}{4 \times 12 \times 10^{-3} \times 10^{3}} = 3$

13.

$a = \frac{F}{m_1 + m_2 + m_3}$

$= \frac{140}{16 + 8 + 4} = \frac{140}{28}$

$\text{F} = 16 \times \frac{140}{28} = 80\ \text{N}$

14. $a = \frac{F}{M+m} \Rightarrow T = ma$

$= M \times \frac{F}{M + m} = \frac{MF}{M + m}$

15. $\text{Volume} = \text{A} \times \text{V} = 20 \times 10^{-4} \times 10$

$= 0.02\ \text{m}^3/\text{sec}$

mass = density × volume

$= 1000 \times 0.02 = 20\ \text{kg/sec}$

change in momentum $= \text{m}(\text{v} - (-\text{v})) = 2\text{mv}$

$= 2 \times 20 \times 10$

$= 400\text{kgm/s}^2$

change in momentum equal to force exerted by water

$\text{F} = 400\ \text{N}$

16. (M) ($u = 0$, for freely falling)

(M) -- After 10sec it attain final velocity (v)

$v = u + gt\ \ v = 10 \times 10 = 100\ \text{m/s}$

(M)

$0 + 0 = 2{:}1$ ratio of masses

M = 3x conservation of momentum

$= 3(x) \times 100 = 2x(200) + x(v')$

$v' = -100$

i.e 100 m/s upwards

17. $F = v\frac{dm}{dt}$

$\text{mg} = \text{v}\frac{\text{dm}}{\text{dt}}$

$\frac{\text{mg}}{\text{v}} = \frac{\text{dm}}{\text{dt}} = \frac{10{,}000 \times 10}{2000} = 50\ \text{kg/sec}$

18. $\text{m} = 20\ \text{g} = 20 \times 10^{-3}\ \text{kg}$

u = 100 m/s
$v = 0$
t = 0.05sec
a =? F =? S = ?
v = u + at

$$u = -at = a = \frac{-u}{t} = \frac{100}{0.05}$$

$= -2000\ m/s^2$
$F = ma = 0.02 \times (-2000) = -40\ N$
-ve sign indicate, that wooden block exet force in the direction, opposite to the directio of motion of the bullet.

25. $\frac{m}{t} = 2\ kg/sec, v = 4\ m/s, F = ?$
$F = ma;\ F = \frac{mv}{t} \Rightarrow F = 2 \times 4 = 8\ N$

26. $F = \frac{2mv}{t} = \frac{2\times50\times40}{0.2} = \frac{4000}{2\times10^{-1}}$
$= 2000 \times 10^1 = 20{,}000\ N$

27. $V = \frac{35\times10^{-3}\times1\times400}{20}\left(\therefore V = \frac{nmv}{M}\right)$
$V = 700 \times 10^{-3} = 0.7\ m/s$

$$F = \frac{mv}{t} = \frac{20 \times 0.7}{1} = 14\ N$$

F = 7X
7X = 14

$$X = \frac{14}{7} = 2$$

# Worksheet – 4

## Level – I

| | | | | |
|---|---|---|---|---|
| 1. B | 3. C | 5. C | 7. D | 9. C |
| 2. A | 4. A | 6. B | 8. A | 10. B |

## Level – II

| | | | | | | |
|---|---|---|---|---|---|---|
| 1) C | 2) C | 3) B | 4) D | 5) B | 6) C | 7) A |
| 8) A | 9) B | 10) B | 11) A | 12) C | 13) B | 14) D |
| 15) B | 16) C | 17) D | 18) B | 19) C | 20) D | 21) A |
| 22) B | 23) a-q;b-q;c-r;d-p | | 24) B,D | 25) B | 26) a-s;b-r;c-r;d-q | |
| 27) B | 28) B | 29) A | | | | |

1. $F = m(g + a) = 0.1(9.8 + 0.2) = 1\ N$
$F = \frac{1}{9.8}\ kg \cdot wt$

3. $a = 2\ m/s^2; m = 0.5\ kg$
from the $FBD$ of block $A$
$mg - N = ma \Rightarrow N = mg - ma \Rightarrow N = m(g - a)$
$N = 0.5(10 - 2) = 4\ N$

4. ' ' will not accelerate w.r.t ground. hence, there is no force on it.

5. $T = mg = \frac{50}{1000} \times 9.8 = 0.49\ N$

6. $v = u + at, \tan\theta = \frac{a}{g}$

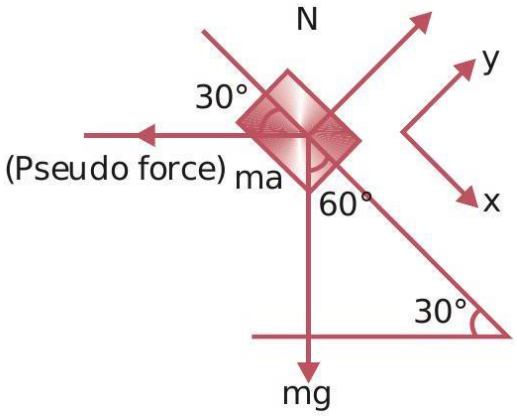
7. For stationary lift $t_1 = \sqrt{\frac{2h}{g}}$ and when the lift is moving up with constant acceleration $t_2 = \sqrt{\frac{2h}{g+a}}$ $\therefore t_1 > t_2$

8. $w_1 = mg + ma; w_2 = mg - ma; w_3 = mg$

9. $\Delta w = 2ma$

10. $T + N - m_1 g = m_1 a; T - N = mg = m_2 a;$
$T = 1125$ N and $N = 375$ N

11. Draw the free body diagram of collar A w.r.t. moving frame




Given that w.r.t. shaft the collar should be at rest.
$\therefore F_{net}$ in any direction is 0 .
Apply $\sum F_x = 0$
$\Rightarrow ma\cos 30° = ma\cos 60°.$
$\Rightarrow a = g\frac{\cos 60°}{\cos 30°} = \frac{g}{\sqrt{3}} = 5.7\ m/s^2.$

12. $\tan\theta = \frac{a + g\sin\alpha}{g\cos\alpha}$

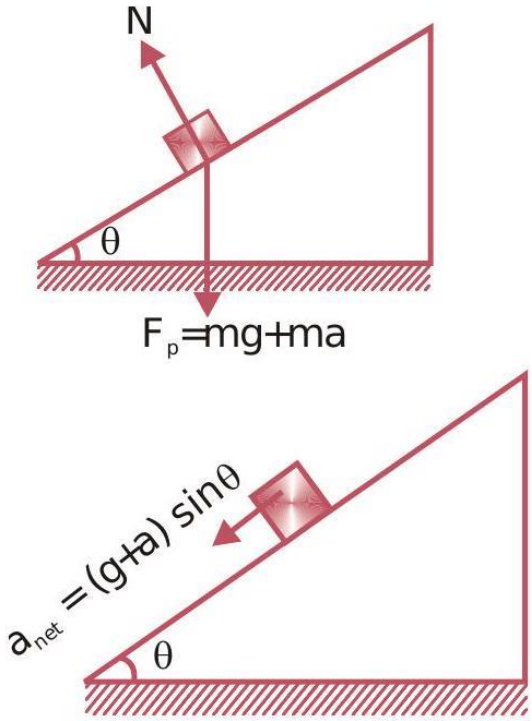
13. $a = g\sin\theta + a\cos\theta$

14. $a = g(\cos\theta - \sin\theta)$

15. $s = \frac{1}{2}at^2 \Rightarrow 2 = \frac{1}{2}a(0.6)^2$
$a = \frac{4}{0.36} = 11.11\ m/s^2$
$T - mg = ma; T = m(g + a)$
$T = 3(9.8 + 11.11) = 62.7$ N

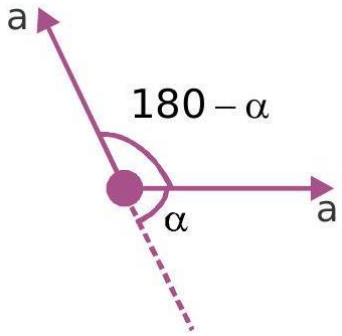
16. $f_{net}$ (downward) $= mg\sin\theta + ma\cos\theta$
$= m(g\sin\theta + a\cos\theta)$
$\therefore g_{eff} = g\sin\theta + a\cos\theta$
Time $(T)$ required to cover 2 L distance along inclined would be
$T = \sqrt{\frac{2L}{g_{eff}}} = \sqrt{\frac{2l}{(g\sin\theta + a\cos\theta)}}$

17. Since, acceleration of block w.r.t. wedge (an accelerating or noninertial frame of reference) is to be find out.



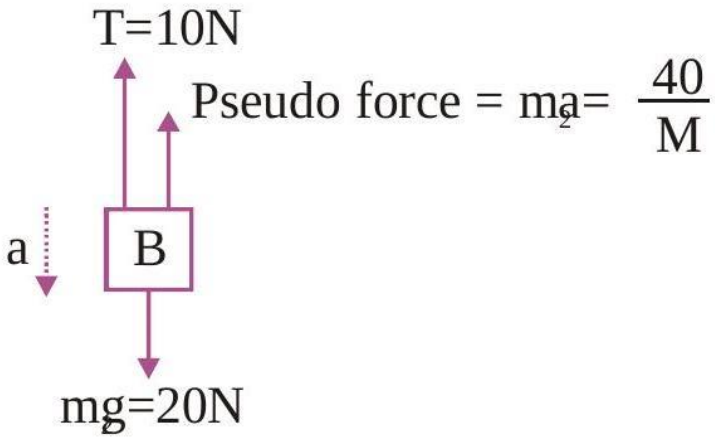

FBD of 'block' w.r.t. 'wedge' is shown in figure.
The acceleration would had been $g$ sin $\theta$ (down the plane) if the lift were stationary or when only weight (i.e., mg) acts downwards.
Here, downward force is $m(g + a)$.
∴ Acceleration of the block (of course w.r.t. wedge) will be $(g + a)$sin $\theta$ down the plane.

18.

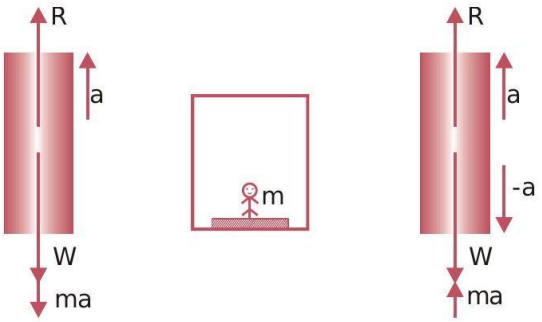



$a_{\text{res}} = \sqrt{a^2 + a^2 + 2a^2(-\cos\ \alpha)}$
$= 2a\sin\left(\frac{\alpha}{2}\right)$

19. Mass $m_1$ moves with constant velocity if tension in the lower string
$T_1 = m_1\ g = (1)(10) = 10$ N (i);
∴ Tension in the upper string is
$T_2 = 2\ T_1 = 20$ N
Acceleration of block M is therefore,
$a = \frac{T_2}{M} = \frac{20}{M}$
This is also the acceleration of pulley 2.

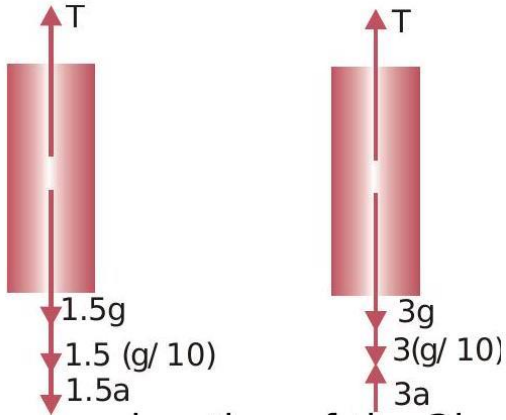



Absolute acceleration of mass $m_1$ is zero. Thus, acceleration of $m_1$ relative to pulley 2 is a upwards or acceleration of $m_2$ with respect to pulley 2 is a downwards. Drawing free body diagram of $m_2$ with respect to pulley 2 .
Equation of motion gives
$20 - \frac{40}{M} - 10 = 2a = \frac{40}{M}$.
Solving this we get $M = 8$ kg 20,21 .




When the elevator is accelerating upwards, maximum weight will berecorded.
$R - (W + ma) = 0$
$\Rightarrow R = W + ma = m(g + a)$maxwt
When decelerating upwards, maximum weight will be recorded.
$R + ma - W = 0$
$\Rightarrow R = W - ma = m(g - a)$maxwt
So, $m(g + a) = 72 \times 9.9$
$m(g - a) = 60 \times 9.9$
Now, $mg + ma = 72 \times 9.9$
$\Rightarrow mg - ma = 60 \times 9.9$
$\Rightarrow 2mg = 1306.8 \Rightarrow m = \frac{1306.8}{2 \times 9.9} = 66$ kg
So, the true weight of the man is 66 kg $\Rightarrow a = \frac{72 \times 9.9 - 66 \times 9.9}{66} = \frac{9.9}{11} = 0.9$ m/s$^2$

22.




Let the acceleration of the 3 kg mass relative to the elevator is ' $a$ ' in the downward direction.

As shown in the free body diagram $T - 1.5\ g - 1.5(\ g/10) - 1.5a = 0$ from figure (1) and $T - 3g - 3(g/10) + 3a = 0$ from figure (2)

$\Rightarrow T = 1.5\ g + 1.5(\ g/10) + 1.5a$

and $T = 3\ g + 3(\ g/10) - 3a$

Equation (i) $\times 2$

$\Rightarrow 3\ g + 3(\ g/10) + 3a = 2\ T$

Equation (ii) $\times 1$

$$\Rightarrow 3g + 3(g/10) - 3a = T$$

Subtracting the above two equations

We get, $T = 6a$

Subtracting $T = 6a$ in the equation (ii); $6a = 3g + 3(g/10) - 3a$.

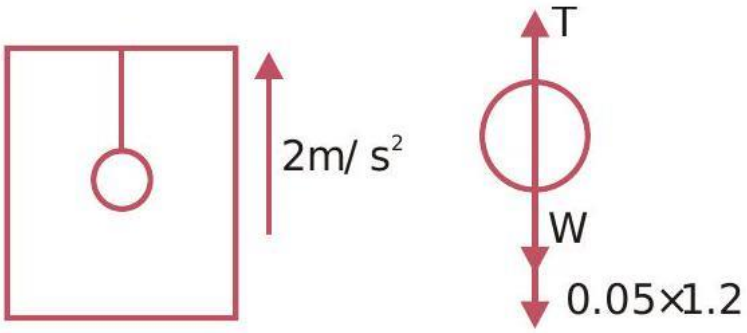
$\Rightarrow 9a = \frac{33\ g}{10} \Rightarrow a = \frac{(9.8)33}{10} = 32.34$

$\Rightarrow a = 3.59$

$\therefore T = 6a = 6 \times 3.59 = 21.55$

Mass $= \frac{wt}{g} = \frac{43.1}{9.8} = 4.39 = 4.4$ kg

23. a) The tension in the string is found out for the different conditions from the free body diagram as shown below.

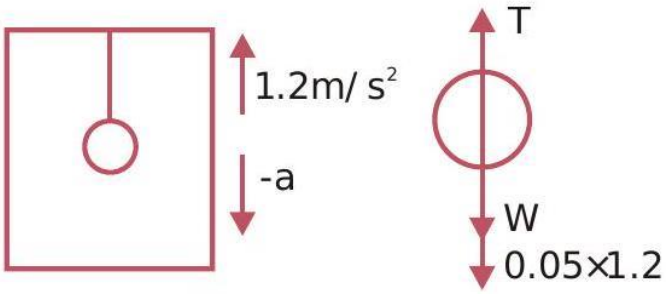



$T - (W + 0.06 \times 1.2) = 0$

$\Rightarrow T = 0.05 \times 9.8 + 0.05 \times 1.2 = 0.55$ N

$\Rightarrow T = 0.05 \times 9.8 + 0.05 \times 1.2 = 0.55$ N

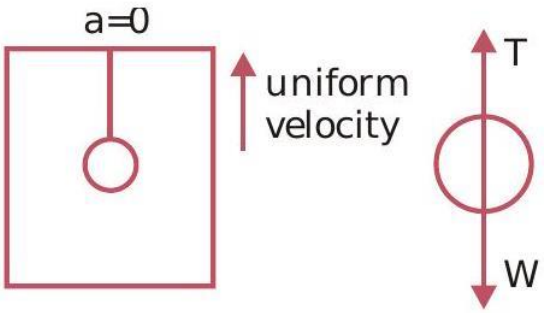
b)




$T - (W + 0.05 \times 1.2) = 0$

$\Rightarrow T = W + 0.05 \times 1.2 = 0.55$ N

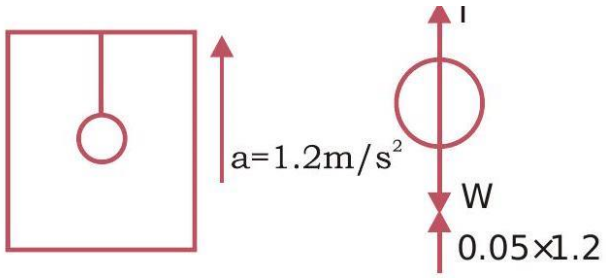
c)




When the elevator makes unifrom motion $T - W = 0$

$\Rightarrow T = W = 0.05 \times 9.8 = 0.49$ N

d)




$T + 0.05 \times 1.2 - W = 0$

$\Rightarrow T = W - 0.05 \times 1.2 = 0.43$ N

24. 2,4

25. 2

26. $\vec{a}_A = \frac{d\vec{v}_A}{dt} = (2\vec{i})\text{m/s}^2; \vec{a}_B = \frac{d\vec{v}_B}{dt} = 0;$

$\vec{a}_c = \frac{d\vec{v}_c}{dt} = (4\vec{j})\text{m/s}^2$

Now, pseudo force is in opposite direction of acceleration of frame from where object is observed.

27. $m(g + a_0)\sin\theta = ma$

28. $S = ut + \frac{1}{2}at^2; u = 0;$

$s = L\cos\theta; t = \sqrt{\frac{2\ L}{a\cos\theta}}$

29. $v$ = constant, $a_0 = 0;$

$a = g\sin\theta; t = \sqrt{\frac{2L}{g\sin\theta \cdot \cos\theta}}$

## Worksheet – 5

# Level – I

| | | | | |
|---|---|---|---|---|
| 1. B | 3. A | 5. C | 7. C | 9. D |
| 2. A | 4. B | 6. C | 8. A | |

# Level – II

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| 1) C | 2) B | 3) D | 4) C | 5) B | 6) C | 7) A | 8) D |
| 9) B | 10) C | 11) B | 12) B | 13) A | 14) A | 15) C | 16) C |
| 17) A,B | 18) A,B,C,D | 19) B,C | 20) A | 21) a-s, b-r, c-p, d-q | | 22) D | 23) D |
| 24) C | 25) C | 26) C | 27) C | 28) C | 29) B | 30) C | |

1. $N = mg\cos\theta$
2. $a = \frac{P}{m+M}; f = ma \Rightarrow f = \frac{PM}{m+M}$
3. $m_1 = 2$ kg, $m_2 = 1$ kg, $F = 3$ N
   $a = \frac{F}{m_1 + m_2} \Rightarrow a = \frac{3}{2+1} = 1\ m/s^2$
   $F = m_2 a \Rightarrow F = 1 \times 1 = 1$ N
4. $m_1 = 10$ kg, $m_2 = 6$ kg, $m_3 = 4$ kg, $F = 40$ N
   $$a = \frac{F}{m_1 + m_2 + m_3} = \frac{40}{10+6+4} = \frac{40}{20} = 2\ m/s^2$$
   $F = (m_1 + m_2)a \Rightarrow F = (10+6)2 = 32$ N
5. $m_1 = 10$ kg, $m_2 = 6$ kg, $m_3 = 4$ kg, $F = 40$ N
   $a = \frac{F}{m_1 + m_2 + m_3} \Rightarrow a = \frac{40}{20} = 2\ m/s^2$
   $F = m_1 a \Rightarrow F = 10 \times 2 = 20$ N
6. $m_1 = 4$ kg, $m_2 = 2$ kg, $F = 20$ N,
   $a = \frac{10}{3} m/s^2$
   $F = m_2 a \Rightarrow F = 2 \times \frac{10}{3} = \frac{20}{3}$ N
7. $F_{net} = 4\ kg \times a \Rightarrow F_{net} = 4 \times \frac{10}{3} = \frac{40}{3}$ N
8. $m_1 = 16$ kg, $m_2 = 8$ kg, $m_3 = 4$ kg,
   $F = 140$ N
   $a = \frac{F}{m_1 + m_2 + m_3} \Rightarrow a = \frac{140}{16+8+4} = \frac{140}{28}$
   $= 5\ m/s^2$
   $F_{16} = 16\ kg \times a \Rightarrow F_{16} = 16 \times 5 = 80$ N
9. $m_1 = 16$ kg, $m_2 = 8$ kg,
   $m_3 = 4$ kg, $F = 140$ N, $a = 5\ m/s^2$
   $F = (m_2 + m_3)a \Rightarrow F = (8+4) \times 5 = 60$ N
10. $m_1 = 16$ kg, $m_2 = 8$ kg,
    $m_3 = 4$ kg, $F = 140$ N, $a = 5\ m/s^2$
    $F = m_3 a \Rightarrow F = 4 \times 5 = 20$ N
11. $a = \frac{F}{m_1+m_2+m_3} \Rightarrow a = \frac{50}{1+4+2} = \frac{50}{7}\ m/s^2$
12. $F_{12} = m_1 a \Rightarrow 1 \times \frac{50}{7} = \frac{50}{7} N$
13. $50 - N = 2 \times \frac{50}{7} \Rightarrow N = 50 - 2 \times \frac{50}{7}$

$$\Rightarrow N = \frac{350 - 100}{7} = \frac{250}{7} N$$

14. $F_{resultant} = 120 - 50 = 70$ N

$m_1 = 1$ kg, $m_2 = 4$ kg, $m_3 = 2$ kg

$a = \frac{F}{m_1 + m_2 + m_3} \Rightarrow a = \frac{70}{1 + 4 + 2} = \frac{70}{7}$

$= 10$ m/s$^2$

15. $(120 - N) = mg \Rightarrow 120 - N = 1 \times 10 = 110$ N

16. $F_{2\ kg} = m_3 a \Rightarrow F_{2\ kg} = 2 \times 10 = 20$ N

17. Gravitational and electro magnetic.

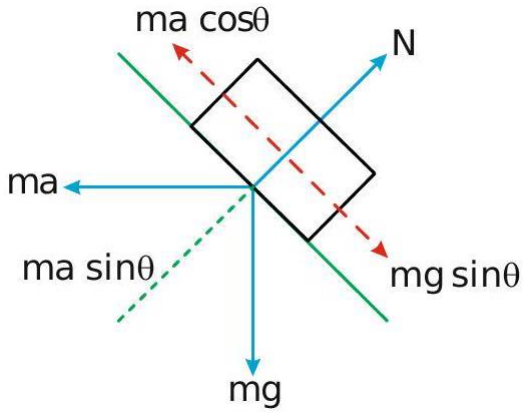
18. $a = \frac{F}{m_1 + m_2 + m_3}$

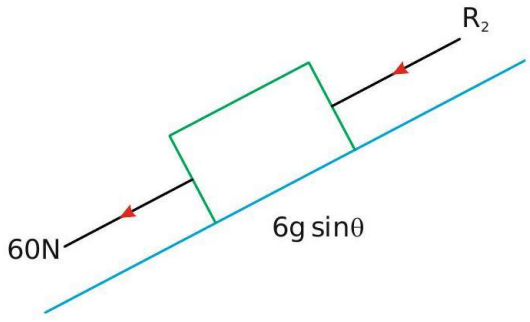
$F - f_{12} = m_1 a \Rightarrow f_{12} = F - m_1 a$

$= -m_1 \left(\frac{F}{m_1 + m_2 + m_3}\right) = \frac{(m_2 + m_3)F}{(m_1 + m_2 + m_3)}$

$f_{23} = m_3 a = f_{23} = \frac{m_3 F}{m_1 + m_2 + m_3}$

22. $a = \frac{F}{m_1 + m_2 + m_3} \Rightarrow a = \frac{14}{4+2+1} = 2$ m/s$^2$ $F_{AB} = (m_2 + m_3)a \Rightarrow F_{AB} = (2 + 1)2 = 6$ N




$60 - R_2 - 30 \times \frac{1}{2} = 6 \Rightarrow R_2 = 39N$

# Worksheet – 6

## Level – I

1. 3
2. 1
3. 2
4. 4
5. 1
6. 4
7. 2

5. $\text{Tan}\ \theta = \frac{ma}{mg}$

$T = m\sqrt{a^2 + g^2}$

6. $\mathrm{T}\cos\theta = \mathrm{mg}$

$$\frac{\mathrm{T}}{\mathrm{mg}} = \frac{1}{\cos\theta}$$

$1 : \cos\theta$

7. In equilibrium

$\mathrm{T}\sin\theta = \mathrm{F}, \mathrm{T}\cos\theta = \mathrm{mg}$

$\mathrm{F} = \mathrm{mg}\tan\theta$

$\mathrm{T} = \sqrt{\mathrm{F}^2 + (\mathrm{mg})^2}$

$x$ is the displacement

$$\frac{x}{F} = \frac{I}{T} = \frac{\sqrt{I^2 - x^2}}{mg}$$

$$\frac{x}{\mathrm{I}} = \frac{\mathrm{F}}{\mathrm{T}} = \frac{\sqrt{\mathrm{I}^2 - \mathrm{x}^2}}{\mathrm{mg}}$$

# Level – II

| | | | | | | |
|---|---|---|---|---|---|---|
| 1) C | 2) B | 3) C | 4) C | 5) D | 6) B | 7) A |
| 8) C | 9) B | 10) B | 11) B | 12) C | 13) B | 14) D |
| 15) a-q, b-q, c-r, d-p | | 16) C | 17) B | 18) A | 19) C | 20) C |
| 21) A | 22) B | 23) A | 24) B | 25) C | 26) D | |

1. $\mathrm{T} = \sqrt{\mathrm{F}^2 + (\mathrm{mg})^2}$

$\mathrm{T} = \sqrt{4^2 + (2)^2} = \sqrt{16 + 4} = \sqrt{20}$ N

2. $a = \frac{F}{(m_1 + m_2)}$

$$a = \frac{20}{(2+2)} = \frac{20}{(4)} = 5 \text{ m/s}^2$$

3. $a = \frac{F}{(m_1 + m_2)} = \frac{30}{6} = 5 \text{ m/s}^2$

$\mathrm{T} = \mathrm{m}_2(\mathrm{a}) \Rightarrow 3(5) = 15$ N

5. $T = g(m_1 + m_2)$

$T = 10(0.2 + 0.3)$

$= 10(0.5) = 5$ N

$T_2 = m_2 g = 0.3 \times 10 = 3N$

6. $F = T\sin\theta$

$mg = T\cos\theta$

$$\frac{(1)}{(2)} = \frac{F}{mg} = \frac{T\sin\theta}{T\cos\theta}$$

$F = mg\tan\theta$

$F = (0.5)(9.8)\tan 60^\circ = 4.9\sqrt{3}N$

7. For equilibrium of weight

$T_1\cos 60^\circ + w = T_2\sin 45^\circ$

$T_1 + 2w = T_2\sqrt{2}$

$T_1\sin 60^\circ = T_2\cos 45^\circ$

$\sqrt{3}T_1 = \sqrt{2}T_2$
From equation 1&2
$\sqrt{3}T_1 - T_1 = 2w$
$T_1 = \dfrac{2w}{\sqrt{3}-1}$
From equation 1 and 2
$T_2 = \dfrac{\sqrt{6}w}{\sqrt{3}-1}$

8. $T_1 = m_1 a = 10(1) = 10$ N
$F - T_2 = m_3 a$
$60 - T_2 = (30)1$
$T_2 = 30N$

9. $a = \frac{60-2mg\sin\theta}{m_1+m_2}$
$a = \dfrac{60 - 2(2)(10)\sin 30^\circ}{2+2}$
$a = \dfrac{60-20}{4}$
$a = \dfrac{40}{4} = 10$ m/s$^2$
$a = 10$ m/s$^2$
$T - mg\sin\theta = ma$
$T - 2(10)\sin 30^\circ = 2(10)$
$T - 10 = 20$
$T = 30N$

10. $a = g\frac{(2-3\sin\theta)}{2+3}$
$a = 10 \times \dfrac{(2-3\sin 30^\circ)}{2+3}$
$= 10^2 \times \dfrac{(2-1.5)}{2+3} = 2(0.5)$
$a = 1$ m/s$^2$

11. $T = m_1 g - m_1 a$ $T = m_1(g-a)$
$T = 2(10-1)$
$T = 2(9)$
$T = 18N$

12. $w_1\cos 53^\circ + w_3\cos 37^\circ = w_2$
$w_1\sin 53^\circ = w_3\sin 37^\circ$
$4w_1 = 3w_3$
from equation (1) and (2)
$\dfrac{3w_1}{5} + \dfrac{4w_1}{5} = 400$
$3w_1 + 4w_1 = 2000$
$3\left(\dfrac{3w_3}{4}\right) + 4w_3 = 2000$
$W_3 = 320$ N
and $w_1 = 240$ N

13. For equilibrium of weight, we use
$T_1\sin 30^\circ + T_2\sin 45^\circ = w$
$\dfrac{T_1}{2} + \dfrac{T_2}{\sqrt{2}} = w$

$T_1 + \sqrt{2}T_2 = 2w$ and $T_1 \cos 30^\circ = T_2 \cos 45^\circ$

$\sqrt{3}T_1 = \sqrt{2}T_2$

From equation (1) and (2)

$T_1 + \sqrt{3}T_1 = 2w$

$T_1 = \dfrac{2w}{1+\sqrt{3}}, T_2 = \dfrac{\sqrt{6}w}{1+\sqrt{3}}$

14. $\mathrm{T} = \mathrm{F}\left(1 - \frac{\mathrm{I}}{\mathrm{L}}\right)$

$\mathrm{T} = 10\left(1 - \dfrac{2}{5}\right)$

$\mathrm{T} = 10\left(\dfrac{3}{5}\right) = 6\ \mathrm{N}$

$\dfrac{\mathrm{T}}{\mathrm{F}} = \dfrac{6}{10} = \dfrac{3}{5} = 3.5$

15. a) $\mathrm{T} = \mathrm{m(a+g)}$

$= 0.05(9.8 + 1.2) = 0.55\ \mathrm{N}$

b) $\mathrm{T} = \mathrm{m(a+g)}$

$= 0.05(9.8 + 1.2) = 0.55\ \mathrm{N}$

c) $\mathrm{T} = 0.05 \times 9.8 = 0.49\ \mathrm{N}$

d) $\mathrm{T} + (0.05 \times 1.2) - \mathrm{W} = 0$

$\mathrm{T} = (0.05 \times 1.2) - \mathrm{W} = 0.49 - (0.05 \times 1.2)$

$\mathrm{T} = 0.43\ \mathrm{N}$

16. $a = \frac{F}{m_1+m_2+m_3} = \frac{F}{3m}$

17. $T_1 = (2m)a = 2m\frac{F}{3m}$

$\mathrm{T}_1 = \dfrac{2\ \mathrm{F}}{3}\ \mathrm{N}$

18. Net force of block $A$

$\mathrm{F}_{\mathrm{net}} = \mathrm{ma}$

$\mathrm{m}\dfrac{\mathrm{F}}{3\ \mathrm{m}} = \dfrac{\mathrm{F}}{3}$

19. $\mathrm{T}\cos\theta = \mathrm{mg}$

$\mathrm{T}\cos\theta = 1$

$T = \dfrac{1}{\cos\theta}$

$T\sin\theta = F$

$T\sin\theta = 2$

$T = \dfrac{2}{\sin\theta}$

$T = \sqrt{F^2 + (mg)^2} = \sqrt{2^2 + 1^2} = \sqrt{5}N$

20. $T_1 = mg\sin 30^\circ = 5 \times 10 \times \frac{1}{2} = 25N$ $T_2 = mg\sin 60^\circ = 5 \times 10 \times \frac{\sqrt{3}}{2} = 25\sqrt{3}N$

21. $T_1\cos\theta = w = 30N$

$T_1\sin\theta = 40N$

$(1^2 + 2^2)$

$T_1 = \sqrt{30^2 + 40^2} = 50N$

$\dfrac{T_1}{T_2} = \dfrac{50}{30} = \dfrac{5}{3} = 5:3$

22. $a = \frac{F}{m_1+m_2+m_3}$

$a = \dfrac{14}{7} = 2\ \mathrm{m/s^2}$

$T_2 = 4a = 4(2) = 8N$
$T_2 = 8N$
$F - T_1 = 1(a)$
$F - T_1 = 2 \Rightarrow 14 - 2 = T_1$
$T_1 = 12$
$\frac{T_1}{T_2} = \frac{12}{8} = \frac{3}{2}$

23. $m_1 g - T_1 = m_1 a$
$T_1 - T_2 = m_2 a$
$T_1 - m_3 = m_3 a$
$a = \frac{(M_1 - M_3)g}{M_1 + M_2 + M_3}$
$T_1 = m_1 a$
$T_1 = \frac{M_1 g(2M_3 + M_2)}{M_1 + M_2 + M_3}$

24. $AC = 0.5M, BC = 0.3M\ AB = 0.4M$
$\cos\ \theta = \frac{4}{5}, \sin\ \theta = \frac{3}{5}$
According to Iami's theorem
$\frac{F}{\sin(180 - \theta)} = \frac{8}{\sin(90 + \theta)} = \frac{T}{\sin\ 90°}$
$\frac{F}{\sin\ \theta} = \frac{8}{\cos\ \theta} = \frac{T}{1}$
$T = \frac{8}{\cos\ \theta} = \frac{8}{\frac{4}{5}} = 10N\ \left(\cos\ \theta = \frac{4}{5}\right)$
$F = \frac{8\sin\ \theta}{\cos\ \theta} = \frac{8\left(\frac{3}{5}\right)}{\left(\frac{4}{5}\right)} = 6$ N

25. Lower wire mass = 0.1 kg
$T_1 - 2g = 2(a) \Rightarrow T_1 = 2(g + a)$
$= 2(9.8 + 0.2) = 20N$

26. $T_2 - 5g = 5a$
$T_2 = 5(g + a) = 5(9.8 + 0.2) = 50$ N

# Worksheet – 7

## Level – I

1. D
2. B
3. C
4. A
5. C
6. B

## Level – II

| | | | | | | |
|---|---|---|---|---|---|---|
| 1) 4 | 2) 3 | 3) 4 | 4) 4 | 5) 3 | 6) 4 | 7) 2 |
| 8) 2 | 9) 4 | 10) 1 | 11) 3 | 12) 4 | 13) 1 | 14) 1,4 |
| 15) a-r, b-s, c-p, d-q | | 16) 4 | 17) 2 | 18) 3 | 19) 1 | 20) 4 |
| 21) 3 | 22) 1 | 23) 1 | 24) 1 | 25) 2 | 26) 3 | 27) 2 |

1 $m_1 = 10$ kg; $m_2 = 5$ kg $\Rightarrow 5$ g – T = 5a

$T - 10\ g\sin\ 30° = 10a$

$(1) + (2)5a = 0 \Rightarrow a = 0$

2. $5\ g - T = 5a$

(1)

$T - 8\ g\sin\ \theta = 8a$

$(1) + (2)$

$\rightarrow 5(10) - 8(10) \cdot \frac{1}{2} = 13(a)$

$= 10 = 13(a)$

$a = 10/13\ m/s^2$

3 $(10)(10) \cdot \sin\ 30° - T = 10(a)$

$T - 4(10) = 4(a)$

(1)-(2)

100. $\frac{1}{2} - T = 109$

$T - 4 = 49$

$50 - 40 = 149$

$a = \frac{10}{14} = \frac{5}{7}\ m/s^2$

$50 - T = 10 - \frac{5}{7}$

$T = 50 - \frac{50}{7}$

$T = \frac{350 - 50}{7} = \frac{300}{7}\ N$

4 $\sin\ \theta = \frac{1}{x}$

opposite $= 1$

Hypothenious $= x$

adjacent $= \sqrt{x^2 - 1}$

$\tan\ \theta = \frac{opp}{Adj} = \frac{1}{\sqrt{x^2 - 1}}$

5 Acceleration of block and wedge

$a = \frac{F}{(m + M)}$

FBD of block w.r.t ground (apply real force)

$\sum Fy = 0$

$N\cos\ \theta = mg$

$\sum F_n = ma$

$ma = N\sin\ \theta$

form equation (1) and (2) $a = g\tan\ \theta$ $F = (M + m)a$

$F = (M + m)g\tan\theta$

6

$a = g\frac{(m_1\sin\ \alpha - m_2\sin\ \beta)}{m_1 + m_2}$

$a = g\frac{\left(2\left(\frac{4}{5}\right) - 2\left(\frac{3}{5}\right)\right)}{2 + 2}$

$a = \frac{g}{4} \times 2\left(\frac{4}{5} - \frac{3}{5}\right)$

2

$a = \frac{g}{10}\ m/s^2$

7 $4\ g - 12 - T = 4(5.5)$

$4\ g - 12 - T = 22$

$T = 4\ g - 12 - 22 \Rightarrow T = 4(9.8) - 34$

$T = 39.2 - 34 \Rightarrow T = 5.2\ N$

8 $ma = mg\sin\theta + ma\cos\theta$

$a = g\sin\theta + a\cos\theta$

9 $mg\cos\theta - mg\sin\theta = ma$

$a = g(\cos\theta - \sin\theta)$

10

$T = \frac{m_1 m_2 g}{m_1 + m_2}(\sin\alpha + \sin\beta)$

$= \frac{2 \times 2 \times 10}{2+2}(\sin 30 + \sin 30°) = 10\ N$

$T_R = \sqrt{T^2 + T^2 + 2\ T^2 \cos(180 - (\alpha + \beta))}$

$= \sqrt{T^2 + T^2 - 2\ T^2 \sin 30°}$

$= \sqrt{T^2 + T^2 - 2T^2} \Rightarrow T = 10N$

11 $F_{net}$ ( downword)

$= mg\sin\theta + ma\cos\theta - m(g\sin\theta + a\cos\theta)$

$g_{eff} = g\sin\theta + a\cos\theta$

$T = \sqrt{\frac{2L}{g_{eff}}} = \sqrt{\frac{2L}{g\sin\theta + a\cos\theta}}$

12 Acceleration of the block w.r.t wedge. wille $(g + a)\sin\theta$ down the plane.

13 From FBD (1) $T + Ma - Mg = 0$

From FBD (2)

$T - M^1 a - R\sin\theta = 0$

From FBD (3) $R\sin\theta - ma = 0$

From $FBD$ (4) $R\cos\theta - mg = 0$

By eliminating $T$ and $R$ from above equations

We get $M = \frac{M^1 + m}{\cot\theta - 1}$

14 $a = \frac{4-3}{7} \times 9.8$

$= \frac{98}{70} = 1.4\ m/s^2$

$T = \frac{2(3)(4)}{7}$

$= \frac{24 \times 9.8}{7} = 33.6\ N$

15. FBD for 6 kg

$F - 6g\sin 30° N_1 = 6a$

$80 - 6 \times 10 \times \frac{1}{2} - N_1 = 6a$

$50 - N_1 = 6a$

FBD for 4 kg

$N_1 - 4 \times \frac{1}{2} \times 10 - N_2 = 4a$

$N_1 - 20 - N_2 = 4a$

FBD for 2 kg

$N_2 - 2 \times 10 \times \frac{1}{2} = 2a$

$N_2 - 10 = 2a$

$(1) + (2) + (3)$

a)

$50 - X_1 = 6a$

$NN_1 - 20 - NV_2 = 4a$

$\frac{N_2}{50} - \frac{10}{30} = \frac{2a}{12}$

$a = \frac{20}{12}$

b) Net force on 6 kg block

$F = 6 \times \frac{20}{12} = 10$ N

c) Normal reaction between 4 kg and 2 kg

$50 - N_1 = 6a$

10

$50 - N_1 = \emptyset \times \frac{20}{12}$

2

$N_1 = 50 - 10 = 40$ N

d. Normal reaction between 6 kg and 4 kg

$N_1 = 20 - N_2 = 4a$

$40 - 20 - N_2 = 4 \times \frac{20}{12}$

3

$20 - N_2 = \frac{20}{3}$

$N_2 = \frac{20}{3} = \frac{60 - 20}{3} = \frac{40}{3}$ N

$16a = g\frac{(m_1 - m_2)\sin\theta}{m_1 + m_2}$

and $a_B = a_A \sin\theta$

$T = m_1 g - m_1 a$ sub 'a' value

$T = \frac{m_1 m_2 g(1 + \sin\theta)}{m_1 + m_2}$

17. $R\sin\theta = ma$

$R\cos\theta = mg$

from (1) and (2)

$\tan\theta = \frac{a}{g}$

$a = g\tan\theta$

18. (a) Force to applied on the wedge.

$F = (M + m)a$

$F = (M + m)g\tan\theta$

19. Force excerted by the wedge on the block.

$R\cos\theta = mg$ (or)

$R = \frac{mg}{\cos\theta} = mg\sec\theta$

(iii) II method

$R = mg\cos\theta + ma\sin\theta$

$= mg\cos\theta + m(g + \tan\theta)\sin\theta$

$= mg\cos\theta + mg \cdot \frac{\sin^2\theta}{\cos\theta}$

$= mg\frac{(\cos^2\theta + \sin^2\theta)}{\cos\theta}$

$= \frac{mg}{\cos\theta} = mg\sec\theta$

20. $f = \sqrt{\frac{2l}{g\sin\theta}} =$

$= \sqrt{\frac{2\times 10}{10\times\frac{1}{2}}} = \sqrt{4} = 2\text{sec}$

21. $t = \sqrt{\frac{4b}{g}} = \sqrt{\frac{4(10)}{10}} = 2\text{sec}$

22. $a = g\sin\theta$

$f_1 + m_2 g\sin\theta = m_2 a$

from equation

(1) and (2)

$f_1 - f_2 + 3g = 6a$

as $f_1 = f_2$

$f_2 = m_1 g\sin\theta - m_1 a$

$f_2 = m_1 g\cdot\frac{1}{2} - m_1 g\sin 30^\circ$

$= \frac{mg}{2} - \frac{m\cdot g}{2} = 0$

23. $m_2\, g - T = 0$

$T - m_1\, g\sin\theta = 0$

from equation ( 1 ) and (2)

$\sin\theta = \frac{m_2}{m_1}$

$\theta = \sin^{-1}\left(\frac{m_2}{m_1}\right)$

$\therefore N = m_1\, g\cos\theta$

$= m_1 g\sqrt{1-\sin^2\theta}$

$N = m_1 g\sqrt{1-\left(\frac{m_2}{m_1}\right)^2}$

24. $T = mg\sin\theta$

$T = 100\times 10\times\sin 53^\circ$

$= 100\times 10\times\frac{4}{5}$

$= 800\ N$

for man $T - mg = ma$

$T = m(g+a)$

$800 = 5p(10+a)$

$a = 6\ m/s^2$

25. resultant acceleration $a = g$

$ma\cos\theta - mg\sin\theta = mga$

$a\cos\theta - mg\sin\theta = mg\ a\cos\theta = g\sin\theta + g$

$a = g\left(\frac{\sin\theta+1}{\cos\theta}\right)$

$a = g(\tan\theta + \sec\theta)$

26. In the time the wedge moves a distance $x$ towards left with an acceleration ' ' the block falls through a height ' $h$ ' with an acceleration ' $g$ '

$$\frac{x=\frac{1}{2}at^2}{h-\frac{1}{2}gt^2}=\frac{x}{h}=\frac{a}{g}$$

$$=\cot\theta=\frac{a}{g}\Rightarrow a=g\cot\theta$$

27. $10\sqrt{3}\cos 30^\circ = 10\sqrt{3}\cdot\frac{\sqrt{3}}{2}$

$= 5(3) = 15\ \text{m/s}^2$

$A = 15 - 5 = 10\ \text{m/s}^2$

$S = \frac{1}{2}att^2$

$1 = \frac{1}{2}(10)t^2 \Rightarrow t = \frac{1}{\sqrt{5}}\ s$

# Worksheet – 8

## Level – I

1) C 2) C 3) A 4)C 5) B 6) D

6. $a = \frac{F}{(M+m_1+m_2)}T = (M+m_2)a$

$$T=\frac{(M+m_2)F}{(M+m_1+m_2)}$$

## Level – II

1) B 2) C 3) B 4) D 5) C 6) A 7) A 8) A
9) C 10) A,C 11) a-s, b-t, c-p, d-r 12) B 13) B 14) A

1. $T = F\left(1-\frac{1}{n}\right)$

$T = 20\left(1-\frac{1}{4}\right)$

$T = 20\left(\frac{3}{4}\right)$

$T = 15$ N.

2. $T\cos\theta = mg -$

$T\sin\theta = F$

$\frac{(1)}{(2)}=\frac{T\cos\theta}{T\sin\theta}=\frac{mg}{F}$

$F = mg\tan\theta$

$F = (3)(10)\tan 60^\circ$

$F = 30\sqrt{3}.\text{N}$

3. $T_P = \frac{1}{2}mg\cot\theta$

$T_P = \frac{1}{2}(4)(10)\cot 45^\circ$

$T_P = \frac{1}{2}\times 40$

$T_P = 20$ N

4. $a = \frac{F}{m} = \frac{F}{\int C}$

$T = 2ma$

$= 2\ m\left(\frac{F}{5\ m}\right)$

$T_1 = F - \frac{2}{5}\ F$

$T_1 = \frac{3}{5}\ F$

$T_1 = \frac{3}{5} \times 20$

$T_1 = 12\ N$

from, $T_2 = F - \frac{3}{5}F$

$= F\frac{2}{5}$

$= 20 \times \frac{2}{5} = 8\ N.$

$T_1 : T_2 = 12 : 8$

$T_1 : T_2 = 3 : 2$

5. $M = 6KG, L = 60CM$

$\sigma = \frac{m}{l} = \frac{6\ kg}{0.6} = 10\ kg/m$

$a = \frac{F_{net}}{m} = \frac{45 - 21}{6} = 4\ m/s^2$

$m_{45} = (45 \times 10^{-2})(10)$

$= 4.5\ kg$

$F - T = 4.5(a)$

$45 - T = 18$

$T = 45 - 18$

$T = 27\ N$

6. $a = \frac{100-80}{2+2+48} = \frac{20}{8} = 2.5\ m/s^2$

7. $100 - (2.5)2 = T_A + 20$

$100 - 5 = T_A + 20$

$T_A = 100 - 25 = 75\ N.$

8. $T_B - 40 = 4(2.5)$

$T_B = 40 + 10$

$T_B = 50\ N.$

9. $T_C = (1 + 4)g + 5(2.5)$

$T_C = 50 + 12.5$

$T_C = 62.5\ N.$

10. $a = \frac{8}{4} = 2\ m/s^2$

$F - T_A = 2a$

$F - T_A = 2(2)$

$8 - T_A = 4\ T_A = 4\ N$

$F - T_B = 1.a$

$8 - T_B = 1(2)$

$8 - T_B = 1(2)$

$T_B = 6\ N$

11. a) $T_1 = (1.9 + 0.1)(g + a)$

$= 2(9.8 + 0.2)$

$= 2(10)$

$= 20\ N$

b) $T_2 = (1.9 + 0.2 + 2.90.05)(g + a)$
$T_2 = (5.05)(9.8 + 0.2)$
$T_2 = (5.05)(10)$
$T_2 = 50.5$ N
C) $F_{(seudo)} = (1.9 + 0.2 + 2.9 + 0.1)(g + a)$
$= (5.1)(10) = 51$ N
d) $\frac{T_1}{T_2} = \frac{200}{505}$

12. $a = \frac{F_{net}}{m} = \frac{32-20}{3} = \frac{12}{3} = 4$ m/s² density $= \frac{3}{0.3} = 10$ kg/m
$m_1 = 10(0.1) = 1$ kg
$m_2 = 10(0.2) = 2$ kg
$F - T = 2a$
$32 - T = 2(4)$
$T = 32 - 8 = 24$ N

13. $T = \left(m + \frac{3m}{4}\right) a$
$a = \frac{F}{2m}$
$T = m\left(\frac{7}{4}\right)\left(\frac{F}{2\ m}\right)$
$T = \frac{7\ F}{8}$

14. $T_A = (m_2 + m_3)a$
$T_A = (4 + 2)5 = 30$ N
$T_B = \left(m_3 + \frac{m_2}{2}\right) a$
$T_B = (4 + 1)5 = 25$ N
$T_C = m_3 a$
$T_C = (4)(5)$
$T_C = 20$ N